Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 1 — Algebraic methods
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第1章 · 代数方法
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P2-C01-R01 · P2 · Chapter 1

$f(x) = 2x^3 - 3x^2 - 11x + 6.$

(a) Show that $(x - 3)$ is a factor of $f(x)$. (2)

(b) Factorise $f(x)$ completely and hence solve $f(x) = 0$. (4)

(Total for Question 1 is 6 marks)

$f(x) = x^3 + ax + b$. When divided by $(x - 1)$ the remainder is 4, and when divided by $(x + 2)$ the remainder is -11.

(a) Find a, b. (3)

(b) Find the remainder when $f(x)$ is divided by $(2x - 1)$. (2)

(Total for Question 2 is 5 marks)

3. P2-C01-R03 · P2 · Chapter 1

$f(x) = x^4 - 5x^2 + 4$.

(a) Factorise $f(x)$ completely. (3)

(b) Find all real solutions of $f(x) = 4$. (2)

(Total for Question 3 is 5 marks)

Let $f(x) = x^3 + px^2 + qx - 12$. Both $(x - 2)$ and $(x + 3)$ are factors.

(a) Find p, q. (3)

(b) Find the remaining linear factor. (2)

(Total for Question 4 is 5 marks)

5.

For an integer n, consider $N = n^2 + n + 1$.

(a) Prove that N is odd for every integer n. (3)

(b) Disprove the claim that N is prime for every positive integer n. (2)

(Total for Question 5 is 5 marks)

$f(x) = 3x^3 + 2x^2 - 5x + 7.$

(a) Find the quotient and remainder when $f(x)$ is divided by $(x + 1)$. (3)

(b) Find the value of $f(-1)$ without substituting into the original cubic, and state the theorem used. (2)

(Total for Question 6 is 5 marks)

7.

A polynomial f leaves remainder 5 on division by $(x-2)$. Define $g(x) = f(x) + 3x - 1$.

(a) Find the remainder when $g(x)$ is divided by $(x-2)$. (2)

(b) Find k such that $(x-2)$ is a factor of $f(x) + kx$. (2)

(Total for Question 7 is 4 marks)

8. P2-C01-R08 · P2 · Chapter 1

$f(x) = x^3 - 4x^2 + x + 6.$

(a) Show that $f(x) = (x + 1)(x - 2)(x - 3).$ (3)

(b) Find the roots of $f(2t - 1) = 0.$ (2)

(Total for Question 8 is 5 marks)

9. P2-C01-R09 · P2 · Chapter 1

An integer n satisfies $1 \leq n \leq 6$.

(a) By exhaustion, prove that $n^2 + 2$ is divisible by 3 if and only if n is not divisible by 3, for these six integers. (4)

(b) Give a reason why checking only $n = 1, 2, 3$ would not by itself be a proof by exhaustion over the stated domain. (2)

(Total for Question 9 is 6 marks)

10.

The polynomial $f(x) = 2x^3 + ax^2 + bx - 6$ is divisible by $(x - 1)(x + 2)$.

(a) Find a, b. (3)

(b) Write $f(x)$ as a product of three linear factors. (2)

(Total for Question 10 is 5 marks)

11. P2-C01-R11 · P2 · Chapter 1

For real x, $F(x) = (x-2)^3 - (x-2)$.

(a) Factorise $F(x)$ completely. (3)

(b) Find all real values of x for which $F(x) = 0$, and show that their sum is 6. (2)

(Total for Question 11 is 5 marks)

$f(x) = x^3 + 2x^2 + cx + d$. When divided by $(x - 1)$ and $(x + 1)$, the remainders are equal.

(a) Show that $c = -1$. (3)

(b) Given that $(x - 2)$ is a factor, find d and the common remainder. (4)

(Total for Question 12 is 7 marks)

A student claims that if a polynomial has equal values at $x = 1$ and $x = -1$, it must contain only even powers of x.

(a) Give a polynomial counterexample of degree 3. (3)

(b) Find the remainder when your polynomial is divided by $x^2 - 1$. (2)

(Total for Question 13 is 5 marks)

$f(x) = x^3 - 2x^2 - 5x + 6.$

(a) Divide $f(x)$ by $x - 1$ and hence factorise it completely. (4)

(b) Find the exact roots of $f(\sqrt{t}) = 0$ for $t \geq 0$. (3)

(Total for Question 14 is 7 marks)

$f(x) = ax^3 + bx^2 + cx + d$ is identically equal to $(x + 1)(2x^2 - 3x + 4) + 5$.

(a) Find a, b, c, d. (3)

(b) Find $f(-1)$ and the remainder on division by $x - 2$. (3)

(Total for Question 15 is 6 marks)

16. P2-C01-R16 · P2 · Chapter 1

For integer n, let $E = (2n+1)^2 - 1$.

(a) Prove that E is divisible by 8. (3)

(b) Is E necessarily divisible by 16? Justify your answer with a specific integer value. (2)

(Total for Question 16 is 5 marks)

$P(x) = x^4 + ax^2 + b$. Its remainders on division by $x - 1$ and $x - 2$ are 0 and 9, respectively.

(a) Find a, b. (3)

(b) Find the quotient when P is divided by $x^2 - 1$. (2)

(Total for Question 17 is 5 marks)

The polynomial $f(x)$ has $(x + 1)$ as a factor. Let $g(x) = f(2x - 3)$.

(a) Find a linear factor of $g(x)$. (3)

(b) Given also that $f(3) = 8$, find the remainder when g is divided by $x - 3$. (2)

(Total for Question 18 is 5 marks)

19.

A cubic polynomial with leading coefficient 2 has roots -1, 2 and 3.

(a) Find the polynomial in expanded form. (3)

(b) Find the remainder on division by $2x + 1$. (3)

(Total for Question 19 is 6 marks)

Consider the statement: “For every real x, $x^2 \geq x$.”

(a) Find the complete set of values of x for which the statement is true. (2)

(b) Give a counterexample to the statement and explain why checking only integer values would miss every counterexample. (3)

(Total for Question 20 is 5 marks)

$f(x) = 4x^3 - 4x^2 - 11x + 6.$

(a) Show that $(2x - 1)$ is a factor and find the quotient. (3)

(b) Hence solve $f(x) = 0$ exactly. (2)

(Total for Question 21 is 5 marks)

A polynomial $P(x)$ satisfies the identity $P(x) = (x - 2)Q(x) + 3x + 1$.

(a) Find the remainder when P is divided by $x - 2$. (2)

(b) Write the identity in division form with a constant remainder. (3)

(Total for Question 22 is 5 marks)

For real x, $H(x) = x^3 + 3x^2 - 4$.

(a) Show that $H(x) = (x - 1)(x + 2)^2$. (3)

(b) Find the number of distinct real solutions of $H(x) = 0$ and state them. (2)

(Total for Question 23 is 5 marks)

$f(x) = x^3 + ax^2 + bx + 4$. The remainder on division by $x - 1$ is twice the remainder on division by $x + 1$. Also $f(2) = 12$.

(a) Find a, b. (4)

(b) Find the remainder on division by x. (2)

(Total for Question 24 is 6 marks)

For integers $m, n,$ consider the claim: “If mn is even, then both m and n are even.”

(a) Disprove the claim. (3)

(b) Prove that if m and n are both odd, then mn is odd. (3)

(Total for Question 25 is 6 marks)

$f(x) = x^4 - 2x^3 - 7x^2 + 8x + 12.$

(a) Divide $f(x)$ by $x^2 - 4$. (3)

(b) Hence factorise $f(x)$ completely. (3)

(Total for Question 26 is 6 marks)

A cubic polynomial f satisfies $f(0) = 0$, $f(1) = 0$, $f(2) = 0$ and $f(3) = 12$.

(a) Find $f(x)$. (3)

(b) Find the remainder when $f(x)$ is divided by $x + 1$. (2)

(Total for Question 27 is 5 marks)

Let n be an integer.

(a) Prove that if n is odd, then $n^2 + 2n$ is odd. (2)

(b) A student claims that if $n^2 + 2n$ is odd, then n is odd.

(i) Prove that this converse is true. (2)

(ii) Prove that, when n is even, $n^2 + 2n$ is divisible by 8. (2)

(Total for Question 28 is 6 marks)

A number n is an integer greater than 1.

(a) Prove that the product $(n-1)n(n+1)$ is divisible by 6. (3)

(b) A student claims that $(n-1)n(n+1)$ is divisible by 24 for every integer $n > 1$.

(i) Give a counterexample. (1)

(ii) Prove that the claim becomes true if n is odd. (2)

(Total for Question 29 is 6 marks)

The polynomial f is defined by $f(x) = x^3 + px^2 + qx - 12$, where p, q are constants. When $f(x)$ is divided by $x - 1$ the remainder is 6. Also, $x + 2$ is a factor of $f(x)$.

(a) Find p and q. (3)

(b) Hence

(i) write $f(x)$ as a linear factor multiplied by a quadratic factor with integer coefficients. (2)

(ii) solve $f(x) = 0$, giving exact answers. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C01-R01 · 6 marks

(a) · 2 marks

$f(3) = 54 - 27 - 33 + 6 = 0.$

按因式定理检验指定零点；必须展示代入后为0。

M1	正确代入3
A1	算得0并引用因式定理

(b) · 4 marks

$f(x) = (x-3)(2x-1)(x+2)$; roots $3, 1/2, -2$.

除x−3得$2x^2+3x-2$，再分解(2x−1)(x+2)。

M1	正确商式
M1	分解二次式
A1	完全因子
A1	三根

Question 2

P2-C01-R02 · 5 marks

(a) · 3 marks

$a = 2, b = 1$.

f(1)=4给a+b=3；f(−2)=−11给−2a+b=−3，联立。

M1	两个余式条件
M1	联立消元
A1	a与b

(b) · 2 marks

$17/8$.

代x=1/2得1/8+1+1。

M1	用线性除式零点
A1	精确余式

Question 3

P2-C01-R03 · 5 marks

(a) · 3 marks

$(x-2)(x+2)(x-1)(x+1)$.

先令u=x^2分解$(u-1)(u-4)$，再两次平方差。

M1	关于x平方分解
M1	平方差
A1	四个一次因子

(b) · 2 marks

$x=0,\pm\sqrt{5}$.

原式减4后$x^2(x^2-5)=0$；零根只计一次。

M1	正确移项并因式分解
A1	三实根

Question 4

P2-C01-R04 · 5 marks

(a) · 3 marks

$p = 3, q = -4.$

f(2)=0给2p+q=2；f(−3)=0给3p−q=13；相加5p=15，故p=3、q=−4。

M1	两因式条件
M1	解联立
A1	两参数

(b) · 2 marks

$(x + 2).$

常数−12及前两因子乘积(x−2)(x+3)常数−6，第三因子常数2，首项系数1。

M1	用常数或多项式除法
A1	剩余因子

Question 5

P2-C01-R05 · 5 marks

(a) · 3 marks

$n(n+1)$ is even, so $N = n(n+1) + 1$ is odd.

连续两整数必一偶，乘积为偶；加1为奇。

M1	连续整数中有偶数
M1	乘积为偶
A1	奇数结论

(b) · 2 marks

$n = 4$ gives $N = 21 = 3 \times 7$.

一个满足正整数条件且得合数的反例足以否定全称命题。

M1	合法反例值
A1	展示合数分解

Question 6

P2-C01-R06 · 5 marks

(a) · 3 marks

Quotient $3x^2 - x - 4$; remainder 11.

长除法商如上；展开$(x+1)(3x^2-x-4)+11$复核原式。

M1	正确长除法步骤
A1	商
A1	余式

(b) · 2 marks

$f(-1) = 11$ by the remainder theorem.

在除法恒等式中$x=-1$使除式项为0，只剩余式。

M1	从除法式读取
A1	值和定理

Question 7

P2-C01-R07 · 4 marks

(a) · 2 marks

10.

f(2)=5，g(2)=5+6−1=10。

M1	使用f(2)条件
A1	g余式

(b) · 2 marks

$k = -5/2$.

新多项式在2处为5+2k，要因式故设0。

M1	因式条件
A1	k

Question 8

P2-C01-R08 · 5 marks

(a) · 3 marks

$(x-2)(x-3)=x^2-5x+6$; multiplying by $x+1$ gives $f(x)$.

直接展开证明恒等式；不能仅列三个数为零而不说明首项和次数。

M1	展开两个因子
M1	乘第三因子
A1	与原式一致

(b) · 2 marks

$t=0,3/2,2.$

输入2t−1必须等于原根−1、2、3，分别解一次方程。

M1	正确三个输入条件
A1	全部t值

Question 9

P2-C01-R09 · 6 marks

(a) · 4 marks

$n^2+2=3,6,11,18,27,38$ for $n=1,2,3,4,5,6$; divisible at $1,2,4,5$ only.

穷举所有六个允许值，列结果并与n被3整除情况逐一比较。

M1	列齐六种情况
A1	六个值
M1	检查整除
A1	双向结论

(b) · 2 marks

The permitted cases $n=4,5,6$ would remain unchecked.

有限域穷举必须覆盖每一种允许情况；本题未另证明周期推广。

M1	明确遗漏哪些情况
A1	穷举覆盖理由

Question 10

P2-C01-R10 · 5 marks

(a) · 3 marks

$a=5, b=-1.$

f(1)=0给a+b=4；f(−2)=0给4a−2b=22；解a=5,b=−1。

M1	两零点条件
M1	联立
A1	参数

(b) · 2 marks

$(x-1)(x+2)(2x+3).$

先两因子乘x^2+x-2，商为2x+3；展开复核。

M1	正确商因子
A1	完整三因子

Question 11

P2-C01-R11 · 5 marks

(a) · 3 marks

$(x-2)(x-3)(x-1)$.

提x−2后余因子$(x-2)^2-1$，再用平方差。

M1	提公共因子
M1	平方差
A1	完整分解

(b) · 2 marks

$x=1,2,3$; sum 6.

各一次因子分别为0，求和。

M1	三根
A1	全部值及和

Question 12

P2-C01-R12 · 7 marks

(a) · 3 marks

$f(1)-f(-1)=2+2c=0\Rightarrow c=-1.$

两个余式分别3+c+d与1−c+d，相等后d抵消。

M1	两余式条件
M1	正确相减
A1	c

(b) · 4 marks

$d=-14$; common remainder -12.

f(2)=8+8−2+d=0给d=−14；f(1)=3−1−14=−12。

M1	因式条件
A1	d
M1	回代求余式
A1	余式

Question 13

P2-C01-R13 · 5 marks

(a) · 3 marks

One possible example: $f(x) = x^3 - x$ has $f(1) = f(-1) = 0$ but contains odd powers.

提供一个三次多项式并逐一验证两个值相同，同时指出奇次项仍存在。

M1	合法三次多项式
M1	验证相等值
A1	指出违反结论

(b) · 2 marks

0 for $f(x) = x^3 - x = x(x^2 - 1)$.

把(a)中的明确例子作分解，除式是其因式。 这里针对(a)所给示例。一般若所选多项式在$x = 1, -1$的共同值为c，除以$x^2 - 1$的余式是常数c，并不必为零；任何符合条件的反例及相应余式均应认可。

M1	使用自己的同一个例子
A1	正确给出自己所选多项式的余式

Question 14

P2-C01-R14 · 7 marks

(a) · 4 marks

Quotient x^2-x-6; $f=(x-1)(x-3)(x+2)$.

长除法得二次商且余式0，再分解二次商。

M1	正确长除法
A1	商和零余式
M1	分解商
A1	三因子

(b) · 3 marks

$t=1,9$.

$\sqrt{\text{t}}$只能非负，原根-2须排除；输入为1或3后平方得到t。

M1	使用非负输入限制
M1	根映射回t
A1	完整集合

Question 15

P2-C01-R15 · 6 marks

(a) · 3 marks

$a = 2, b = -1, c = 1, d = 9.$

展开乘积得$2x^3-x^2+x+4$，再加5；比较系数。

M1	正确展开
M1	常数相加
A1	四系数

(b) · 3 marks

$f(-1) = 5$; remainder 23.

由恒等式x=−1使乘积为0；x=2时3(8−6+4)+5=23。

B1	f负1
M1	代2求余式
A1	余式

Question 16

P2-C01-R16 · 5 marks

(a) · 3 marks

$E = 4n(n+1)$ and $n(n+1)$ is even.

展开后提4，连续整数乘积含2，因此整体含8。

M1	展开因式分解
M1	连续整数偶性
A1	整除8

(b) · 2 marks

No; $n = 1$ gives $E = 8$.

选允许整数1，得到8不被16整除，构成反例。

M1	合法整数例
A1	明确不整除16

Question 17

P2-C01-R17 · 5 marks

(a) · 3 marks

$a=-2, b=1.$

1+a+b=0，16+4a+b=9；相减15+3a=9给a=−2,b=1。

M1	两余式方程
M1	联立
A1	两参数

(b) · 2 marks

$x^2-1.$

$P=x^4-2x^2+1=(x^2-1)^2$。

M1	识别完全平方
A1	商

Question 18

P2-C01-R18 · 5 marks

(a) · 3 marks

$(x-1)$.

使输入$2x-3=-1$，得$x=1$，故$g(1)=0$。

M1	映射已知根
M1	因式定理
A1	因式

(b) · 2 marks

8.

$g(3)=f(6-3)=f(3)=8$。

M1	正确复合输入
A1	余式

Question 19

P2-C01-R19 · 6 marks

(a) · 3 marks

$2x^3-8x^2+2x+12$.

先写2(x+1)(x−2)(x−3)，展开。

M1	正确三个因子及系数
M1	展开
A1	正确四项

(b) · 3 marks

$35/4$.

除式零点−1/2，因子式代入得2(1/2)(−5/2)(−7/2)=35/4。

M1	使用正确除式根
M1	代入因式式
A1	余式

Question 20

P2-C01-R20 · 5 marks

(a) · 2 marks

$x \leq 0$ or $x \geq 1$.

$x^2-x=x(x-1)\geqslant 0$，取两根外侧。

M1	建立不等式并分解
A1	完整解集

(b) · 3 marks

$x = 1/2$ gives $1/4 < 1/2$; all counterexamples lie in $(0, 1)$, which contains no integers.

引用(a)的补集解释为何整数检验不足；不仅只说检查不充分。

M1	合法反例及计算
M1	确定反例区间
A1	整数检验局限

Question 21

P2-C01-R21 · 5 marks

(a) · 3 marks

$f(1/2)=0$; quotient $2x^2-x-6$.

代入1/2检验；长除2x−1得到二次式。

M1	正确零点检查
M1	长除法
A1	商及零余式

(b) · 2 marks

$x=1/2, 2, -3/2$.

二次商(2x+3)(x−2)，和已知因子共同给三根。

M1	正确分解商
A1	全部三根

Question 22

P2-C01-R22 · 5 marks

(a) · 2 marks

7.

当前3x+1次数不小于除式，不能直接当余式；令x=2求P(2)=7。

M1	正确代除式根
A1	余式7

(b) · 3 marks

$P(x)=(x-2)(Q(x)+3)+7.$

把3x+1重写成3(x−2)+7，再合并商。

M1	重写线性剩余项
M1	合并商
A1	合法除法形式

Question 23

P2-C01-R23 · 5 marks

(a) · 3 marks

$(x+2)^2 = x^2 + 4x + 4$; multiplying by $x - 1$ gives $x^3 + 3x^2 - 4$.

逐项展开，注意x项抵消。

M1	展开平方
M1	乘法并合并
A1	目标式

(b) · 2 marks

Two: $x = -2, 1$.

平方因子给一个不同的根-2，不能把重根算作两个不同解。

M1	从因子找根
A1	两个不同解及数量

Question 24

P2-C01-R24 · 6 marks

(a) · 4 marks

$a=-\frac{1}{7}, b=\frac{2}{7}$.

5+a+b=2(3+a−b)给−a+3b=1；f(2)=12给2a+b=0；解a=−1/7,b=2/7。

M1	正确余式倍数关系
M1	第二条件
M1	解联立
A1	参数

(b) · 2 marks

4.

除x的余式为f(0)，就是常数项，不依赖参数。

M1	使用零点0
A1	余式

Question 25

P2-C01-R25 · 6 marks

(a) · 3 marks

$m=2, n=3$ gives even product 6, but n is odd.

反例必须让前提成立同时结论失败。

M1	正确反例
M1	验证前提
A1	验证结论失败

(b) · 3 marks

$m=2r+1, n=2s+1 \Rightarrow mn=2(2rs+r+s)+1$.

把两奇数写成2整数加1，展开并提出2，括号仍为整数。

M1	正确奇数代数表示
M1	展开整理
A1	奇数结论

Question 26

P2-C01-R26 · 6 marks

(a) · 3 marks

Quotient $x^2 - 2x - 3$, remainder 0.

长除法或展开$(x^2-4)(x^2-2x-3)$复核。

M1	正确首项除法
M1	后续相减
A1	商和余式

(b) · 3 marks

$(x-2)(x+2)(x-3)(x+1)$.

除式平方差，商分解$(x-3)(x+1)$。

M1	分解除式
M1	分解商
A1	完全因式

Question 27

P2-C01-R27 · 5 marks

(a) · 3 marks

$f(x) = 2x(x-1)(x-2)$.

三个已知根给f=kx(x−1)(x−2)，代3得6k=12。

M1	因式定理构造
M1	用非零函数值定k
A1	多项式

(b) · 2 marks

-12.

f(−1)=2(−1)(−2)(−3)=−12。

M1	代正确除式根
A1	余式

Question 28

P2-C01-R28 · 6 marks

(a) · 2 marks

$(2k+1)^2 + 2(2k+1) = 4k^2 + 8k + 3$, odd.

令$n = 2k+1$，其中k为整数。表达式化为$2(2k^2+4k+1)+1$，因此为奇数。

M1	用任意奇整数表示
A1	写成2整数+1并下结论

(b) A student claims that if $n^2 + 2n$ is odd, then n is odd.

(i) · 2 marks

If n were even, $n^2 + 2n$ would be even.

若$n = 2k$，则$n^2 + 2n = 4k(k+1)$为偶数，与已知奇数矛盾。因此n只能为奇数。

M1	检查偶数情形
A1	完整逆命题证明

(ii) · 2 marks

$n^2 + 2n = 4k(k+1)$, a multiple of 8.

相邻整数$k, k+1$中必有一个偶数，所以$k(k+1)$含因子2，乘4后可被8整除。

M1	相邻整数乘积为偶数
A1	整除8结论

Question 29

P2-C01-R29 · 6 marks

(a) · 3 marks

Divisible by both 2 and 3, hence by 6.

三个连续整数中至少一个是偶数，且必有一个是3的倍数；$2,3$互质，因此乘积被6整除。

B1	偶数因子
B1	3的倍数
B1	完整整除结论

(b) A student claims that $(n-1)n(n+1)$ is divisible by 24 for every integer $n>1$.

(i) · 1 marks

$n=2$: product 6.

取$n=2$得$1\times 2\times 3=6$，不是24的倍数，一个反例足以否定全称命题。

B1	合法反例及检验

(ii) · 2 marks

For odd n, the product is divisible by 24.

$n-1,n+1$是相邻的两个偶数，其中一个被4整除，另一个被2整除，所以乘积含因子8。再由(a)含因子3，且$8,3$互质，故含因子24。

M1	两邻偶数贡献8
A1	结合3得24

Question 30

P2-C01-R30 · 7 marks

(a) · 3 marks

$p = 9,\ q = 8.$

$f(1) = 6$给$p + q = 17$；$f(-2) = 0$给$4p - 2q = 20$。联立得$p = 9, q = 8$。

M1	余式条件
M1	因式条件
A1	联立常数

(b) Hence

(i) · 2 marks

$(x + 2)(x^2 + 7x - 6).$

用多项式除法除以$x + 2$得商$x^2 + 7x - 6$；因此$f(x) = (x + 2)(x^2 + 7x - 6)$。

M1	多项式除法
A1	剩余二次式

(ii) · 2 marks

$x = -2,\ (-7 \pm \sqrt{73})/2.$

令线性因子为零，另用二次求根公式解$x^2 + 7x - 6 = 0$。

M1	求根公式
A1	全部精确根

Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 2 — Coordinate geometry in the (x, y) plane
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第2章 · 圆的坐标几何
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1.

The circle C has equation $x^2 + y^2 - 6x + 4y - 12 = 0$.

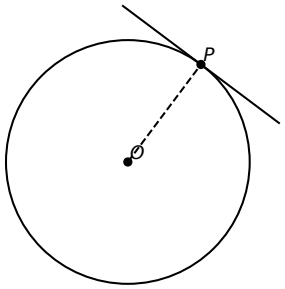


Figure 1

(a) Find its centre and radius. (3)

(b) Find the equation of the tangent at $P = (6, 2)$. (3)

(Total for Question 1 is 6 marks)

$A = (-2, 1)$ and $B = (6, 5)$ are endpoints of a diameter of a circle.

(a) Find an equation of the circle. (3)

(b) Find where the circle meets the line $y = 3$. (2)

(Total for Question 2 is 5 marks)

A circle has centre $(1, 4)$ and passes through $(5, 7)$.

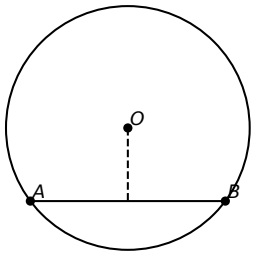


Figure 3

(a) Find its equation. (2)

(b) Find the exact length of the chord cut off by $y = 1$. (3)

(Total for Question 3 is 5 marks)

4.

The circle is $(x+2)^2+(y-1)^2=16$.

(a) Find the intersections with the line $x=0$. (2)

(b) Find equations of both tangents parallel to the y-axis. (2)

(Total for Question 4 is 4 marks)

5.

The line $y = x + k$ is tangent to $x^2 + y^2 = 18$.

(a) Find both possible values of k. (3)

(b) Find the point of contact for $k = 6$. (2)

(Total for Question 5 is 5 marks)

6. P2-C02-R06 · P2 · Chapter 2

A circle has equation $x^2 + y^2 + 2gx + 2fy = 0$. It passes through $(4, 0)$ and $(0, 6)$.

(a) Find g, f and the centre. (3)

(b) Show that the origin and $(4, 6)$ are opposite ends of a diameter. (3)

(Total for Question 6 is 6 marks)

The circle C is $(x-3)^2+(y+1)^2=25$, and $P=(9,7)$.

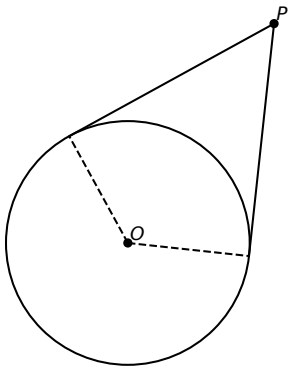


Figure 7

(a) Determine whether P is inside, on or outside C. (2)

(b) Find the exact length of a tangent from P to C. (3)

(Total for Question 7 is 5 marks)

8. P2-C02-R08 · P2 · Chapter 2

A circle has centre $O = (2, -1)$. A chord has endpoints $A = (-1, 3)$ and $B = (5, 3)$.

(a) Find the radius of the circle. (2)

(b) Find the midpoint of AB and verify that the line from O to this midpoint is perpendicular to AB. (3)

(Total for Question 8 is 5 marks)

The circles are $C_1 : x^2 + y^2 = 25$ and $C_2 : (x - 6)^2 + y^2 = 13$.

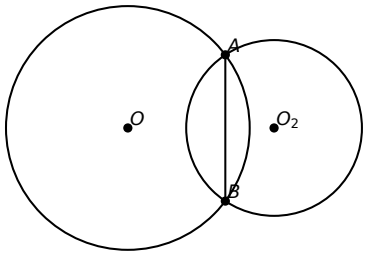


Figure 9

(a) Show that their common chord lies on $x = 4$. (2)

(b) Find the intersection points and the length of the common chord. (3)

(Total for Question 9 is 5 marks)

C is $x^2 + y^2 - 8x - 6y + 16 = 0$.

(a) Find its centre and radius. (3)

(b) Find all tangents to C parallel to the x-axis, including their contact points. (3)

(Total for Question 10 is 6 marks)

The circle $(x - a)^2 + y^2 = 25$ passes through $(1, 3)$. It is given that $a > 0$.

(a) Find a. (3)

(b) Find the equation of the tangent at $(1, 3)$. (3)

(Total for Question 11 is 6 marks)

The line $y = 2$ cuts the circle $x^2 + y^2 = 20$ at A, B.

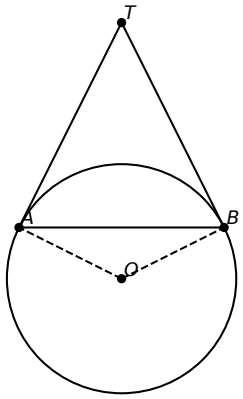


Figure 12

(a) Find A, B and the length AB. (3)

(b) Tangents at A, B meet at T. Find T. (3)

(Total for Question 12 is 6 marks)

13.

A circle has centre $(h, 2)$ and is tangent to the y-axis. It passes through $(4, 4)$, and $h > 0$.

(a) Find h and its radius. (3)

(b) Find its other intersection with the horizontal line $y = 4$. (2)

(Total for Question 13 is 5 marks)

14.

The circle C has centre $(2, 3)$ and is tangent to $y = 7$.

(a) Find an equation of C. (2)

(b) Find the exact coordinates where C meets $y = x + 1$. (3)

(Total for Question 14 is 5 marks)

A circle has diameter endpoints $A = (1, 2)$ and $B = (7, 10)$.

(a) Find its equation and determine whether $P = (0, 6)$ lies on it. (3)

(b) Find an equation of the tangent at A. (3)

(Total for Question 15 is 6 marks)

16. P2-C02-R16 · P2 · Chapter 2

The circle C is $(x+1)^2+(y-2)^2=10$. The line L is $y=x+4$.

(a) Find both points where L meets C. (3)

(b) Find the midpoint of the chord and show that it lies on the perpendicular from the centre to L. (3)

(Total for Question 16 is 6 marks)

A circle with centre $(3, -2)$ is tangent to the line $x = 8$.

(a) Find its equation and the point of contact. (3)

(b) Find the exact length of the chord on the line $y = 1$. (3)

(Total for Question 17 is 6 marks)

18. P2-C02-R18 · P2 · Chapter 2

The circle is $x^2 + y^2 + 4x - 10y + k = 0$. Its radius is 6.

(a) Find k and the centre. (3)

(b) Determine whether the circle cuts the x-axis. Find any intersections exactly. (3)

(Total for Question 18 is 6 marks)

The line $y = mx$ is tangent to $(x - 5)^2 + y^2 = 9$.

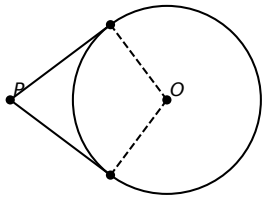


Figure 19

(a) Find the two possible values of m. (3)

(b) Find both contact points. (3)

(Total for Question 19 is 6 marks)

$A = (2, 0)$ and $B = (0, 4)$ lie on a circle whose centre lies on $y = x$.

(a) Find the centre. (3)

(b) Find the circle equation and its intersections with $y = x$. (4)

(Total for Question 20 is 7 marks)

The circles $C_1 : (x-1)^2 + y^2 = 9$ and $C_2 : (x-6)^2 + y^2 = 4$ are given.

(a) Show that the circles touch externally. (3)

(b) Find the contact point and the common tangent there. (3)

(Total for Question 21 is 6 marks)

Circle C has centre $(0, 0)$ and radius 5. Point $P = (4, 3)$ lies on it.

(a) Find the tangent L at P. (3)

(b) Find the area of the triangle enclosed by L and the coordinate axes. (3)

(Total for Question 22 is 6 marks)

The circle C is $(x-2)^2+(y+3)^2=13$.

(a) Find the points on C whose x-coordinate is 4. (2)

(b) Find the equations of the tangents at these points and their intersection. (4)

(Total for Question 23 is 6 marks)

A circle has centre $(1, 2)$ and is tangent to the line $y = x + 6$.

(a) Find the foot of the perpendicular from the centre to the line. (3)

(b) Find an equation of the circle. (2)

(Total for Question 24 is 5 marks)

The circle C has diameter endpoints $A = (-1, 2)$ and $B = (5, 10)$. Point $P = (6, 3)$ is given.

(a) Find the equation of C. (3)

(b) Show that P lies on C and that angle APB is 90° using gradients. (3)

(Total for Question 25 is 6 marks)

26.

The circle C is $x^2 + y^2 - 2x - 8y + 8 = 0$. The point $A = (1, 4)$ is given.

(a) Find the centre and radius and explain why A is not on C. (3)

(b) Find all points on C lying on the line $y = 2x + 2$. (3)

(Total for Question 26 is 6 marks)

The circle C is $x^2 + y^2 = 50$. The chord AB has midpoint $M = (3, 1)$.

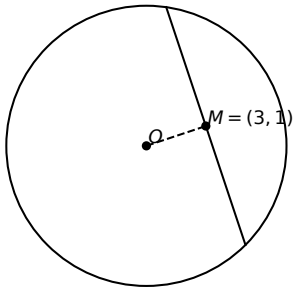


Figure 27

(a) Find an equation of the chord. (3)

(b) Find the exact chord length. (4)

(Total for Question 27 is 7 marks)

28.

A circle has equation $x^2 + y^2 + Dx + Ey + F = 0$ and passes through $(0, 0)$, $(6, 0)$ and $(0, 8)$.

(a) Find D, E, F and the equation in centre-radius form. (4)

(b) Find the tangent at the origin. (3)

(Total for Question 28 is 7 marks)

The circle C has equation $x^2 + y^2 - 6x + 4y - 12 = 0$. The line l has equation $y = x + k$.

(a) Find the centre and radius of C. (2)

(b) The line l is tangent to C.

(i) Find both possible values of k. (2)

(ii) For $k = -5 + 5\sqrt{2}$, find the exact coordinates of the point of contact. (2)

(Total for Question 29 is 6 marks)

A circle has diameter AB, where $A(-1, 2)$ and $B(7, 6)$. A horizontal line cuts the circle at P and Q, and $PQ = 6$.

(a) Find an equation for the circle. (2)

(b) Find

(i) the two possible equations of the horizontal line. (2)

(ii) the coordinates of P and Q for the higher line. (1)

(Total for Question 30 is 5 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C02-R01 · 6 marks

(a) · 3 marks

Centre $(3, -2)$, radius 5.

配方得$(x-3)^2+(y+2)^2=25$。

M1	两坐标配方
A1	中心
A1	半径

(b) · 3 marks

$3x + 4y = 26$.

半径向量$(3,4)$，斜率$4/3$，切线$-3/4$，过P整理。

M1	半径梯度
M1	垂直切线过点
A1	方程

Question 2

P2-C02-R02 · 5 marks

(a) · 3 marks

$(x-2)^2+(y-3)^2=20.$

中心为中点(2,3)，半径平方$AB^2/4=(64+16)/4=20$。

M1	中心
M1	半径平方
A1	圆方程

(b) · 2 marks

$(2-2\sqrt{5},3),(2+2\sqrt{5},3).$

线通过中心，代y=3得$(x-2)^2=20$。

M1	正确代入
A1	两交点

Question 3

P2-C02-R03 · 5 marks

(a) · 2 marks

$(x-1)^2+(y-4)^2=25.$

半径平方$4^2+3^2=25$。

M1	两点距离平方
A1	方程

(b) · 3 marks

8.

代$y=1$得$(x-1)^2=16$，端点$x=-3,5$，同高距离8。

M1	联立求两横坐标
M1	取横差
A1	弦长

Question 4

P2-C02-R04 · 4 marks

(a) · 2 marks

$(0, 1 \pm 2\sqrt{3})$.

代x=0后$(y-1)^2=12$。

M1	正确平方关系
A1	两坐标

(b) · 2 marks

$x = -6$ and $x = 2$.

竖直切线位于中心横坐标±半径，即-2 ± 4。

M1	使用圆的水平极端位置
A1	两竖直方程

Question 5

P2-C02-R05 · 5 marks

(a) · 3 marks

$k = \pm 6$.

代入圆得$2x^2+2kx+k^2-18=0$；判别式$4k^2-8(k^2-18)=0$。

M1	代线入圆
M1	判别式等零
A1	两k

(b) · 2 marks

$(-3, 3)$.

相切二次式重根$x=-k/2=-3$，$y=x+6=3$。

M1	求重根
A1	接触点

Question 6

P2-C02-R06 · 6 marks

(a) · 3 marks

$g=-2, f=-3$; centre $(2,3)$.

点(4,0)给16+8g=0；(0,6)给36+12f=0，中心为(−g,−f)。

M1	两点代入
A1	两参数
B1	中心

(b) · 3 marks

Their midpoint is $(2,3)$ and both lie on the circle.

原点由圆式常数0可见在圆上；(4,6)代入16+36−16−36=0；中点为中心。

M1	检查另一点在圆上
M1	计算中点
A1	直径结论

Question 7

P2-C02-R07 · 5 marks

(a) · 2 marks

Outside, since $CP^2 = 6^2 + 8^2 = 100 > 25$.

比较点到中心距离平方与半径平方，无需先开方。

M1	正确距离平方
A1	分类与比较

(b) · 3 marks

$5\sqrt{3}$.

切点处半径垂直切线，勾股长度 $\sqrt{(CP^2 - r^2)} = \sqrt{75}$。

M1	使用直角切线三角形
M1	正确平方差
A1	根式

Question 8

P2-C02-R08 · 5 marks

(a) · 2 marks

5.

OA坐标差$-3,4$，长度5。

M1	半径两点距离
A1	半径

(b) · 3 marks

Midpoint $(2,3)$; AB horizontal and the centre-to-midpoint line is $x=2$, vertical.

先取中点，再比较水平弦和竖直中垂线，不能用无定义的竖线斜率相乘。

M1	中点
M1	两方向
A1	垂直验证

Question 9

P2-C02-R09 · 5 marks

(a) · 2 marks

Subtracting gives $-12x + 36 = -12$, so $x = 4$.

展开第二圆再与第一圆相减，消去x^2+y^2。

M1	展开并相减
A1	公共弦所在直线

(b) · 3 marks

$(4, 3), (4, -3)$; length 6.

$x=4$代第一圆得$y^2=9$，弦竖直用纵差6。

M1	回代解两个y
A1	两交点
A1	弦长

Question 10

P2-C02-R10 · 6 marks

(a) · 3 marks

Centre $(4, 3)$; radius 3.

配方得$(x-4)^2+(y-3)^2=9$。

M1	完成配方
A1	中心
A1	半径

(b) · 3 marks

$y = 0$ at $(4, 0)$ and $y = 6$ at $(4, 6)$.

水平切线在中心正上方和正下方，各相距$r=3$。

M1	正确极端位置
A1	两线
A1	接触点

Question 11

P2-C02-R11 · 6 marks

(a) · 3 marks

$a = 5.$

$(1-a)^2=16$给a=5或−3；正值限制选5。

M1	代点解平方方程
M1	使用正值限制
A1	a

(b) · 3 marks

$4x - 3y = -5.$

半径从(5,0)到(1,3)斜率−3/4，切线4/3；y−3=(4/3)(x−1)。

M1	半径斜率
M1	切线负倒数
A1	方程

Question 12

P2-C02-R12 · 6 marks

(a) · 3 marks

$(\pm 4, 2)$; $AB = 8$.

$x^2+4=20$，$x=\pm 4$，弦水平。

M1	联立
A1	两点
A1	长度

(b) · 3 marks

$T = (0, 10)$.

切线为$4x+2y=20$和$-4x+2y=20$，联立$x=0$,$y=10$；这些方程可由半径垂直推得。

M1	两条正确切线
M1	联立
A1	T

Question 13

P2-C02-R13 · 5 marks

(a) · 3 marks

$h = 5/2$; radius $5/2$.

到y轴的距离为h，故r=h；$(4-h)^2+4=h^2$给$20-8h=0$。

M1	切y轴的半径关系
M1	点条件
A1	中心参数和半径

(b) · 2 marks

$(1, 4)$.

关于中心竖直线x=5/2对称，两个交点横坐标和5；另一x=1，或联立求根。

M1	对称或圆线联立
A1	另一点

Question 14

P2-C02-R14 · 5 marks

(a) · 2 marks

$(x-2)^2+(y-3)^2=16$.

中心到水平切线距离7−3=4，即半径。

M1	半径
A1	方程

(b) · 3 marks

$(2\pm2\sqrt{2},3\pm2\sqrt{2})$, with matching signs.

直线过中心，代入得$2(x-2)^2=16$，$y=x+1$。

M1	正确联立简化
M1	求两根
A1	两配对点

Question 15

P2-C02-R15 · 6 marks

(a) · 3 marks

$(x-4)^2+(y-6)^2=25$; P gives $16\neq 25$, so P does not lie on it.

中心(4,6)，r^2=25；给定点(0,6)距中心4并非5。

M1	由直径求圆
M1	检查点
A1	正确结论

(b) · 3 marks

$3x+4y=11$.

半径斜率4/3，切线−3/4，过(1,2)得3x+4y=11。

M1	半径方向
M1	切线过点
A1	方程

Question 16

P2-C02-R16 · 6 marks

(a) · 3 marks

$\left(\frac{-3-\sqrt{19}}{2}, \frac{5-\sqrt{19}}{2}\right)$ and $\left(\frac{-3+\sqrt{19}}{2}, \frac{5+\sqrt{19}}{2}\right)$.

代线入圆得$(x+1)^2+(x+2)^2=10$，即$2x^2+6x-5=0$；二次公式给$x=(-3\pm\sqrt{19})/2$，回代$y=x+4$。

M1	代入建立二次方程
M1	解方程
A1	两点

(b) · 3 marks

Midpoint $(-3/2, 5/2)$, on $y=-x+1$.

二次方程两根平均为$-3/2$，直线给$y=5/2$；中心$(-1,2)$向斜率1直线的垂线为$y=-x+1$。

M1	中点
M1	中心垂线
A1	验证

Question 17

P2-C02-R17 · 6 marks

(a) · 3 marks

$(x-3)^2+(y+2)^2=25$; contact $(8,-2)$.

半径为8−3=5，接触点与中心同纵坐标。

M1	半径
A1	方程
B1	接触点

(b) · 3 marks

8.

距中心竖直3，半弦 $\sqrt{(25-9)}=4$，整弦8。

M1	正确中心距与半弦勾股
M1	从半弦到整弦
A1	长度

Question 18

P2-C02-R18 · 6 marks

(a) · 3 marks

$k = -7$; centre $(-2, 5)$.

配方得$(x+2)^2+(y-5)^2=29-k$；$29-k=36$给$k=-7$。

M1	正确配方
M1	半径条件
A1	k和中心

(b) · 3 marks

It cuts at $(-2 \pm \sqrt{11}, 0)$.

设$y=0$得$(x+2)^2=11>0$，故有两个不同实交点。

M1	联立x轴
M1	判断两实根
A1	两坐标

Question 19

P2-C02-R19 · 6 marks

(a) · 3 marks

$m = \pm 3/4$.

代入得$(1+m^2)x^2-10x+16=0$，判别式$100-64(1+m^2)=0$。

M1	圆线联立
M1	重根条件
A1	两斜率

(b) · 3 marks

$(16/5, 12/5)$ and $(16/5, -12/5)$.

重根$x=10/[2(1+9/16)]=16/5$；$y=m\,x$。

M1	重根横坐标
M1	对应纵坐标
A1	两点

Question 20

P2-C02-R20 · 7 marks

(a) · 3 marks

$(3,3)$.

设中心(h,h)，$OA^2=(h-2)^2+h^2$，$OB^2=h^2+(h-4)^2$；相等得h=3。

M1	中心参数化
M1	两半径平方相等
A1	中心

(b) · 4 marks

$(x-3)^2+(y-3)^2=10$; $(3\pm\sqrt{5},3\pm\sqrt{5})$.

$r^2=1+9=10$；联立y=x得$2(x-3)^2=10$。

M1	半径平方
A1	圆方程
M1	联立过心线
A1	两点

Question 21

P2-C02-R21 · 6 marks

(a) · 3 marks

Centre distance 5 equals the sum of radii $3+2$.

两中心在同一水平线上，距离6−1=5，与半径和相等，所以外切。

M1	中心距与两半径
M1	比较半径和
A1	外切结论

(b) · 3 marks

Contact $(4,0)$; tangent $x=4$.

接触点在连心线上，距第一中心向右3；半径水平所以切线竖直。

M1	定位接触点
A1	点
A1	切线

Question 22

P2-C02-R22 · 6 marks

(a) · 3 marks

$4x + 3y = 25$.

半径斜率$3/4$，切线$-4/3$；过$(4,3)$。

M1	正确垂直斜率
M1	过点
A1	方程

(b) · 3 marks

$625/24$ square units.

截距$25/4$和$25/3$，面积为其积的一半。

M1	两截距
M1	正确面积式
A1	精确面积

Question 23

P2-C02-R23 · 6 marks

(a) · 2 marks

$(4, 0)$ and $(4, -6)$.

代x=4得$(y+3)^2=9$。

M1	正确平方式
A1	两点

(b) · 4 marks

$2x + 3y = 8$ and $2x - 3y = 26$; intersection $(17/2, -3)$.

半径向量分别(2,3)、(2,−3)，切线用垂直关系；两线相加4x=34。

M1	第一切线
M1	第二切线
M1	联立
A1	交点

Question 24

P2-C02-R24 · 5 marks

(a) · 3 marks

$(-3/2, 9/2)$.

垂线斜率-1，过中心为y=−x+3，与y=x+6联立。

M1	中心垂线
M1	联立
A1	垂足

(b) · 2 marks

$(x-1)^2+(y-2)^2=25/2$.

半径平方=$(-5/2)^2+(5/2)^2=25/2$。

M1	垂足到中心距离平方
A1	圆方程

Question 25

P2-C02-R25 · 6 marks

(a) · 3 marks

$(x-2)^2+(y-6)^2=25$.

中点(2,6)，半径长度5。

M1	直径中点
M1	半径
A1	方程

(b) · 3 marks

$(6-2)^2+(3-6)^2=25$; $m_{PA}=1/7$, $m_{PB}=-7$, product -1.

先代圆式验证点，再计算过P的两边斜率，乘积−1。

M1	点在圆上验证
M1	两个斜率
A1	垂直结论

Question 26

P2-C02-R26 · 6 marks

(a) · 3 marks

Centre $A = (1, 4)$; radius 3; the centre is not on a circle of positive radius.

配方为$(x-1)^2+(y-4)^2=9$，A恰为中心而非周界点。

M1	配方
A1	中心与半径
B1	排除A在圆上的理由

(b) · 3 marks

$(1 \pm 3/\sqrt{5}, 4 \pm 6/\sqrt{5})$, with matching signs.

直线过中心，用$u=x-1$得$y-4=2u$，圆上$5u^2=9$。

M1	正确联立简化
M1	解正负u
A1	两点

Question 27

P2-C02-R27 · 7 marks

(a) · 3 marks

$3x + y = 10$.

OM斜率1/3，弦垂直OM故斜率−3且过M，y−1=−3(x−3)。

M1	弦与中心连线垂直
M1	用中点过线
A1	方程

(b) · 4 marks

$4\sqrt{10}$.

$OM^2=10$，半弦平方$r^2-OM^2=40$，整弦$2\sqrt{40}=4\sqrt{10}$。

M1	中心距平方
M1	半弦勾股
M1	乘2
A1	最简长度

Question 28

P2-C02-R28 · 7 marks

(a) · 4 marks

$D=-6, E=-8, F=0$; $(x-3)^2+(y-4)^2=25$.

原点给F=0，两个轴点给D=−6,E=−8，再配方。

B1	F
M1	求D和E
M1	配方
A1	标准式

(b) · 3 marks

$3x+4y=0$.

半径从中心(3,4)至原点，斜率4/3；切线−3/4且过原点。

M1	半径方向
M1	正确垂线
A1	方程

Question 29

P2-C02-R29 · 6 marks

(a) · 2 marks

$(3, -2)$; 5.

配方得$(x-3)^2+(y+2)^2=25$。

M1	正确配方
A1	圆心半径

(b) The line l is tangent to C.

(i) · 2 marks

$k=-5\pm5\sqrt{2}$.

代入得$2x^2+(2k-2)x+k^2+4k-12=0$。相切时判别式为零，化简$k^2+10k-25=0$，得$k=-5\pm5\sqrt{2}$。

M1	联立并用判别式
A1	两个参数

(ii) · 2 marks

$(3-5\sqrt{2}/2, -2+5\sqrt{2}/2)$.

接点在斜率-1、过圆心的半径直线$y=-x+1$上。与$y=x-5+5\sqrt{2}$联立。

M1	圆心至切点垂直切线
A1	精确坐标

Question 30

P2-C02-R30 · 5 marks

(a) · 2 marks

$(x-3)^2+(y-4)^2=20$.

圆心是中点$(3,4)$，半径平方$[(8)^2+4^2]/4=20$。

M1	中点与半径
A1	方程

(b) Find

(i) · 2 marks

$y=4\pm\sqrt{11}$.

水平弦中点横坐标为3，半弦为3；若纵坐标为h，则$9+(h-4)^2=20$。

M1	半弦与半径直角关系
A1	两条直线

(ii) · 1 marks

$(0,4+\sqrt{11})$ and $(6,4+\sqrt{11})$.

弦中点$(3,4+\sqrt{11})$，沿水平方向各移动3。

B1	两端点

Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 3 — Exponentials and logarithms Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第3章 · 指数与对数
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P2-C03-R01 · P2 · Chapter 3

$3^{2x} - 10(3^x) + 9 = 0.$

(a) Using $u = 3^x$, solve for u. (3)

(b) Hence find all real values of x. (2)

(Total for Question 1 is 5 marks)

For $x > 3$, $\log_2(x-1) + \log_2(x-3) = 3$.

(a) Show that $x^2 - 4x - 5 = 0$. (3)

(b) Find x, explaining why one algebraic root must be rejected. (3)

(Total for Question 2 is 6 marks)

3.

The curve is $y = 4(2^x) - 3$.

(a) Find its y-intercept and horizontal asymptote. (2)

(b) Find the exact x-coordinate of its x-intercept, using logarithms. (3)

(Total for Question 3 is 5 marks)

A quantity is modelled by $N = 800(0.85)^t$, where $t \geq 0$ is measured in days.

(a) State the initial value and the percentage decrease per day. (2)

(b) Find the least whole number of days after which $N < 300$. (3)

(Total for Question 4 is 5 marks)

$\log_a 16 = 4$, where $a > 0$ and $a \neq 1$.

(a) Find a. (2)

(b) For this value of a, find

(i) $\log_a(1/8)$, (1)

(ii) $\log_{a^2} 32$. (2)

(Total for Question 5 is 5 marks)

$5^{x+1} = 3^{2x}$.

(a) Express x exactly using common logarithms. (3)

(b) Find x to 3 decimal places and check that it is positive. (3)

(Total for Question 6 is 6 marks)

For $x > 0$, $\log_3 x + \log_3(9x) = 6$.

(a) Find x exactly. (3)

(b) Find $\log_x 27$ for this value of x. (2)

(Total for Question 7 is 5 marks)

8.

A population has model $P = A(1.2)^t$, where t is years after a survey. At $t = 2$, $P = 720$.

(a) Find A. (2)

(b) Find the time at which the model first reaches 1000, to 2 decimal places. (3)

(Total for Question 8 is 5 marks)

9. P2-C03-R09 · P2 · Chapter 3

The curve C has equation $y = \log_2(x + 3) - 1$.

(a) State its domain, vertical asymptote and both axis intersections. (3)

(b) Find the exact coordinates on C where $y = 3$. (2)

(Total for Question 9 is 5 marks)

$4^x - 6(2^x) + 8 = 0$.

(a) Find all real solutions. (3)

(b) Find all real solutions of $4^{-x} - 6(2^{-x}) + 8 = 0$, without solving a new quadratic. (2)

(Total for Question 10 is 5 marks)

The numbers p, q are positive and satisfy $\log p = 2\log q + 1$. Here log means base 10.

(a) Express p in terms of q. (2)

(b) Given also that $p = 40q$, find p, q. (3)

(Total for Question 11 is 5 marks)

An exponential model is $M = Ab^t$, where $A > 0, b > 0$. It gives $M = 18$ at $t = 1$ and $M = 162$ at $t = 3$.

(a) Find A, b. (3)

(b) Find the time when $M = 50$, to 3 decimal places. (3)

(Total for Question 12 is 6 marks)

For $x > 1$, $\log_4(x^2 - 1) - \log_4(x - 1) = 2$.

(a) Simplify the left-hand side to a single logarithm. (3)

(b) Hence find x. (2)

(Total for Question 13 is 5 marks)

The graphs $y = 3^x$ and $y = 9^{1-x}$ intersect.

(a) Find their intersection exactly. (3)

(b) Find the range of x for which $3^x > 9^{1-x}$. (2)

(Total for Question 14 is 5 marks)

Given $\log_2 a = 3$ and $\log_2 b = -2$,

(a) find the exact values of

(i) $\log_2(a^2/b)$, (2)

(ii) $\log_4(ab)$. (3)

(b) Find $a + b$. (2)

(Total for Question 15 is 7 marks)

A substance retains 60% of its mass after 4 hours. Its mass is modelled by $m = m_0 b^t$, where t is in hours and $0 < b < 1$.

(a) Find b in exact form. (2)

(b) Find the time for the mass to fall to 25% of its initial mass, to 2 decimal places. (3)

(Total for Question 16 is 5 marks)

$\log_2(x+2) = 1 + \log_2(3x-1)$.

(a) Find the domain restrictions needed before solving. (2)

(b) Solve the equation and verify the solution is in the domain. (3)

(Total for Question 17 is 5 marks)

18.

A device has output $R = 120 - 80(0.7)^t$, where $t \geq 0$ is in seconds.

(a) Find the initial output and the limiting output predicted by the model. (3)

(b) Find the first time when the output reaches 100, to 2 decimal places. (3)

(Total for Question 18 is 6 marks)

$2^{2x} + 2^{x} - 12 = 0$.

(a) Find all real solutions, explaining why a root of the substituted quadratic is rejected. (4)

(b) Give the solution to 3 decimal places. (2)

(Total for Question 19 is 6 marks)

For $x > 0$, $y = \log_5 x$.

(a) Find the points where $y = -1, 0, 2$. (2)

(b) The graph is translated right by 2 and up by 3. Find its equation and vertical asymptote. (3)

(Total for Question 20 is 5 marks)

The population models for two sites are $A = 300(1.1)^t$ and $B = 500(0.95)^t$, for $t \geq 0$ in years.

(a) Find the time at which the models predict equal populations, to 2 decimal places. (3)

(b) State which site has the larger predicted population after 5 years, supporting your answer with two calculated values. (3)

(Total for Question 21 is 6 marks)

For $a > 0$, $a \neq 1$, $\log_a 9 = \frac{1}{2}\log_a 81$.

(a) Explain why this equality does not determine a. (2)

(b) Given also that $\log_a 3 = 2$, find a. (2)

(Total for Question 22 is 4 marks)

The equation $\log_3 x = 2 - \log_3(x+2)$ is to be solved for real x.

(a) Find the exact admissible solution. (4)

(b) Find the exact value of $\log_3(x(x+2))$ at the solution, without evaluating x numerically. (2)

(Total for Question 23 is 6 marks)

$9^{x-1} = 27^{2x+1}$.

(a) Find x exactly. (3)

(b) Find the common value of the two sides as a power of 3. (2)

(Total for Question 24 is 5 marks)

A bank balance is modelled by $B = 2000(1.04)^n$ after n complete years, with no deposits or withdrawals.

(a) Find the balance after 5 years to the nearest penny. (2)

(b) Find the least complete number of years for the balance to exceed £3000. (3)

(Total for Question 25 is 5 marks)

Positive numbers a, b, c satisfy $a \neq 1$, $\log_a b = 3$ and $\log_a c = 2$.

(a) Find $\log_b(b^2c)$ exactly. (3)

(b) For these numbers,

(i) solve $b^t = c^{t+1}$. (2)

(ii) given also that $b > c$, determine whether $a > 1$ or $0 < a < 1$, giving a reason. (1)

(Total for Question 26 is 6 marks)

The curves are $y = 2^x$ and $y = 8(1/2)^x$.

(a) Find their intersection exactly. (3)

(b) Find all x for which $2^x + 8(1/2)^x = 6$. (4)

(Total for Question 27 is 7 marks)

A model is $Y = K(3)^t + c$. Its horizontal asymptote is $Y = 4$, and it passes through $(0, 6)$.

(a) Find K, c. (3)

(b) Find the value of t for which $Y = 10$, and explain why $Y = 4$ is never reached for finite real t. (4)

(Total for Question 28 is 7 marks)

The intensity I of a beam after passing through x cm of material is modelled by $I = I_0 a^x$, where $I_0 > 0$ and $0 < a < 1$. After passing through 3 cm, the intensity is $0.64I_0$.

(a) Find a to 4 decimal places. (2)

(b) Using the unrounded value of a, find

(i) the thickness which reduces intensity to one quarter of I_0, to 3 significant figures. (2)

(ii) the least whole number of centimetres required to reduce intensity to at most $0.10I_0$. (2)

(Total for Question 29 is 6 marks)

Positive numbers x, y satisfy $\log_2 x + \log_2 y = 5$ and $\log_2(x/y) = 1$.

(a) Find x and y. (3)

(b) Let $z > 0$ satisfy $\log_x z + \log_y z = 5$.

(i) Find $\log_2 z$. (2)

(ii) Find z exactly. (1)

(Total for Question 30 is 6 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C03-R01 · 5 marks

(a) · 3 marks

$u = 1, 9.$

代换后$u^2-10u+9=(u-1)(u-9)=0$，两根均正。

M1	正确代换
M1	分解二次式
A1	两u值

(b) · 2 marks

$x = 0, 2.$

分别解3^x=1及9，不丢失合法正指数值。

M1	映射回指数方程
A1	两x

Question 2

P2-C03-R02 · 6 marks

(a) · 3 marks

$(x-1)(x-3)=8 \Rightarrow x^2-4x-5=0$.

对数相加化为乘积，对数底2转指数8。

M1	对数乘法则
M1	转指数
A1	目标方程

(b) · 3 marks

$x=5$; reject $x=-1$ since $x>3$ is required.

分解(x−5)(x+1)，逐一检查原定义域。

M1	解二次式
M1	原定义域筛选
A1	唯一根

Question 3

P2-C03-R03 · 5 marks

(a) · 2 marks

$(0, 1)$; $y = -3$.

2^0=1；x趋向负无穷时2^x趋0。

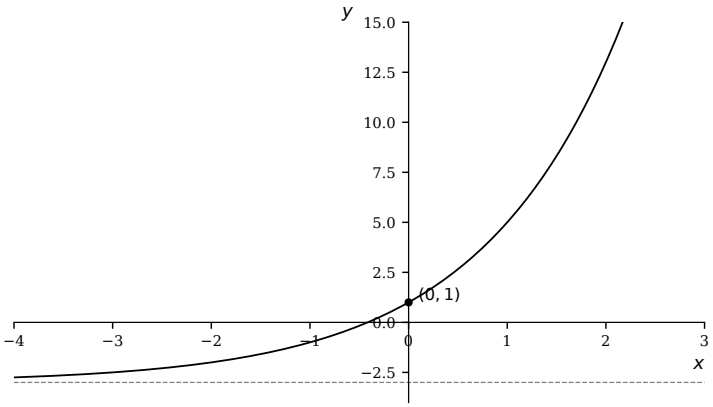
B1	截距
B1	渐近线

(b) · 3 marks

$x = \dfrac{\log(3/4)}{\log 2}$.

令2^x=3/4，两边取同一底的对数；无需自然对数。

M1	正确指数方程
M1	对数化
A1	精确形式



Suggested sketch

Question 4

P2-C03-R04 · 5 marks

(a) · 2 marks

800; 15%.

t=0给800，日保留比例85%，所以减少15%。

B1	初值
B1	百分比

(b) · 3 marks

7 days.

t>log(300/800)/log0.85≈6.035；取最小严格超过阈值的整数7。

M1	正确指数不等式
M1	负对数反向及阈值
A1	整数天

Question 5

P2-C03-R05 · 5 marks

(a) · 2 marks

$a = 2$.

a^4=16，由底正取2。

M1	对数转指数
A1	合法底

(b) For this value of a, find

(i) · 1 marks

-3.

2^{−3}=1/8。

B1	精确值

(ii) · 2 marks

$5/2$.

底a^2=4=2^2，真数32=2^5，比指数5/2。

M1	统一底数
A1	精确值

Question 6

P2-C03-R06 · 6 marks

(a) · 3 marks

$x = \dfrac{\log 5}{2\log 3 - \log 5}$.

取log10得(x+1)log5=2xlog3，收集x。

M1	两边取对数
M1	正确收集未知量
A1	精确商

(b) · 3 marks

$x \approx 2.738 > 0$.

分母log(9/5)>0，分子log5>0；计算精度指定为三位小数。

M1	计算不提前舍入
A1	三位小数
B1	符号理由

Question 7

P2-C03-R07 · 5 marks

(a) · 3 marks

$x = 9$.

log3(9x)=2+log3x，所以2log3x=4。

M1	正确拆对数
M1	线性解对数
A1	x

(b) · 2 marks

$3/2$.

$9=3^2$，$27=3^3$，指数比3/2。

M1	统一底
A1	精确值

Question 8

P2-C03-R08 · 5 marks

(a) · 2 marks

$A = 500$.

720=A×1.44，故A=500。

M1	初值条件
A1	A

(b) · 3 marks

$t = \log 2 / \log 1.2 \approx 3.80$ years.

用已求A设500(1.2)^t=1000，再取同底对数。

M1	正确指数阈值方程
M1	对数求解
A1	精度和单位

Question 9

P2-C03-R09 · 5 marks

(a) · 3 marks

$x > -3$; $x = -3$; $(-1, 0)$ and $(0, \log_2 3 - 1)$.

真数正给定义域；令y=0得x+3=2，令x=0得纵截距。

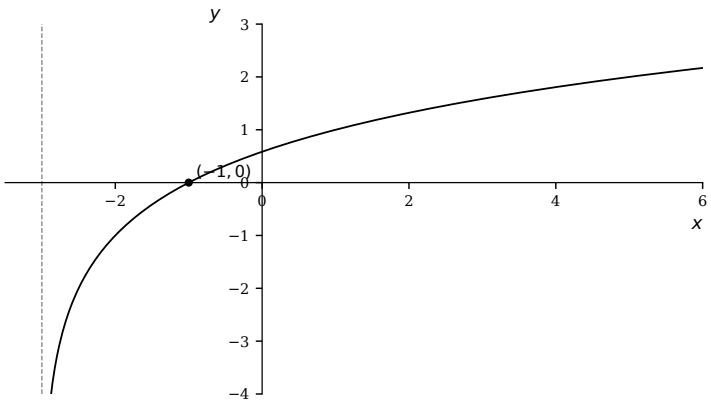
B1	定义域和渐近线
M1	设零解横截距
A1	两截距

(b) · 2 marks

$(13, 3)$.

log2(x+3)=4给x+3=16。

M1	转指数
A1	点



Suggested sketch

Question 10

P2-C03-R10 · 5 marks

(a) · 3 marks

$x = 1, 2.$

令u=2^x>0，4^x=u²，(u−2)(u−4)=0。

M1	正确指数代换
M1	解正u
A1	两x

(b) · 2 marks

$x = -1, -2.$

新方程是把原输入x替换为−x，所以新根是旧根相反数。

M1	识别输入反射
A1	两解

Question 11

P2-C03-R11 · 5 marks

(a) · 2 marks

$p = 10q^2$.

$2\log q = \log q^2$，$1 = \log 10$，再合并。

M1	对数幂与乘积
A1	原数量关系

(b) · 3 marks

$q = 4, p = 160$.

$10q^2 = 40q$；$q > 0$可除q得4，再求p。

M1	联立两个关系
M1	正值排除零
A1	两数

Question 12

P2-C03-R12 · 6 marks

(a) · 3 marks

$b=3, A=6.$

两式相除b^2=9，底正选3，再用Ab=18。

M1	比值消去A
M1	选正底
A1	A和b

(b) · 3 marks

$t=\log(25/3)/\log 3\approx 1.930.$

$6\times 3^{t}=50$，取常用对数解。

M1	正确比值
M1	对数求解
A1	三位小数

Question 13

P2-C03-R13 · 5 marks

(a) · 3 marks

$\log_4(x+1)$.

商法则后$(x^2-1)/(x-1)=x+1$，给定$x>1$保证原两真数为正。

M1	对数商法则
M1	约去合法因子
A1	简式

(b) · 2 marks

$x=15$.

$x+1=4^2=16$，解得15满足定义域。

M1	转指数并检查
A1	解

Question 14

P2-C03-R14 · 5 marks

(a) · 3 marks

$(2/3, \sqrt[3]{9})$.

统一底3得3^x=3^{2−2x}，所以3x=2；纵坐标3^{2/3}。

M1	统一底数
M1	解横坐标
A1	交点

(b) · 2 marks

$x > 2/3$.

底3>1，比较指数保持不等号，x>2−2x。

M1	利用指数单调性
A1	严格范围

Question 15

P2-C03-R15 · 7 marks

(a) find the exact values of

(i) · 2 marks

8.

$2\log a-\log b=6-(-2)$。

M1	正确使用幂和商法则
A1	值8

(ii) · 3 marks

$1/2$.

$\log_2(ab)=3-2=1$，换成底4再除2。

M1	先求底2对数
M1	换底
A1	值1/2

(b) · 2 marks

$33/4$.

$a=8, b=1/4$，相加。

M1	对数还原两数
A1	精确和

Question 16

P2-C03-R16 · 5 marks

(a) · 2 marks

$b = (3/5)^{1/4}$.

$b^4=0.6$，取正四次根。

M1	正确保留比例关系
A1	精确底数

(b) · 3 marks

$t = 4\log(1/4)/\log(3/5) \approx 10.86$ h.

b^t=1/4，使用(a)表达式取对数，初始质量抵消。

M1	正确相对质量方程
M1	对数运算
A1	时间

Question 17

P2-C03-R17 · 5 marks

(a) · 2 marks

$x > 1/3.$

分别要求x+2>0与3x−1>0，两条件交集x>1/3。

M1	两个真数正条件
A1	交集

(b) · 3 marks

$x = 4/5.$

1=log2 2，所以x+2=2(3x−1)，解5x=4；4/5>1/3。

M1	合并常数对数
M1	解一次式
A1	根及域检查

Question 18

P2-C03-R18 · 6 marks

(a) · 3 marks

Initial 40; limiting output 120.

t=0时指数为1；t增大时0.7^t趋0，但任何有限时间仍为正。

B1	初值
M1	指数衰减解释
A1	极限值

(b) · 3 marks

$t = \log(1/4)/\log 0.7 \approx 3.89$ s.

120−80(0.7)^t=100给(0.7)^t=1/4。

M1	正确指数方程
M1	对数求解
A1	时间

Question 19

P2-C03-R19 · 6 marks

(a) · 4 marks

$x = \log 3 / \log 2$; reject $u = -4$ since $u = 2^x > 0$.

$u^2+u-12=(u+4)(u-3)$，只留$u=3$后对数解x。

M1	指数代换及因式
M1	正值筛选
M1	换回x
A1	精确解

(b) · 2 marks

$x \approx 1.585$.

计算$\log 3/\log 2$，不提前舍入对数。

M1	正确数值计算
A1	三位小数

Question 20

P2-C03-R20 · 5 marks

(a) · 2 marks

$(1/5,-1),(1,0),(25,2)$.

对数输出给定时，输入是5的对应次幂。

M1	对数与指数对应
A1	三点

(b) · 3 marks

$y=\log_5(x-2)+3$; $x=2$.

右移用$x-2$，真数正界为$x=2$；上移不改变竖直渐近线。

M1	两方向平移
A1	方程
B1	渐近线

Question 21

P2-C03-R21 · 6 marks

(a) · 3 marks

$t = \log(5/3)/\log(22/19) \approx 3.48$ years.

相除得到$(1.1/0.95)^t=5/3$，再取对数。

M1	消去共同指数并取比例
M1	对数解时间
A1	精度

(b) · 3 marks

$A(5) \approx 483.153$; $B(5) \approx 386.890$; site A.

分别代$t=5$，比较而非仅凭增长或衰减方向猜测。

M1	两模型正确代值
A1	两值
B1	比较结论

Question 22

P2-C03-R22 · 4 marks

(a) · 2 marks

$\log_a 81 = \log_a (9^2) = 2\log_a 9$, so the equality holds for every permitted a.

使用对数幂法则表明等式本身是恒等关系，不是独立底数条件。

M1	正确幂法则
A1	任意合法底均成立

(b) · 2 marks

$a = \sqrt{3}$.

$a^2=3$，底必须正，取$\sqrt{3}$且不等于1。

M1	对数转指数
A1	合法底

Question 23

P2-C03-R23 · 6 marks

(a) · 4 marks

$x = -1 + \sqrt{10}$.

定义域x>0；合并得x(x+2)=9，方程x^2+2x−9=0，取正根。

M1	定义域与对数合并
M1	二次公式
M1	排除负根
A1	精确解

(b) · 2 marks

2.

由原方程已得乘积9，所以log3 9=2。

M1	使用已知乘积
A1	精确值

Question 24

P2-C03-R24 · 5 marks

(a) · 3 marks

$x = -5/4$.

统一底3，指数2x−2=6x+3，解4x=−5。

M1	统一底
M1	指数方程
A1	x

(b) · 2 marks

$3^{-9/2}$.

代入指数2x−2=−9/2。

M1	正确原指数代入
A1	幂形式

Question 25

P2-C03-R25 · 5 marks

(a) · 2 marks

£2433.31.

计算2000×1.04^5，金额最后保留两位小数。

M1	正确复利代值
A1	金额精度

(b) · 3 marks

11 years.

$n>\log 1.5/\log 1.04\approx10.338$，所以最小整数11；可检查10、11两年。

M1	建立严格不等式
M1	对数阈值
A1	最小整数

Question 26

P2-C03-R26 · 6 marks

(a) · 3 marks

$8/3$.

$b=a^3, c=a^2$；换底给$\log_b c=2/3$，所以$\log_b(b^2c)=2+2/3$。

M1	将底关系写清
M1	换底
A1	值

(b) For these numbers,

(i) · 2 marks

$t=2$.

写成$a^{3t}=a^{2t+2}$。因$a>0, a\neq 1$，指数相等，得$t=2$。

M1	相同底与指数方程
A1	解

(ii) · 1 marks

$a>1$.

$b/c=a$；由$b>c>0$直接得到$a>1$。

B1	比值或单调性的正确理由

Question 27

P2-C03-R27 · 7 marks

(a) · 3 marks

$(3/2, 2\sqrt{2})$.

后式2^{3−x}，同底比较得x=3−x。

M1	统一底
M1	解x
A1	精确交点

(b) · 4 marks

$x = 1, 2$.

令u=2^x>0，则u+8/u=6，u^2−6u+8=0，u=2或4。

M1	正确正变量代换
M1	二次求解
M1	映射回x
A1	两值

Question 28

P2-C03-R28 · 7 marks

(a) · 3 marks

$c = 4, K = 2.$

指数在t向负方向时趋0，因此渐近线给c=4，初值K+c=6。

B1	c
M1	初值条件
A1	K

(b) · 4 marks

$t = 1$; $2(3^t) > 0$ for every finite real t.

10=2×3^t+4给3^t=3；严格正指数项使Y始终大于4。

M1	解指数条件
A1	t
M1	指数严格正
A1	不达到渐近值解释

Question 29

P2-C03-R29 · 6 marks

(a) · 2 marks

$a = 0.8618$.

$a^3 = 0.64$，所以$a = 0.64^{1/3} = 0.861773\ldots$。

M1	指数模型的比值
A1	参数精度

(b) Using the unrounded value of a, find

(i) · 2 marks

9.32 cm.

$a^x = 1/4$，取对数$x = \log(1/4)/\log a = 9.318\ldots$。

M1	对数求未知指数
A1	厚度

(ii) · 2 marks

16 cm.

由$a^x \le 0.1$，注意$\log a < 0$，得$x \ge \log 0.1/\log a = 15.47\ldots$；最小整数为$16$。

M1	正确不等号与阈值
A1	取整

Question 30

P2-C03-R30 · 6 marks

(a) · 3 marks

$x = 8,\ y = 4.$

第一式给$xy = 32$，第二式给$x/y = 2$。代入$x = 2y$得$y = 4$（正数），$x = 8$。

M1	正确合并对数
M1	正确比值条件
A1	x,y

(b) Let $z > 0$ satisfy $\log_x z + \log_y z = 5$.

(i) · 2 marks

$6.$

设$t = \log_2 z$。换底后$t/3 + t/2 = 5$，故$t = 6$。

M1	换底
A1	t

(ii) · 1 marks

$64.$

$\log_2 z = 6$即$z = 2^6 = 64$。

B1	精确值

Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 4 — The binomial expansion Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第4章 · 二项式展开
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P2-C04-R01 · P2 · Chapter 4

$f(x) = (1 + 2x)^6$.

(a) Find the first four terms in ascending powers of x. (3)

(b) Find the coefficient of x^3 in $(1 - x)f(x)$. (2)

(Total for Question 1 is 5 marks)

2. P2-C04-R02 · P2 · Chapter 4

The coefficient of x^2 in $(1 + ax)^5$ is 90, where $a > 0$.

(a) Find a. (3)

(b) Find the coefficient of x^4 for this value of a. (2)

(Total for Question 2 is 5 marks)

The expansion of $(2+x)^n$ has constant term 32, where n is a positive integer.

(a) Find n. (2)

(b) For this value of n, find

(i) the coefficient of x^2, (2)

(ii) the coefficient of x^5 in $(2+x)^n(1+3x)$. (2)

(Total for Question 3 is 6 marks)

In the expansion of $(1+x)^n$, where n is an integer and $n \geq 3$, the coefficients of x, x^2 and x^3, in that order, form an arithmetic sequence.

(a) Show that $n^2 - 9n + 14 = 0$. (3)

(b) Hence

(i) find n. (2)

(ii) find the coefficient of x^4 in the expansion. (2)

(Total for Question 4 is 7 marks)

$D = (3 + x)^5 - (3 - x)^5$.

(a) Find D as a polynomial in x. (3)

(b) Hence find the coefficient of x^2 in D/x for $x \neq 0$. (2)

(Total for Question 5 is 5 marks)

6.

The first three terms of $(1 + px)^n$ are $1 + 12x + 60x^2$, where n is a positive integer and $p > 0$.

(a) Form two equations in n, p. (2)

(b) Find n, p. (3)

(Total for Question 6 is 5 marks)

7.

Consider $(1 + 0.02)^5$.

(a) Using the first three terms of a binomial expansion, estimate its value. (3)

(b) Find the exact sum of the omitted terms and state whether the estimate is too large or too small. (3)

(Total for Question 7 is 6 marks)

8.

The coefficient of x in $(3 + bx)^4$ is -216.

(a) Find b. (2)

(b) Find the coefficient of x^3 and explain its sign. (3)

(Total for Question 8 is 5 marks)

$F(x) = (1 + x^2)^5$.

(a) Find the coefficient of x^6. (2)

(b) Find the coefficient of x^7 in $(2 + x)F(x)$. (3)

(Total for Question 9 is 5 marks)

The expansion of $(1+2x)^7$ is $a_0 + a_1x + \cdots + a_7x^7$.

(a) Find $a_0 + a_1 + \cdots + a_7$ without calculating the eight coefficients separately. (2)

(b) Find $a_0 - a_1 + a_2 - a_3 + \cdots - a_7$. (2)

(Total for Question 10 is 4 marks)

11.

In $(1 + x)^n$, the coefficient of x^2 is three times the coefficient of x.

(a) Find the positive integer n. (3)

(b) Find the coefficient of x^5 for this value of n. (2)

(Total for Question 11 is 5 marks)

$F(x) = (2 - x)^6$.

(a) Find the coefficients of x^2 and x^3. (2)

(b) Find k such that the coefficient of x^3 in $(1 + kx)F(x)$ is zero. (2)

(Total for Question 12 is 4 marks)

A cube has side $(10 + h)$ cm, where $h > 0$.

(a) Expand its volume in ascending powers of h. (2)

(b) For $h = 0.1$, find the error if only the constant and linear terms are used to estimate the volume. (3)

(Total for Question 13 is 5 marks)

The expansion of $(a + x)^5$, with $a > 0$, has coefficient of x^3 equal to 40.

(a) Find a. (2)

(b) Find the ratio of the coefficients of x^4 and x^2, in that order. (3)

(Total for Question 14 is 5 marks)

$P(x) = (1 + 3x)^4(1 - x)^2$.

(a) Find the constant and linear terms in $P(x)$. (2)

(b) Find the coefficient of x^2. (3)

(Total for Question 15 is 5 marks)

For a positive integer n, $(1-x)^n = 1 - 9x + ax^2 + bx^3 + \cdots$.

(a) Find n, a, b. (3)

(b) Find the coefficient of x^2 in $(1+x)(1-x)^n$. (2)

(Total for Question 16 is 5 marks)

Consider $(x + 2/x)^6$ for $x \neq 0$.

(a) Find the term independent of x. (3)

(b) Find the coefficient of x^2. (2)

(Total for Question 17 is 5 marks)

18.

The coefficient of x^2 in $(1+ax)^4 + (1-ax)^4$ is 48, where $a > 0$.

(a) Find a. (2)

(b) For this value, find the coefficient of x^4 and explain why the coefficient of x^3 is zero. (3)

(Total for Question 18 is 5 marks)

$F(x) = (3 - 2x)^5$.

(a) Find the coefficient of x^4. (2)

(b) Find the sum of all coefficients of the polynomial $F(2x)$. (3)

(Total for Question 19 is 5 marks)

The polynomial $P(x) = (1 + x)^6$ has coefficients a_k at x^k.

(a) Find a_2 and a_4 and explain why they agree. (3)

(b) Find the coefficient of x^3 in $(1 - x^2)P(x)$. (2)

(Total for Question 20 is 5 marks)

The expansion of $(2 + ax)^3$ has equal coefficients of x and x^2, where $a \neq 0$.

(a) Find a. (3)

(b) For this value of a, find the sum of the constant term and the coefficient of x^3. (2)

(Total for Question 21 is 5 marks)

Use the binomial theorem on $(100 - 1)^4$.

(a) Write the full expansion without evaluating 99^4 directly. (3)

(b) Hence find the final three digits of 99^4, explaining why the first three terms do not affect them. (3)

(Total for Question 22 is 6 marks)

$P(x) = (1 + x)^5(1 - x)^5$.

(a) Simplify $P(x)$ before expanding. (2)

(b) Find the coefficients of x^4 and x^7. (3)

(Total for Question 23 is 5 marks)

The first three terms of $(a + bx)^4$ are $16 + 96x + cx^2$, where $a > 0$.

(a) Find a, b. (3)

(b) Find c and the coefficient of x^4. (3)

(Total for Question 24 is 6 marks)

$F(x) = (1 - 2x)^7$.

(a) Find the first three non-zero terms in ascending powers of x. (3)

(b) Use these terms to estimate 0.98^7, and give the estimate to 4 decimal places. (3)

(Total for Question 25 is 6 marks)

In the expansion of $(x+3)^5$,

(a) find the coefficient of x^2, (2)

(b) find all real k such that the coefficient of x^2 in $(x+k)(x+3)^5$ is zero. (3)

(Total for Question 26 is 5 marks)

For a positive integer n, the coefficients of x and x^2 in $(1+x)^n$ differ by 14, with the coefficient of x^2 larger.

(a) Form and solve an equation for n. (4)

(b) Find the coefficient of x^3 for this value. (2)

(Total for Question 27 is 6 marks)

$P(x) = (1 + 2x)^4 + (1 - 2x)^4 - 2.$

(a) Find $P(x)$ in fully simplified form. (3)

(b) Show that $P(x) > 0$ for every real $x \neq 0$. (3)

(Total for Question 28 is 6 marks)

The expansion of $(1 + ax)^5$, where $a > 0$, has coefficient of x^2 equal to twice the coefficient of x.

(a) Find a. (2)

(b) For this value of a, the expansion of $(1 + ax)^5(1 + bx)$ has no x^3 term.

(i) Find b. (2)

(ii) Find the coefficient of x^5 in the product. (2)

(Total for Question 29 is 6 marks)

A polynomial $P(x)$ is defined by $P(x) = (2 + x)^6 + (2 - x)^6$.

(a) Explain why $P(x)$ contains no odd powers of x, and find its first three non-zero terms in ascending powers of x. (3)

(b) Let $Q(x) = (2 + x)^6 - (2 - x)^6$.

(i) Find the coefficient of x^3 in $Q(x)$. (2)

(ii) Find the coefficient of x^4 in $P(x) + xQ(x)$. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C04-R01 · 5 marks

(a) · 3 marks

$1+12x+60x^2+160x^3$.

用二项式系数1、6、15、20，与(2x)^k相乘。

M1	正确二项式系数
M1	处理2x的各次幂
A1	四项

(b) · 2 marks

100.

所需项为原三次项减原二次项，160−60=100。

M1	识别两个贡献
A1	系数

Question 2

P2-C04-R02 · 5 marks

(a) · 3 marks

$a = 3$.

$10a^2=90$，正值取3。

M1	二次项系数关系
M1	正值选根
A1	a

(b) · 2 marks

405.

系数$5a^4=5\times 81$。

M1	正确项系数
A1	值

Question 3

P2-C04-R03 · 6 marks

(a) · 2 marks

$n = 5$.

常数项2^n=32，得5。

M1	常数项关系
A1	n

(b) For this value of n, find

(i) · 2 marks

80.

$C(5,2)2^3=10\times 8$。

M1	正确二项式项
A1	系数

(ii) · 2 marks

31.

来自原x^5系数1加3倍原x^4系数10，合31。

M1	两项贡献
A1	结果

Question 4

P2-C04-R04 · 7 marks

(a) · 3 marks

$n^2 - 9n + 14 = 0$.

三系数为$n, n(n-1)/2, n(n-1)(n-2)/6$。等差条件为$n(n-1) = n + n(n-1)(n-2)/6$；除以$n > 0$并整理得所求。

M1	三个二项系数
M1	等差条件
A1	代数整理

(b) Hence

(i) · 2 marks

$n = 7$.

$(n-2)(n-7) = 0$，由$n \geq 3$排除2。

M1	分解并用范围
A1	n

(ii) · 2 marks

35.

代入$n = 7$，系数$\binom{7}{4} = 35$。

M1	对应二项系数
A1	值

Question 5

P2-C04-R05 · 5 marks

(a) · 3 marks

$D = 810x + 180x^3 + 2x^5$.

偶次项抵消；奇次项两倍，$2\mathrm{C}(5,1)3^4=810$，$2\mathrm{C}(5,3)3^2=180$。

M1	奇偶项筛选
M1	正确系数与3的幂
A1	三项

(b) · 2 marks

180.

除x使幂次全降1，原三次系数成为二次系数。

M1	正确幂次对应
A1	系数

Question 6

P2-C04-R06 · 5 marks

(a) · 2 marks

$np=12,\ n(n-1)p^2/2=60.$

一次项与二次项分别比较。

M1	一次项条件
A1	二次项条件

(b) · 3 marks

$n=6, p=2.$

将第二式除第一式平方得$(n-1)/(2n)=60/144=5/12$，解$n=6$再求p。

M1	正确消去p
M1	解n
A1	两参数

Question 7

P2-C04-R07 · 6 marks

(a) · 3 marks

$1 + 0.10 + 0.004 = 1.104$.

取$1+5u+10u^2$，$u=0.02$。

M1	正确前三项
M1	代小量
A1	估计

(b) · 3 marks

0.0000808032; underestimate.

余项$10u^3+5u^4+u^5=0.00008+0.0000008+0.0000000032$，全正。

M1	列出全部余项
M1	计算余项
A1	数值及方向

Question 8

P2-C04-R08 · 5 marks

(a) · 2 marks

$b = -2$.

一次系数4×3^{3}b=108b。

M1	正确系数关系
A1	b

(b) · 3 marks

-96; it contains $b^{3} < 0$.

C(4,3)$3b^{3}$=12×(−8)；其余因子正。

M1	正确三次项
A1	系数
B1	负号理由

Question 9

P2-C04-R09 · 5 marks

(a) · 2 marks

10.

需选三次x^2，$C(5,3)=10$。

M1	正确指数索引
A1	系数

(b) · 3 marks

10.

F只有偶次幂，2F没有x^7项，xF的x^7来自F中x^6。

M1	排除不可能幂次
M1	唯一贡献
A1	系数

Question 10

P2-C04-R10 · 4 marks

(a) · 2 marks

2187.

令$x=1$，系数和$=(1+2)^7=3^7$。

M1	选择$x=1$
A1	结果

(b) · 2 marks

-1.

令$x=-1$，值$=(1-2)^7=-1$。

M1	选择x负1
A1	结果

Question 11

P2-C04-R11 · 5 marks

(a) · 3 marks

$n = 7$.

n(n−1)/2=3n，n>0可除n，n−1=6。

M1	系数关系
M1	合法约去n
A1	n

(b) · 2 marks

21.

C(7,5)=C(7,2)=21。

M1	正确组合数
A1	系数

Question 12

P2-C04-R12 · 4 marks

(a) · 2 marks

240 and -160.

$C(6,2)2^4=240$；$C(6,3)2^3(-1)^3=-160$。

M1	正确二项式项
A1	两系数

(b) · 2 marks

$k=2/3$.

三次系数为$-160+240k$，令零。

M1	两个贡献并令零
A1	k

Question 13

P2-C04-R13 · 5 marks

(a) · 2 marks

$1000+300h+30h^2+h^3$ cm^3.

用三次二项式展开，系数1、3、3、1。

M1	正确二项式展开
A1	四项及单位

(b) · 3 marks

0.301 cm^3, underestimate.

遗漏项$30h^2+h^3=0.3+0.001$，均正。

M1	正确遗漏项
M1	代值
A1	误差及方向

Question 14

P2-C04-R14 · 5 marks

(a) · 2 marks

$a = 2$.

$\binom{5}{3}a^2=10a^2=40$，正值取2。

M1	系数条件
A1	a

(b) · 3 marks

$1:8$.

系数分别$5a=10$及$10a^3=80$，约简10:80。

M1	两个正确系数
M1	按顺序取比
A1	比值

Question 15

P2-C04-R15 · 5 marks

(a) · 2 marks

$1+10x$.

常数1；一次项12x−2x=10x。

M1	正确低次展开
A1	两项

(b) · 3 marks

31.

贡献54−24+1=31，分别来自二次×常数、一次×一次、常数×二次。

M1	列齐三种贡献
M1	正确三数
A1	系数

Question 16

P2-C04-R16 · 5 marks

(a) · 3 marks

$n=9, a=36, b=-84.$

一次项−n=−9，二次C(9,2)=36，三次−C(9,3)=−84。

M1	一次项确定n
M1	后续系数
A1	三值

(b) · 2 marks

27.

原二次36加x乘原一次−9，合27。

M1	两项贡献
A1	系数

Question 17

P2-C04-R17 · 5 marks

(a) · 3 marks

160.

第k次选2/x后幂为6−2k；零幂k=3，系数C(6,3)2^3。

M1	正确幂次表达式
M1	求索引
A1	常数项

(b) · 2 marks

60.

6−2k=2给k=2，系数C(6,2)2^2。

M1	幂次索引
A1	系数

Question 18

P2-C04-R18 · 5 marks

(a) · 2 marks

$a = 2$.

两展开式二次系数各$6a^2$，故$12a^2=48$。

M1	偶次项合并
A1	正a

(b) · 3 marks

32; the cubic terms cancel because their signs are opposite.

四次为$2a^4=32$；三次系数$4a^3$与$-4a^3$抵消。

M1	四次贡献
A1	32
B1	三次抵消理由

Question 19

P2-C04-R19 · 5 marks

(a) · 2 marks

240.

$C(5,4)3(-2)^4=5\times3\times16$。

M1	正确项
A1	系数

(b) · 3 marks

-1.

系数和为$F(2)$，即$(3-4)^5=-1$，不是$F(1)$。

M1	选择新多项式的$x=1$
M1	正确原输入2
A1	总和

Question 20

P2-C04-R20 · 5 marks

(a) · 3 marks

$a_2 = a_4 = 15$; $\binom{6}{2} = \binom{6}{4}$.

选择k个x与选择其余6−k个常数具有对称二项式系数。

M1	正确两组合数
A1	值
B1	对称性理由

(b) · 2 marks

14.

原三次20减原一次6，得14。

M1	正确幂次贡献
A1	系数

Question 21

P2-C04-R21 · 5 marks

(a) · 3 marks

$a = 2$.

一次12a，二次6a^2，等式除6a得a=2。

M1	两项系数
M1	使用非零条件
A1	a

(b) · 2 marks

16.

常数8，最高次系数a^3=8，相加16。

M1	正确两端系数
A1	总和

Question 22

P2-C04-R22 · 6 marks

(a) · 3 marks

$100^4 - 4(100)^3 + 6(100)^2 - 4(100) + 1$.

正整数四次展开，负1使符号交替。

M1	正确系数
M1	符号
A1	五项

(b) · 3 marks

601; each of the first three terms is a multiple of 1000, and $-400 + 1 = -399$ leaves final three digits 601.

减399等于减1000再加601，因此后三位601；明确处理负尾数。

M1	识别整千倍数
M1	处理负剩余
A1	三位数字

Question 23

P2-C04-R23 · 5 marks

(a) · 2 marks

$P(x) = (1 - x^2)^5$.

同次幂相乘可合为$[(1+x)(1-x)]^5$，再平方差。

M1	合并同次幂
A1	简式

(b) · 3 marks

10 and 0.

x^4来自选择两次$-x^2$，$C(5,2)=10$；只有偶次幂，x^7不存在。

M1	正确索引
A1	x四次系数
B1	奇次项为零

Question 24

P2-C04-R24 · 6 marks

(a) · 3 marks

$a = 2, b = 3$.

a^4=16取a=2；$4a^3b$=96给32b=96。

M1	常数确定正a
M1	一次项求b
A1	两值

(b) · 3 marks

$c = 216$; highest coefficient 81.

二次$6a^2b^2$=6×4×9，最高次b^4=81。

M1	二次项系数
A1	c
B1	最高次系数

Question 25

P2-C04-R25 · 6 marks

(a) · 3 marks

$1-14x+84x^2$.

系数1、7、21乘$(-2x)^k$。

M1	正确二项式系数
M1	负号和幂
A1	三项

(b) · 3 marks

0.8684.

设$1-2x=.98$得$x=.01$，代$1-.14+.0084=.8684$。

M1	正确选择x
M1	代展开
A1	估计

Question 26

P2-C04-R26 · 5 marks

(a) · 2 marks

270.

选两次x、三次3，$C(5,2)3^3$=270。

M1	正确项
A1	系数

(b) · 3 marks

$k=-3/2$.

来自原一次405与k倍原二次270，405+270k=0。

M1	识别两贡献
M1	正确一次系数
A1	k

Question 27

P2-C04-R27 · 6 marks

(a) · 4 marks

$n(n-1)/2-n=14$ gives $n=7$.

$n^2-3n-28=(n-7)(n+4)=0$，按正整数取7。

M1	正确差的顺序
M1	解二次方程
M1	正整数筛选
A1	n

(b) · 2 marks

35.

$C(7,3)=35$。

M1	正确组合数
A1	系数

Question 28

P2-C04-R28 · 6 marks

(a) · 3 marks

$48x^2 + 32x^4$.

常数2被减去，奇次项抵消，二次$2\times6\times4$，四次2×16。

M1	识别被抵消项
M1	两偶次系数
A1	多项式

(b) · 3 marks

$P(x) = 16x^2(3 + 2x^2) > 0$ for $x \neq 0$.

两个因子都严格正：x不为0使$16x^2$正，另因子至少3。

M1	因式分解
M1	两因子正性
A1	严格结论

Question 29

P2-C04-R29 · 6 marks

(a) · 2 marks

$a = 1$.

系数分别为$10a^2$和$5a$；$10a^2 = 2(5a)$，由$a > 0$取$a = 1$。

M1	正确两个二项系数
A1	合法参数

(b) For this value of a, the expansion of $(1 + ax)^5(1 + bx)$ has no x^3 term.

(i) · 2 marks

$b = -1$.

x^3系数为$10 + 10b$，令其为零得$b = -1$。

M1	两项贡献
A1	b

(ii) · 2 marks

-4.

x^5系数为$1 + 5b = 1 - 5 = -4$，不能忽略来自$bx \cdot 5x^4$的贡献。

M1	卷积的两项
A1	系数

Question 30

P2-C04-R30 · 7 marks

(a) · 3 marks

$128 + 480x^2 + 120x^4$.

展开中奇次项在两式中符号相反而抵消。偶次项：$2(2^6) = 128$，$2\binom{6}{2}2^4 = 480$，$2\binom{6}{4}2^2 = 120$。

B1	解释抵消
M1	二项系数
A1	三个非零项

(b) Let $Q(x) = (2 + x)^6 - (2 - x)^6$.

(i) · 2 marks

320.

差式保留并加倍奇次项，$2\binom{6}{3}2^3 = 320$。

M1	识别加倍奇次项
A1	系数

(ii) · 2 marks

440.

来自P的x^4项120及xQ的x^4项320，相加为440。

M1	追踪移位次数
A1	总系数

Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 5 — Sequences and series
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第5章 · 数列与级数
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1.

An arithmetic sequence has fifth term 17 and twelfth term 45.

(a) Find its first term and common difference. (3)

(b) Find the sum of its first 20 terms. (2)

(Total for Question 1 is 5 marks)

A geometric sequence has first term 12 and common ratio $-1/2$.

(a) Find its fourth and fifth terms. (2)

(b) Find the sum to infinity, explaining why it exists. (3)

(Total for Question 2 is 5 marks)

3.

A theatre has 16 rows. The first row has 18 seats, and each later row has 2 more seats than the preceding row.

(a) Find the number of seats in the final row and the total number of seats. (4)

(b) Every seat in the first 6 rows costs £20, and all remaining seats cost £15. Find

(i) the number of £20 seats, (2)

(ii) the total ticket income when all seats are sold. (3)

(Total for Question 3 is 9 marks)

The first three terms of an arithmetic sequence are k, $2k - 3$ and $4k - 11$.

(a) Find k. (2)

(b) Find the least n for which the nth term exceeds 100. (3)

(Total for Question 4 is 5 marks)

5.

A geometric sequence has second term 6 and fifth term 162. Its common ratio is positive.

(a) Find its first term and common ratio. (3)

(b) Find the sum of the first 6 terms. (2)

(Total for Question 5 is 5 marks)

6.

The sum of the first n terms of an arithmetic sequence is $S_n = 3n^2 + 2n$.

(a) Find its first term and common difference. (3)

(b) Find the least positive integer n such that $S_n > 500$. (3)

(Total for Question 6 is 6 marks)

7.

A bouncing ball is dropped from 5 m. Each rebound reaches $3/5$ of the preceding maximum height.

(a) Find the height reached on the fourth rebound. (2)

(b) Find the total vertical distance travelled from its initial release, including the first fall, over infinitely many rebounds under the model. (3)

(Total for Question 7 is 5 marks)

8.

An arithmetic sequence has $a = 40$ and $d = -3$.

(a) Find the last positive term and its position. (3)

(b) Find the sum of all positive terms. (2)

(Total for Question 8 is 5 marks)

9.

The first term of a geometric series is 8 and its sum to infinity is 20.

(a) Find the common ratio. (2)

(b) Find the sum of the first 4 terms and the exact sum of the remaining terms. (4)

(Total for Question 9 is 6 marks)

A sequence is defined by $u_1 = 4$ and $u_{n+1} = 2u_n - 3$.

(a) Find u_2, u_3, u_4. (2)

(b) Let $v_n = u_n - 3$. Show that (v_n) is geometric and find u_n in terms of n. (4)

(Total for Question 10 is 6 marks)

11.

A runner increases the daily distance by 0.5 km each day, starting at 3 km on day 1.

(a) Find the distance on day 12 and the total distance over the first 12 days. (4)

(b) Find the first day on which the cumulative distance exceeds 100 km. (3)

(Total for Question 11 is 7 marks)

For $r > 0$, the numbers $r - 1$, $r + 1$, $2r + 2$ are consecutive terms of a geometric sequence.

(a) Find r. (3)

(b) Find the sum of the first 5 terms when these are the first three terms. (3)

(Total for Question 12 is 6 marks)

$$S = \sum_{k=1}^{20} (3k - 2).$$

(a) Find S. (3)

(b) Find $\sum_{k=6}^{20} (3k - 2)$. (3)

(Total for Question 13 is 6 marks)

The sum of the first 10 terms of an arithmetic sequence is 180, and the sum of the first 20 terms is 660

.

(a) Find the first term and common difference. (3)

(b) Find the sum of terms 11 to 20 inclusive. (2)

(Total for Question 14 is 5 marks)

A geometric series has positive first term a and common ratio r with $0 < r < 1$. The sum of its first two terms is 18 and its sum to infinity is 24.

(a) Find r. (4)

(b) Find a and the least n for which $S_n > 23$. (4)

(Total for Question 15 is 8 marks)

A charity receives £100 in week 1, then £15 more each week than in the preceding week.

(a) Find the total received in 8 weeks. (2)

(b) From week 9 onwards the weekly amount stays equal to the week 8 amount. Find

(i) the fixed weekly amount, (2)

(ii) the first week by the end of which the cumulative total exceeds £2000. (3)

(Total for Question 16 is 7 marks)

The first three terms of a sequence are $2, 5, 8$.

(a) Assuming the sequence is arithmetic, find its twentieth term. (2)

(b) The formula $u_n = 3n - 1 + (n-1)(n-2)(n-3)$ also gives these first three terms. Find u_4 and explain why three listed terms alone do not prove that a sequence is arithmetic. (4)

(Total for Question 17 is 6 marks)

18.

A geometric sequence has first term 81 and common ratio $2/3$.

(a) Find its fifth term and the sum of its first five terms. (4)

(b) Find the sum to infinity of the sequence formed by its even-numbered terms. (4)

(Total for Question 18 is 8 marks)

19.

An arithmetic sequence has first term a and common difference $d \neq 0$. Its fourth, seventh and thirteenth terms form a geometric sequence in that order.

(a) Show that $a = 0$. (4)

(b) Given that the fourth term is 6, find d and the common ratio of these three selected terms. (3)

(Total for Question 19 is 7 marks)

An account receives £200 at the beginning of each year. At the end of each year its balance is multiplied by 1.05. There are no other transactions.

(a) Find the balance at the end of year 3. (3)

(b) Write a formula for the balance at the end of year n. (3)

(Total for Question 20 is 6 marks)

$u_n = 7 - 2n$ for positive integers n.

(a) Find the first negative term and its position. (2)

(b) Find $\sum_{n=4}^{12} u_n$. (3)

(Total for Question 21 is 5 marks)

A geometric series has first term 5 and ratio r, with $|r| < 1$. Its sum to infinity is $10/3$.

(a) Find r. (2)

(b) Find the sum of its first 6 terms and state whether it is above or below the infinite sum. (4)

(Total for Question 22 is 6 marks)

A sequence is defined by $u_1 = 2$ and $u_{n+1} = u_n + 3n$.

(a) Find u_2, u_3, u_4. (2)

(b) By summing the increments, show that $u_n = 2 + \frac{3}{2}n(n-1)$. (3)

(Total for Question 23 is 5 marks)

The third term of a geometric sequence is 12 and the fifth is 48.

(a) Find all possible real common ratios and corresponding first terms. (3)

(b) For which of these sequences is the sum of the first four terms negative? Find that sum. (3)

(Total for Question 24 is 6 marks)

The first n terms of an arithmetic sequence have sum $n(2n + 3)$.

(a) Find a formula for the nth term. (3)

(b) Find the sum of terms 21 to 40 inclusive. (3)

(Total for Question 25 is 6 marks)

26.

A machine produces 1200 items in its first hour. Each subsequent hour it produces 90% as many as in the previous hour. Production is modelled without rounding the output in individual hours.

(a) Find the total production in the first 8 hours, to the nearest item. (3)

(b) Find the greatest total production predicted over infinitely many hours and explain why it is finite. (3)

(Total for Question 26 is 6 marks)

The sum of the first n terms of a sequence is $S_n = 2n^2 + 5n$, for positive integers n.

(a) Show that the nth term is $u_n = 4n + 3$. (2)

(b) The sum of the terms from u_{n+1} to u_{2n} inclusive is 650.

(i) Show that $n(6n + 5) = 650$. (2)

(ii) Find n and the value of u_{2n}. (2)

(Total for Question 27 is 6 marks)

28.

A workshop produces 80 parts in its first week. Under plan A, production increases by 12 parts each subsequent week. Under plan B, production increases by 10% each subsequent week. Production under plan B is modelled without rounding to whole parts.

(a) Find the total production over the first 8 weeks under plan A. (2)

(b) For plan B,

(i) find the total production over the first 8 weeks to the nearest whole part. (2)

(ii) find the first week in which weekly production exceeds 200 parts. (2)

(Total for Question 28 is 6 marks)

29.

A geometric sequence has positive terms. Its second term is 18 and its fifth term is $243/4$.

(a) Find the first term a and common ratio r. (2)

(b) The first n terms have sum S_n.

(i) Show that $S_{n+1} = \frac{3}{2}S_n + 12$. (3)

(ii) Find the least n for which $S_n > 1000$. (2)

(Total for Question 29 is 7 marks)

At the end of each year, a trust pays a grant. The first grant is £5000, and each subsequent grant is 4% less than the preceding grant. No other payments are made.

(a) Find the total of all grants over the first 12 years, to the nearest pound. (2)

(b) The trust has £125000 available for these grants.

(i) Show that the total value of all grants in the infinite model is £125000. (3)

(ii) Find the least number of complete years after which the money not yet paid is less than £50000. (3)

(Total for Question 30 is 8 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C05-R01 · 5 marks

(a) · 3 marks

$a=1, d=4$.

a+4d=17，a+11d=45，相减7d=28。

M1	两项条件
M1	消元
A1	a与d

(b) · 2 marks

780.

S20=20[2+19×4]/2=780。

M1	正确等差和式
A1	总和

Question 2

P2-C05-R02 · 5 marks

(a) · 2 marks

$-3/2$ and $3/4$.

第n项12(−1/2)^{n−1}，注意符号交替。

M1	正确项指数
A1	两项

(b) · 3 marks

8; $|r| = 1/2 < 1$.

无穷和12/[1−(−1/2)]=8，负公比仍可收敛。

B1	收敛条件
M1	无穷和公式
A1	和

Question 3

P2-C05-R03 · 9 marks

(a) · 4 marks

Final row 48; total 528.

$a_{16}=18+15\times2$；$S_{16}=\frac{16(18+48)}{2}$。

M1	最后一行项数
A1	48
M1	等差求和
A1	528

(b) Every seat in the first 6 rows costs £20, and all remaining seats cost £15. Find

(i) · 2 marks

138.

$S_6=\frac{6[36+5\times2]}{2}=138$。

M1	前六行等差和
A1	席位

(ii) · 3 marks

£8610.

$20\times138+15\times(528-138)=2760+5850$。

M1	两票价分组
M1	后组席位
A1	总收入

Question 4

P2-C05-R04 · 5 marks

(a) · 2 marks

$k = 5.$

两差相等k−3=2k−8，得5。

M1	等差条件
A1	k

(b) · 3 marks

$n = 49.$

a=5,d=2，5+2(n−1)>100给n>48.5，最小49。

M1	正确通项
M1	严格不等式与整数
A1	n

Question 5

P2-C05-R05 · 5 marks

(a) · 3 marks

$a = 2, r = 3.$

ar=6，ar^4=162，除得r^3=27取3，a=2。

M1	项比消元
M1	求公比
A1	a与r

(b) · 2 marks

$728.$

$2(3^6-1)/(3-1)=728$。

M1	正确有限等比和
A1	总和

Question 6

P2-C05-R06 · 6 marks

(a) · 3 marks

$a = 5, d = 6.$

a=S1=5；a2=S2−S1=16−5=11，差6；或比较和式系数。

M1	从和求第一项
M1	求第二项或比较系数
A1	a与d

(b) · 3 marks

$n = 13.$

$3n^2+2n-500>0$，或检查S12=456，S13=533，且正n上S递增。

M1	正确阈值
M1	相邻整数验证及增长
A1	n

Question 7

P2-C05-R07 · 5 marks

(a) · 2 marks

$81/125 = 0.648$ m.

第k次反弹高5(3/5)^k，第四次5×81/625。

M1	正确反弹索引
A1	高度

(b) · 3 marks

20 m.

初降5，加两倍所有反弹高度和：5+2[3/(1−3/5)]=20。

M1	初次下落单独计入
M1	上下双程等比和
A1	路程

Question 8

P2-C05-R08 · 5 marks

(a) · 3 marks

Last positive term $a_{14} = 1$.

40−3(n−1)>0给n<43/3，最大整数14，代入1。

M1	正通项条件
M1	最大整数索引
A1	位置和项

(b) · 2 marks

287.

S14=14(40+1)/2=287。

M1	使用正确末项及项数
A1	总和

Question 9

P2-C05-R09 · 6 marks

(a) · 2 marks

$r = 3/5$.

8/(1−r)=20给1−r=2/5；结果满足绝对值小于1。

M1	无穷和关系
A1	r

(b) · 4 marks

$S_4 = 2176/125$; remaining sum $324/125$.

S4=20[1−(3/5)⁴]=2176/125；用20减S4得324/125。

M1	有限等比和
A1	S4
M1	从无穷和相减
A1	余和

Question 10

P2-C05-R10 · 6 marks

(a) · 2 marks

$5, 7, 11$.

按递推逐次计算，下一项依赖上一项而非索引n。

M1	正确逐项递推
A1	三项

(b) · 4 marks

$v_{n+1} = 2v_n$, $v_1 = 1$; $u_n = 3 + 2^{n-1}$.

两边减3得u_{n+1}−3=2(u_n−3)，所以v为公比2等比，再还原u。

M1	正确改写递推
M1	确定首项和公比
M1	还原u
A1	通项

Question 11

P2-C05-R11 · 7 marks

(a) · 4 marks

8.5 km; 69 km.

$a_{12}=3+11\times.5$，$S_{12}=12(3+8.5)/2$。

M1	通项索引
A1	第12天
M1	求和
A1	总距离

(b) · 3 marks

Day 16.

$S_n=n[6+.5(n-1)]/2=.25n^2+2.75n$；$S_{15}=97.5$，$S_{16}=108$。

M1	累计和表达式
M1	检查相邻整数
A1	日数

Question 12

P2-C05-R12 · 6 marks

(a) · 3 marks

$r = 3$.

$(r+1)^2=(r-1)(2r+2)$，化为$(r+1)(r-3)=0$；$r>0$排除-1。

M1	等比中项平方条件
M1	解并筛选
A1	r

(b) · 3 marks

62.

首项2、公比2，$S5=2(2^5-1)=62$。

M1	正确a与公比
M1	有限和
A1	总和

Question 13

P2-C05-R13 · 6 marks

(a) · 3 marks

590.

所加项为1、4、…、58，20项等差和20(1+58)/2。

M1	正确首末项与项数
M1	等差和
A1	S

(b) · 3 marks

555.

先五项和为5(1+13)/2=35，从590减去35。

M1	正确去除前五项
M1	首五和
A1	结果

Question 14

P2-C05-R14 · 5 marks

(a) · 3 marks

$a = 9/2, d = 3$.

2a+9d=36，2a+19d=66，相减10d=30，a=4.5。

M1	两个和条件
M1	消元
A1	a与d

(b) · 2 marks

480.

该分段为S20−S10=660−180。

M1	正确部分和差
A1	结果

Question 15

P2-C05-R15 · 8 marks

(a) · 4 marks

$r = 1/2$.

a(1+r)=18与a/(1−r)=24相除得$1-r^2=3/4$，$r^2=1/4$取正。

M1	两和条件
M1	消去a
M1	用正公比条件
A1	r

(b) · 4 marks

$a = 12$, least $n = 5$.

Sn=24(1−2^{−n})>23给2^n>24，n最小5。

M1	a
M1	正确有限和不等式
M1	整数比较
A1	n

Question 16

P2-C05-R16 · 7 marks

(a) · 2 marks

£1220.

S8=8[200+7×15]/2。

M1	等差求和式
A1	总额

(b) From week 9 onwards the weekly amount stays equal to the week 8 amount. Find

(i) · 2 marks

£205.

a8=100+7×15。

M1	正确索引
A1	金额

(ii) · 3 marks

Week 12.

额外k周需1220+205k>2000，k>780/205≈3.805，取4，再加8。

M1	正确分段累计不等式
M1	取额外整数周
A1	总周数

Question 17

P2-C05-R17 · 6 marks

(a) · 2 marks

59.

a=2,d=3，a20=2+19×3。

M1	识别a,d
A1	结果

(b) · 4 marks

$u_4 = 17$; an alternative rule matches the first three terms but not the fourth arithmetic term 11.

代4得12−1+6=17；共同前三项不足唯一规定后续规则。

M1	正确代n=4
A1	u4
M1	比较不同后续
A1	说明前提作用

Question 18

P2-C05-R18 · 8 marks

(a) · 4 marks

16; 211.

$a_5=81(2/3)^4=16$；$S_5=81[1-(2/3)^5]/(1/3)=211$。

M1	正确通项
A1	第五项
M1	有限和
A1	总和

(b) · 4 marks

$486/5$.

偶数项首项54，新公比$(2/3)^2=4/9$，无穷和$54/(1-4/9)$。

M1	正确新首项
M1	正确新公比
M1	无穷和
A1	精确结果

Question 19

P2-C05-R19 · 7 marks

(a) · 4 marks

$(a+6d)^2=(a+3d)(a+12d)$ gives $-3ad=0$, hence $a=0$.

两边展开，a^2和$36d^2$抵消，剩$12ad=15ad$；d非零才可推出a零。

M1	等比条件
M1	展开化简
M1	使用d非零
A1	a

(b) · 3 marks

$d=2$; ratio 2.

首项0，第四项$3d=6$，三选项6、12、24。

M1	正确d
M1	求三选项
A1	比值

Question 20

P2-C05-R20 · 6 marks

(a) · 3 marks

£662.03 to the nearest penny.

三笔分别积累3、2、1年，余额$200(1.05^3+1.05^2+1.05)=662.025$。

M1	正确三笔计息指数
M1	相加
A1	金额精度

(b) · 3 marks

$B_n = 210\dfrac{1.05^n-1}{0.05}$.

按计息年数排列为$210+210(1.05)+\cdots+210(1.05)^{n-1}$，共n项等比和。

M1	正确等比首项及项数
M1	有限和
A1	公式

Question 21

P2-C05-R21 · 5 marks

(a) · 2 marks

$u_4 = -1$.

7−2n<0给n>3.5，首个正整数4。

M1	负项条件
A1	项与索引

(b) · 3 marks

-81.

共有9项，首−1末−17，等差和9(−18)/2=−81。

M1	正确项数
M1	首末项和
A1	结果

Question 22

P2-C05-R22 · 6 marks

(a) · 2 marks

$r = -1/2$.

5/(1−r)=10/3给1−r=3/2。

M1	无穷和方程
A1	r

(b) · 4 marks

$S_6 = 105/32$; below $10/3$.

S6=(10/3)(1−1/64)=105/32；偶数幂余项正被减去。

M1	正确有限和
A1	精确值
M1	比较余项符号
A1	方向

Question 23

P2-C05-R23 · 5 marks

(a) · 2 marks

$5, 11, 20.$

依次加3、6、9，而不是每步固定加3。

M1	正确索引递推
A1	三项

(b) · 3 marks

$u_n = 2 + 3\sum_{k=1}^{n-1} k = 2 + 3n(n-1)/2.$

从u1到un需要n−1次增量，增量构成3、6、…、3(n−1)的等差和。

M1	正确增量项数
M1	等差求和
A1	公式

Question 24

P2-C05-R24 · 6 marks

(a) · 3 marks

$r = 2$ or $r = -2$; $a = 3$ in both cases.

两项相除$r^2=4$，保留两符号；$ar^2=12$给$a=3$。

M1	项比条件
M1	两符号
A1	a与公比

(b) · 3 marks

$r = -2$; $S_4 = -15$.

$r=-2$时$3-6+12-24=-15$；$r=2$所有项正。

M1	分析两公比符号
M1	负公比四项或和式
A1	公比与和

Question 25

P2-C05-R25 · 6 marks

(a) · 3 marks

$a_n = 4n + 1$.

Sn−S(n−1)=2n²+3n−[2(n−1)²+3(n−1)]=4n+1，n=1也一致。

M1	相邻和差
M1	展开化简
A1	通项

(b) · 3 marks

2460.

S40−S20=40×83−20×43=3320−860。

M1	正确部分和差
M1	代入
A1	总和

Question 26

P2-C05-R26 · 6 marks

(a) · 3 marks

6834 items.

S8=1200(1−.9⁸)/.1=6834.39348，最后舍入。

M1	正确等比和
M1	不提前舍入
A1	整数产量

(b) · 3 marks

12000 items; $|r| = 0.9 < 1$.

无穷等比和1200/(1−.9)，各小时产量趋零；只是模型极限。

M1	无穷和
A1	数值
B1	收敛理由

Question 27

P2-C05-R27 · 6 marks

(a) · 2 marks

$u_n = 4n + 3$.

对$n \geq 2$，$u_n = S_n - S_{n-1} = 2[n^2 - (n-1)^2] + 5 = 4n + 3$；另核对$n = 1$有$u_1 = S_1 = 7$，公式同样成立。

M1	相邻部分和作差
A1	化简并核对首项

(b) The sum of the terms from u_{n+1} to u_{2n} inclusive is 650.

(i) · 2 marks

$S_{2n} - S_n = 6n^2 + 5n = 650$.

此段有n项，和为$S_{2n} - S_n = (8n^2 + 10n) - (2n^2 + 5n) = 6n^2 + 5n$。

M1	正确上下限对应部分和
A1	所得方程

(ii) · 2 marks

$n = 10$; $u_{20} = 83$.

$6n^2 + 5n - 650 = (n - 10)(6n + 65) = 0$。由$n$为正整数取$10$，所以$u_{20} = 4(20) + 3 = 83$。

M1	解二次方程
A1	合法整数与末项

Question 28

P2-C05-R28 · 6 marks

(a) · 2 marks

976 parts.

等差数列首项80、公差12，$S_8 = 8[2(80) + 7(12)]/2 = 976$。

M1	等差和
A1	总数

(b) For plan B,

(i) · 2 marks

915 parts.

等比和$80(1.1^8 - 1)/(1.1 - 1) = 914.871\ldots$，最终取整为$915$。

M1	等比和
A1	正确最终取整

(ii) · 2 marks

Week 11.

$80(1.1)^{n-1} > 200$，得$n - 1 > \log 2.5 / \log 1.1 = 9.613\ldots$，故$n = 11$；这比较的是单周产量而非累计产量。

M1	对数解指数阈值
A1	首个周次

Question 29

P2-C05-R29 · 7 marks

(a) · 2 marks

$a = 12, r = 3/2$.

$ar = 18, ar^4 = 243/4$。相除$r^3 = 27/8$，所以$r = 3/2, a = 12$。

M1	两项条件相除
A1	a,r

(b) The first n terms have sum S_n.

(i) · 3 marks

$S_n = 24[(3/2)^n - 1]$, hence the relation.

有限等比和$S_n = 12[(3/2)^n - 1]/(1/2) = 24[(3/2)^n - 1]$；乘$3/2$再加$12$得到$24[(3/2)^{n+1} - 1]$。

M1	写出部分和
M1	对比n与n+1
A1	关系成立

(ii) · 2 marks

$n = 10$.

$24[(3/2)^n - 1] > 1000$即$(3/2)^n > 128/3$。取对数$n > 9.257067\ldots$，所以最小正整数为10。

M1	正确和的阈值
A1	最小整数

Question 30

P2-C05-R30 · 8 marks

(a) · 2 marks

£48411.

首项5000、公比0.96，前12项和$5000(1-0.96^{12})/0.04 = 48411.\ldots$，四舍五入到英镑。

M1	有限等比和
A1	金额

(b) The trust has £125000 available for these grants.

(i) · 3 marks

$5000/(1-0.96) = 125000$.

因为$|0.96| < 1$，无穷等比和存在，值为125000。

B1	收敛条件
M1	无穷和
A1	金额

(ii) · 3 marks

23 years.

$125000 - S_n = 125000(0.96)^n$。要求$(0.96)^n < 0.4$，得$n > \log 0.4 / \log 0.96 = 22.446\ldots$；至少23年。

M1	用无穷和减有限和
M1	对数不等式
A1	年数

Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 6 — Trigonometric identities and equations
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第6章 · 三角恒等式与方程
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

For angles θ for which all the expressions are defined,

(a) prove that $\tan\theta + \dfrac{1}{\tan\theta} = \dfrac{1}{\sin\theta\cos\theta}$. (2)

(b) For $0^\circ < \theta < 90^\circ$, $\tan\theta + 1/\tan\theta = 4$.

(i) find all values of θ to 1 decimal place. (2)

(ii) find $\sin\theta\cos\theta$ without finding the two angles separately. (1)

(Total for Question 1 is 5 marks)

For $0^\circ \leq x < 360^\circ$, $2\sin^2 x + \sin x - 1 = 0$.

(a) Find all exact solutions. (4)

(b) State how many solutions lie in $180^\circ < x < 360^\circ$. (1)

(Total for Question 2 is 5 marks)

$3\cos^2 x - 5\cos x + 2 = 0$, where $0 \leq x \leq 2\pi$.

(a) Find all solutions, giving non-exact angles to 3 decimal places in radians. (4)

(b) Find the least strictly positive solution. (2)

(Total for Question 3 is 6 marks)

For $0^\circ \leq x < 360^\circ$, $\sin x \cos x = \frac{1}{2} \sin x$.

(a) Solve the equation without losing any solutions. (4)

(b) Identify the solutions lost by dividing the original equation by $\sin x$. (2)

(Total for Question 4 is 6 marks)

5.

For $-\pi < x \leq \pi$, $\tan x = \sqrt{3}$.

(a) Find every exact solution. (3)

(b) For each solution, state the signs of $\sin x$ and $\cos x$. (2)

(Total for Question 5 is 5 marks)

For $0^\circ \leq \theta \leq 180^\circ$, $\sin(2\theta) = \frac{\sqrt{3}}{2}$.

(a) Find all values of θ. (3)

(b) Find all solutions if the domain is extended to $0^\circ \leq \theta \leq 360^\circ$. (3)

(Total for Question 6 is 6 marks)

Prove the following identities wherever both sides are defined.

(a) $\sin x \tan x + \cos x = \dfrac{1}{\cos x}$. (3)

(b) $\dfrac{\sin x}{1 - \cos x} = \dfrac{1 + \cos x}{\sin x}$. (3)

(Total for Question 7 is 6 marks)

8. P2-C06-R08 · P2 · Chapter 6

For $0 \leq x < 2\pi$, $2\cos^2 x = 3\sin x$.

(a) Rewrite the equation as a quadratic in $\sin x$. (2)

(b) Hence find all exact solutions. (4)

(Total for Question 8 is 6 marks)

9.

For $0^\circ \leq x \leq 360^\circ$, $\cos(3x) = 0$.

(a) Find all solutions. (3)

(b) State the spacing between successive solutions and explain it using the input factor 3. (2)

(Total for Question 9 is 5 marks)

$\sin x = \cos x$ for $-180^\circ \leq x \leq 180^\circ$.

(a) Find all solutions. (3)

(b) Find the exact common value of $\sin x$ and $\cos x$ at each solution. (3)

(Total for Question 10 is 6 marks)

For $0 \leq x \leq \pi$, $4\sin^2 x = 3$.

(a) Find the exact solutions. (3)

(b) Find all solutions in $-\pi \leq x \leq \pi$. (2)

(Total for Question 11 is 5 marks)

The expression $F(x) = \dfrac{1 - \sin x}{\cos x}$ is defined where $\cos x \neq 0$.

(a) Show that $F(x) = \dfrac{\cos x}{1 + \sin x}$ on their common domain. (3)

(b) Find the exact value of $F(30^\circ)$. (3)

(Total for Question 12 is 6 marks)

For $0^\circ \leq \theta < 360^\circ$, $\tan^2 \theta = 3$.

(a) Find all exact solutions. (3)

(b) Find the solutions that also satisfy $\cos \theta < 0$. (2)

(Total for Question 13 is 5 marks)

The graph of $y = \sin x$ and the horizontal line $y = k$ are considered for $0 \leq x \leq 2\pi$.

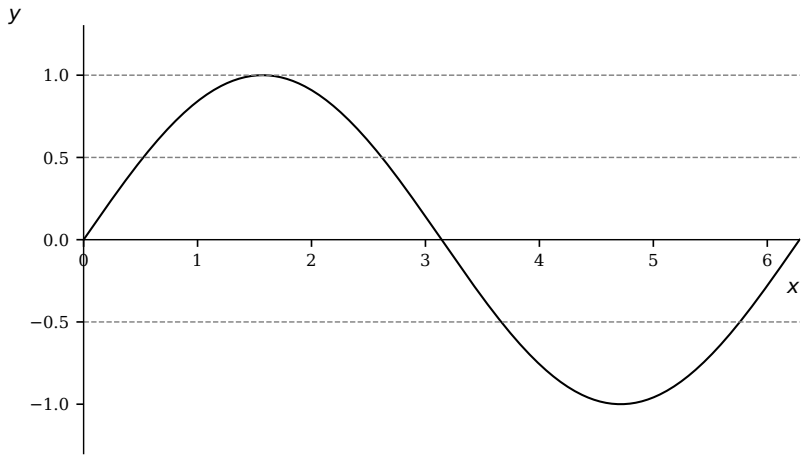


Figure 14

(a) State the number of distinct intersections when $k = 0$, $k = 1/2$ and $k = 1$. (3)

(b) Find every real k giving exactly two distinct intersections. (3)

(Total for Question 14 is 6 marks)

For $0^\circ \leq x \leq 180^\circ$, $\cos(2x - 30^\circ) = 1/2$.

(a) Find all solutions. (4)

(b) Find the least positive solution if the interval is extended to all $x > 0$. (2)

(Total for Question 15 is 6 marks)

For all x for which it is defined, $H(x) = \dfrac{\tan^2 x}{1 + \tan^2 x}$.

(a) Show that $H(x) = \sin^2 x$. (3)

(b) Solve $H(x) = 1/4$ for $0^\circ \leq x < 360^\circ$. (4)

(Total for Question 16 is 7 marks)

For $0 \leq x < 2\pi$, $\cos^2 x = \sin^2 x + 1/2$.

(a) Find $\sin^2 x$. (2)

(b) Hence find all exact solutions. (3)

(Total for Question 17 is 5 marks)

A point on a circle has coordinates $P = (5\cos\theta, 5\sin\theta)$, with $0^\circ \leq \theta < 360^\circ$.

(a) Show that $x^2 + y^2 = 25$ for every such point. (2)

(b) The point also lies on $y = \frac{5}{2}$. Find

(i) all possible values of θ, (2)

(ii) the exact corresponding x-coordinates. (2)

(Total for Question 18 is 6 marks)

19.

The equation $\sin^2 x = \sin x$ is considered.

(a) Find all solutions in $0^\circ \leq x \leq 360^\circ$. (4)

(b) A student replaces the equation by $\sin x = 1$. State the missing solutions and the incorrect operation used. (2)

(Total for Question 19 is 6 marks)

For $-\pi/2 < x < \pi/2$, $3\tan x + 1 = 0$.

(a) Find x in radians to 3 decimal places. (3)

(b) Find the exact value of $\cos x$. (3)

(Total for Question 20 is 6 marks)

For $0^\circ \leq x < 360^\circ$, $4\sin x = 5\cos^2 x$.

(a) Find the exact possible values of $\sin x$. (4)

(b) Find all solutions x to 1 decimal place. (2)

(Total for Question 21 is 6 marks)

The graphs $y = \cos x$ and $y = 2\sin^2 x - 1$ are considered on $0 \leq x \leq 2\pi$.

(a) Find their intersection x-coordinates exactly. (4)

(b) Find the y-coordinate at each intersection. (2)

(Total for Question 22 is 6 marks)

For $0^\circ \leq x \leq 360^\circ$, $\sin(3x + 20^\circ) = 0$.

(a) Find all solutions, giving exact fractional degree values where needed. (4)

(b) Find the number of these solutions in $0^\circ \leq x \leq 180^\circ$. (2)

(Total for Question 23 is 6 marks)

Given $\sin\alpha = 3/5$ and $90^\circ < \alpha < 180^\circ$,

(a) find exactly

(i) $\cos\alpha$ and $\tan\alpha$, (3)

(ii) $\dfrac{1+\sin\alpha}{1-\sin\alpha}$. (2)

(b) Find the exact value of $\sin(180^\circ - \alpha)$ using the graph symmetry of sine. (2)

(Total for Question 24 is 7 marks)

The expression $G(x) = \sin^2 x + 4\cos^2 x$ is defined for all real x.

(a) Show that $G(x) = 1 + 3\cos^2 x$ and find its minimum and maximum possible values. (3)

(b) Find all x in $0^\circ \le x < 360^\circ$ for which $G(x) = 7/4$. (3)

(Total for Question 25 is 6 marks)

For $0 \leq x \leq 2\pi$, $\tan x = 2\sin x$, with $\cos x \neq 0$.

(a) Find all exact solutions. (4)

(b) State the number of solutions if both endpoints are excluded. (2)

(Total for Question 26 is 6 marks)

The equation $2\cos^2 x + \cos x = 0$ holds for $0^\circ \leq x < 360^\circ$.

(a) Find all solutions. (4)

(b) Find the sum of the solutions in degrees. (2)

(Total for Question 27 is 6 marks)

The graphs $y = \sin(2x)$ and $y = 1/2$ are considered for $0 \leq x \leq \pi$.

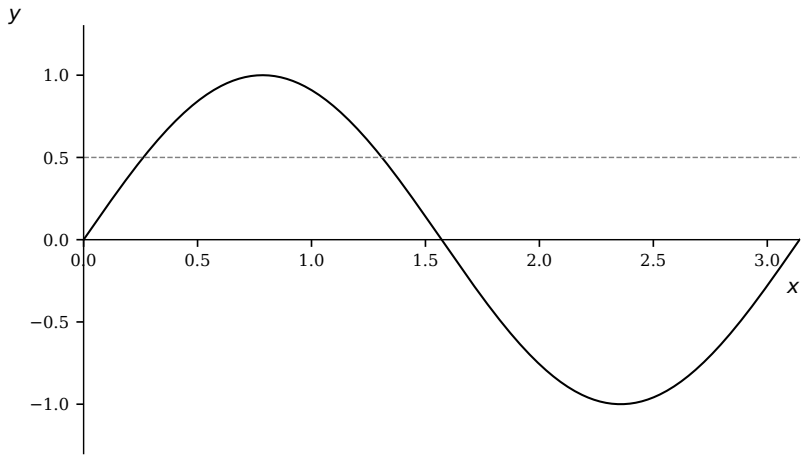


Figure 28

(a) Find the exact intersection coordinates. (3)

(b) Find the total length of the set of x-values for which $\sin(2x) \geq 1/2$ in this interval. (3)

(Total for Question 28 is 6 marks)

For $0^\circ \leq \theta < 360^\circ$, $3\cos^2\theta + \sin\theta = 1$.

(a) Show that the equation can be written as $(3\sin\theta + 2)(\sin\theta - 1) = 0$. (2)

(b) Hence

(i) find all values of θ, to 1 decimal place where necessary. (2)

(ii) find all solutions of $3\cos^2(2x) + \sin(2x) = 1$ for $0^\circ \leq x < 180^\circ$. (2)

(Total for Question 29 is 6 marks)

The expression $F(\theta)$ is defined by $F(\theta) = \dfrac{1 - \cos^2\theta}{\sin\theta}$ wherever the denominator is non-zero.

(a) Simplify $F(\theta)$ and state all excluded values of θ in $0^\circ \leq \theta \leq 360^\circ$. (2)

(b) For the original expression F,

(i) solve $2F(\theta) = 1$ on this interval. (2)

(ii) explain why $F(\theta) = 0$ has no solutions on this interval. (1)

(Total for Question 30 is 5 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C06-R01 · 5 marks

(a) · 2 marks

Stated identity.

左边通分为$(\sin^2\theta+\cos^2\theta)/(\sin\theta\cos\theta)$，平方和为1。原式需$\sin\theta\neq0$且$\cos\theta\neq0$。

M1	改写并通分
A1	基本恒等式与结果

(b) For $0^\circ<\theta<90^\circ$, $\tan\theta+1/\tan\theta=4$.

(i) · 2 marks

$15.0^\circ, 75.0^\circ$.

设$t=\tan\theta>0$，得$t^2-4t+1=0$，$t=2\pm\sqrt{3}$。分别用计算器取反正切，得到$15, 75$度；无需引用P3和差角公式。

M1	二次代换求t
A1	两角

(ii) · 1 marks

$1/4$.

直接由(a)恒等式，$1/(\sin\theta\cos\theta)=4$，所以乘积$1/4$。

B1	直接复用恒等式

Question 2

P2-C06-R02 · 5 marks

(a) · 4 marks

$x = 30^\circ, 150^\circ, 270^\circ$.

(2sinx−1)(sinx+1)=0，所以sinx=1/2或−1，分别列角。

M1	正确二次分解
M1	两种三角值
M1	对应全部角
A1	完整集合

(b) · 1 marks

One, 270°.

筛选(a)中的值，端点未产生额外解。

B1	数量和角

Question 3

P2-C06-R03 · 6 marks

(a) · 4 marks

$x = 0, \arccos(2/3), 2\pi - \arccos(2/3), 2\pi$, approximately $0, 0.841, 5.442, 2\pi$.

分解(3cosx−2)(cosx−1)；cos=1在两个包含的端点都取到。

M1	正确分解
M1	余弦两支
M1	端点完整
A1	弧度精度

(b) · 2 marks

$\arccos(2/3) \approx 0.841$.

0不严格正，按从小到大选择第二项。

M1	正确排除0
A1	最小正角

Question 4

P2-C06-R04 · 6 marks

(a) · 4 marks

$x = 0^\circ, 60^\circ, 180^\circ, 300^\circ$.

因式sinx(cosx−1/2)=0；不应直接除sinx而丢0、180。

M1	因式分解
M1	两组条件
M1	全部象限
A1	解集

(b) · 2 marks

0° and 180°.

除以某因子将剔除该因子零点，需回原式验证这两点确为解。

M1	原式核查零因子
A1	两角

Question 5

P2-C06-R05 · 5 marks

(a) · 3 marks

$x = -2\pi/3, \pi/3$.

正切周期π，参考角π/3；从π/3+kπ筛选指定区间。

M1	周期解
M1	区间筛选
A1	两精确角

(b) · 2 marks

Both negative at $-2\pi/3$; both positive at $\pi/3$.

负角−120°同终边240°为第三象限；π/3第一象限。

M1	象限定位
A1	两组符号

Question 6

P2-C06-R06 · 6 marks

(a) · 3 marks

$30^\circ, 60^\circ$.

输入2θ走0°到360°，正弦为√3/2在60°和120°，再除2。

M1	扩大输入范围
M1	全部输入解
A1	θ值

(b) · 3 marks

$30^\circ, 60^\circ, 210^\circ, 240^\circ$.

输入范围0°到720°，加一周期的420°、480°后除2。

M1	处理扩展输入范围
M1	周期补解
A1	完整集合

Question 7

P2-C06-R07 · 6 marks

(a) · 3 marks

$\sin x \tan x + \cos x = (\sin^2 x + \cos^2 x)/\cos x = 1/\cos x$.

把tan写成sin/cos再通分，用平方和1。

M1	代入正切商
M1	通分平方和
A1	目标式

(b) · 3 marks

Multiply the left numerator and denominator by $1 + \cos x$ and use $1 - \cos^2 x = \sin^2 x$.

共轭乘式将分母化为$\sin^2 x$，再约去一个$\sin$；保留原域限制。

M1	共轭乘式
M1	平方恒等式与约分
A1	恒等式

Question 8

P2-C06-R08 · 6 marks

(a) · 2 marks

$2\sin^2 x + 3\sin x - 2 = 0.$

$\cos^2=1-\sin^2$，代入后移项。

M1	正确平方恒等式
A1	二次式

(b) · 4 marks

$x = \pi/6, 5\pi/6.$

$(2\sin x-1)(\sin x+2)=0$；$\sin=-2$非法，只留$1/2$。

M1	因式分解
M1	排除范围外三角值
M1	对应角
A1	解集

Question 9

P2-C06-R09 · 5 marks

(a) · 3 marks

$30^\circ, 90^\circ, 150^\circ, 210^\circ, 270^\circ, 330^\circ$.

3x在0°到1080°，余弦零点90°+180°k共六个，除3。

M1	正确输入区间
M1	列齐输入零点
A1	六x

(b) · 2 marks

60°; cosine zeros are 180° apart in input, so x spacing is $180^\circ/3$.

区别函数周期120°与零点间距60°。

M1	输入零点间距
A1	正确缩放和结论

Question 10

P2-C06-R10 · 6 marks

(a) · 3 marks

$x = -135^\circ, 45^\circ$.

解处cos不能为0，否则sin=±1不相等；可除cos得tanx=1，再筛选。

M1	除法合法说明或图像
M1	周期角
A1	两解

(b) · 3 marks

$-\sqrt{2}/2$ at -135°; $\sqrt{2}/2$ at 45°.

两比相等且平方和1，大小为$\sqrt{2}/2$，符号按象限。

M1	平方和或参考角
M1	象限符号
A1	两值

Question 11

P2-C06-R11 · 5 marks

(a) · 3 marks

$x = \pi/3, 2\pi/3.$

$\sin = \pm\sqrt{3}/2$，但该区间正弦非负，只选正值的两角。

M1	平方开根两符号
M1	结合区间筛选
A1	两角

(b) · 2 marks

$x = -2\pi/3, -\pi/3, \pi/3, 2\pi/3.$

扩展负半区间有负正弦对应的两个角，或平方函数关于0对称。

M1	新区间的负支
A1	全部角

Question 12

P2-C06-R12 · 6 marks

(a) · 3 marks

Multiply by $(1+\sin x)/(1+\sin x)$ and use $1-\sin^2 x=\cos^2 x$.

分子平方差变为$\cos^2$x，约分得到所给式；只在共同定义域讨论。

M1	共轭式
M1	平方恒等式
A1	正确简式

(b) · 3 marks

$\sqrt{3}/3$.

(1−1/2)/(√3/2)=1/√3，有理化。

M1	代精确三角值
M1	有理化
A1	值

Question 13

P2-C06-R13 · 5 marks

(a) · 3 marks

$60^\circ, 120^\circ, 240^\circ, 300^\circ$.

tan=±√3，共四象限的参考角60°。

M1	保留正负
M1	正切周期
A1	四解

(b) · 2 marks

$120^\circ, 240^\circ$.

余弦负在第二、第三象限，与(a)求交。

M1	正确象限筛选
A1	两角

Question 14

P2-C06-R14 · 6 marks

(a) · 3 marks

3, 2, 1, respectively.

零线包含区间两端和π共3点；中间正高度两点；最高点一。

B1	k零时
B1	中间值
B1	顶值

(b) · 3 marks

$-1 < k < 0$ or $0 < k < 1$.

正负内部高度均两点；k=0因端点变3，±1各1，范围外0。

M1	按高度范围分类
M1	处理边界与零
A1	完整区间

Question 15

P2-C06-R15 · 6 marks

(a) · 4 marks

$x = 45^\circ, 165^\circ$.

输入范围−30°到330°，cos为1/2的输入60°、300°；各加30再除2。

M1	正确输入范围
M1	筛选输入角
M1	还原x
A1	两解

(b) · 2 marks

45°.

最近的正输入解由−60°给x=−15°非法，60°给45°，故45为最小正。

M1	检查临近周期支
A1	最小值

Question 16

P2-C06-R16 · 7 marks

(a) · 3 marks

Use $\tan x = \sin x / \cos x$ and $\sin^2 x + \cos^2 x = 1$.

分子分母同乘$\cos^2 x$，分母变为$\sin^2 x+\cos^2 x=1$；原$\cos$不得为0。

M1	正切替换
M1	清分母及平方和
A1	目标式

(b) · 4 marks

$30^\circ, 150^\circ, 210^\circ, 330^\circ$.

$\sin^2=1/4$给$\sin=\pm 1/2$，四个角都不使$\cos=0$，所以均合法。

M1	平方开根
M1	全象限
M1	原域复核
A1	四解

Question 17

P2-C06-R17 · 5 marks

(a) · 2 marks

$1/4$.

用$\cos^2=1-\sin^2$，$1-\sin^2=\sin^2+1/2$。

M1	平方和替换
A1	结果

(b) · 3 marks

$\pi/6, 5\pi/6, 7\pi/6, 11\pi/6$.

$\sin=\pm 1/2$，在整周期列齐四个角。

M1	两符号
M1	弧度角筛选
A1	四解

Question 18

P2-C06-R18 · 6 marks

(a) · 2 marks

$x^2+y^2=25(\cos^2\theta+\sin^2\theta)=25.$

平方两坐标并用基本恒等式。

M1	正确平方相加
A1	常数25

(b) The point also lies on $y=\frac{5}{2}$. Find

(i) · 2 marks

$30^\circ, 150^\circ.$

5sinθ=5/2得sinθ=1/2，两解。

M1	正确三角方程
A1	两角

(ii) · 2 marks

$\pm 5\sqrt{3}/2.$

分别cos30=√3/2、cos150=−√3/2，乘5。

M1	象限符号
A1	两值

Question 19

P2-C06-R19 · 6 marks

(a) · 4 marks

$0^\circ, 90^\circ, 180^\circ, 360^\circ$.

sinx(sinx−1)=0，分别零点与最大值点；360端点包含。

M1	正确因式
M1	两种三角值
M1	端点
A1	解集

(b) · 2 marks

Missing $0^\circ, 180^\circ, 360^\circ$; division by $\sin x$ without checking its zeros.

除以可能为零的因子丢去整组零正弦解。

M1	指出除法问题
A1	全部漏解

Question 20

P2-C06-R20 · 6 marks

(a) · 3 marks

$x = -\arctan(1/3) \approx -0.322$.

tan在这一分支一一对应，负比值给负角，勿加π跑出域。

M1	正确反正切
M1	分支筛选
A1	精度

(b) · 3 marks

$3/\sqrt{10} = 3\sqrt{10}/10$.

参考直角三角形两直角边1、3，斜边$\sqrt{10}$；该区间cos正。

M1	参考边与勾股
M1	cos正号
A1	精确值

Question 21

P2-C06-R21 · 6 marks

(a) · 4 marks

$\sin x = (-2+\sqrt{29})/5$ only.

用cos²=1−sin²得5s²+4s−5=0，s=(−2±√29)/5；负号根小于−1须排。

M1	正确二次方程
M1	二次公式
M1	三角值范围筛选
A1	可行值

(b) · 2 marks

42.6° and 137.4°.

正弦值约0.677033，取arcsin及补角，按角度范围筛。

M1	反正弦与补角
A1	两角

Question 22

P2-C06-R22 · 6 marks

(a) · 4 marks

$x = \pi/3, \pi, 5\pi/3$.

令c=cosx，右边=1−2c²，得2c²+c−1=(2c−1)(c+1)=0。

M1	平方和化为余弦
M1	二次分解
M1	全部角
A1	三x

(b) · 2 marks

$1/2, -1, 1/2$ in that order.

在对应角上代cosx。

M1	使用同一图像函数值
A1	正确配对

Question 23

P2-C06-R23 · 6 marks

(a) · 4 marks

$x = 160/3, 340/3, 520/3, 700/3, 880/3, 1060/3$ degrees.

输入20°到1100°，正弦零点180°、360°、540°、720°、900°、1080°；减20再除3。

M1	正确输入区间
M1	完整输入零点
M1	还原x
A1	六解

(b) · 2 marks

3.

首三项约53.33、113.33、173.33；第四233.33已超出。

M1	正确筛选
A1	数量

Question 24

P2-C06-R24 · 7 marks

(a) find exactly

(i) · 3 marks

$-4/5$ and $-3/4$.

平方和给cos大小4/5，第二象限负；tan为二比之商。

M1	求cos大小
M1	象限负号
A1	两比

(ii) · 2 marks

4.

直接代(1+3/5)/(1−3/5)=(8/5)/(2/5)。

M1	正确分式代入
A1	值

(b) · 2 marks

$3/5$.

正弦在补角下相等，此为P1图像对称，不必用和角公式。

M1	补角对称
A1	精确值

Question 25

P2-C06-R25 · 6 marks

(a) · 3 marks

$1+3\cos^2 x$; minimum 1, maximum 4.

以$1-\cos^2$替换$\sin^2$，利用$0\leqslant\cos^2\leqslant 1$。

M1	正确替换
M1	平方比范围
A1	极值

(b) · 3 marks

$60^\circ, 120^\circ, 240^\circ, 300^\circ$.

$1+3\cos^2=7/4$给$\cos^2=1/4$，四角。

M1	解平方三角值
M1	两符号与全象限
A1	四解

Question 26

P2-C06-R26 · 6 marks

(a) · 4 marks

$0, \pi/3, \pi, 5\pi/3, 2\pi.$

合法乘cosx得sinx(1−2cosx)=0；sin零给0、π、2π，cos1/2给π/3、5π/3，都满足原域。

M1	合法清分母
M1	因式分解
M1	全部角及原域
A1	解集

(b) · 2 marks

3.

删0和2π，保留三个内部解。

M1	端点删除
A1	数量

Question 27

P2-C06-R27 · 6 marks

(a) · 4 marks

$90^\circ, 120^\circ, 240^\circ, 270^\circ$.

cosx(2cosx+1)=0，分别cos0和−1/2。

M1	正确分解
M1	两种余弦值
M1	象限解
A1	完整集合

(b) · 2 marks

720°.

直接相加90+120+240+270。

M1	使用完整四解
A1	和

Question 28

P2-C06-R28 · 6 marks

(a) · 3 marks

$(\pi/12, 1/2)$ and $(5\pi/12, 1/2)$.

输入$2x\in[0,2\pi]$，等于$\pi/6$或$5\pi/6$，横坐标再除2。

M1	正确输入角
M1	横坐标缩放
A1	两点

(b) · 3 marks

$\pi/3$.

满足区间$[\pi/12,5\pi/12]$，长度$4\pi/12=\pi/3$。

M1	图像确定闭区间
M1	求区间长度
A1	结果

Question 29

P2-C06-R29 · 6 marks

(a) · 2 marks

$(3\sin\theta + 2)(\sin\theta - 1) = 0.$

用$\cos^2\theta = 1 - \sin^2\theta$，整理得$3\sin^2\theta - \sin\theta - 2 = 0$，再分解。

M1	正确使用基本恒等式
A1	因式形式

(b) Hence

(i) · 2 marks

$90^\circ, 221.8^\circ, 318.2^\circ.$

$\sin\theta = 1$给90°；$\sin\theta = -2/3$的参考角$41.810\ldots^\circ$，得第三、四象限两解。

M1	选取所有分支
A1	全部角

(ii) · 2 marks

$45^\circ, 110.9^\circ, 159.1^\circ.$

$2x$遍历原区间，故将前问的三个角各除以2；不需要另创方程。

M1	输入范围正确
A1	三个角

Question 30

P2-C06-R30 · 5 marks

(a) · 2 marks

$F(\theta) = \sin\theta$; excluded $0^\circ, 180^\circ, 360^\circ$.

分子为$\sin^2\theta$；约去$\sin\theta$只在非零分母条件下成立，所以必须保留原来的三个排除点。

M1	化简
B1	所有排除角

(b) For the original expression F,

(i) · 2 marks

$30^\circ, 150^\circ$.

在合法定义域内$2\sin\theta = 1$，两解均不在排除点中。

M1	解基本正弦方程
A1	合法解

(ii) · 1 marks

Every zero of the simplified sine expression is excluded from the original domain.

$\sin\theta = 0$仅在$0, 180, 360$度，但原分母在这些点全部为零，不能作为原式解。

B1	准确关联零点与原定义域

Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 7 — Differentiation
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第7章 · 微分应用
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

1. P2-C07-R01 · P2 · Chapter 7

The curve C has equation $y = \frac{1}{4}x^4 - x^3$.

(a) Find the coordinates of all stationary points. (3)

(b) Using the sign of the first derivative,

(i) determine which stationary point is a minimum and explain why the other is not a turning point. (2)

(ii) state the intervals on which the curve is decreasing. (1)

(Total for Question 1 is 6 marks)

$f(x) = x^4 - 8x^2 + 3$.

(a) Find and classify all stationary points. (4)

(b) Find the intervals on which f is strictly increasing. (2)

(Total for Question 2 is 6 marks)

3.

A rectangle has perimeter 40 m. Its width is x m, with $0 < x \leq 10$.

(a) Find its area $A(x)$ and stationary value. (3)

(b) Use differentiation to justify that this is the maximum on the permitted domain. (3)

(Total for Question 3 is 6 marks)

4.

For $x > 0$, $C(x) = x^2 + 16/x$ is a cost model.

(a) Find the stationary point. (3)

(b) Show that this gives the global minimum for $x > 0$. (3)

(Total for Question 4 is 6 marks)

$f(x) = 2x^3 - 9x^2 + 12x - 1$ on $[0, 3]$.

(a) Find its stationary points. (3)

(b) Find the greatest and least values on the stated closed interval. (3)

(Total for Question 5 is 6 marks)

The curve $C : y = x^3 + ax^2 + bx$ has stationary points with $x = -1$ and $x = 3$.

(a) Find a, b. (3)

(b) Find and classify the two stationary points. (3)

(Total for Question 6 is 6 marks)

A box without a lid is made by removing squares of side x cm from each corner of a 12 cm by 18 cm sheet and folding up the sides.

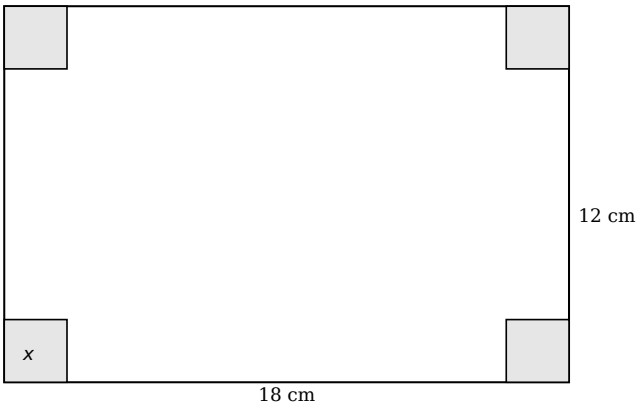


Figure 7

(a) Show that $V = 4x^3 - 60x^2 + 216x$ and state its physical domain. (4)

(b) Find the cut size that maximises volume, giving an exact answer and a maximum-volume justification. (4)

(Total for Question 7 is 8 marks)

8. P2-C07-R08 · P2 · Chapter 7

$f(x) = x^3 - 3x^2 + 3x + 4.$

(a) Find its stationary point. (2)

(b) Use the sign of f' on either side to show that this point is neither a local maximum nor a local minimum. (3)

(Total for Question 8 is 5 marks)

9. P2-C07-R09 · P2 · Chapter 7

For $x > 0$, $A(x) = 12x^{1/2} - x^{3/2}$.

(a) Find the stationary value of x. (3)

(b) Show that the corresponding value is a maximum and find it. (3)

(Total for Question 9 is 6 marks)

A product sells for £p per item. Weekly demand is modelled by $q = 300 - 20p$, for $0 < p < 15$.

(a) Find the revenue in terms of p and the price giving maximum revenue. (3)

(b) At this price, find

(i) the weekly revenue, (2)

(ii) the demand and a reason why the stationary revenue is a maximum. (3)

(Total for Question 10 is 8 marks)

$C : y = x^3 - 12x + 1.$

(a) Find the intervals where C has negative gradient. (2)

(b) Find the exact maximum and minimum values of y on $[-3, 3]$. (3)

(Total for Question 11 is 5 marks)

A cylinder has fixed volume 54π cm^3, radius $r > 0$ and height h. The total surface area of a closed cylinder is $2\pi r^2 + 2\pi r h$.

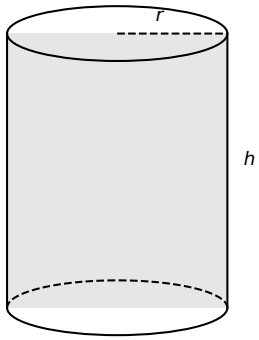


Figure 12

(a) Show that $S(r) = 2\pi r^2 + 108\pi / r$. (3)

(b) Find the radius and height giving minimum surface area and justify the minimum. (4)

(Total for Question 12 is 7 marks)

$f(x) = x^4 + ax^2 + b$. It is claimed to have a stationary point at $(2, -3)$ and to satisfy $f(0) = 5$.

(a) Determine whether these three conditions are consistent. (4)

(b) If $(2, -3)$ is replaced by $(2, -11)$, find and classify all stationary points. (4)

(Total for Question 13 is 8 marks)

14.

A rectangular pen is built against a straight wall. Fencing of length 30 m is used for three sides only.

(a) Form an area function in the width x m perpendicular to the wall, and state its domain. (3)

(b) Find the maximum area and the corresponding dimensions. (4)

(Total for Question 14 is 7 marks)

15. P2-C07-R15 · P2 · Chapter 7

The curve C has equation $y = x^2 + 4/x$ for $x > 0$.

(a) Find its stationary point exactly. (4)

(b) Find the minimum gradient of C on $[1, 2]$. (3)

(Total for Question 15 is 7 marks)

$f(x) = x^3 - 6x^2 + cx$. It has a local maximum at $x = 1$.

(a) Find c and verify the local maximum condition. (4)

(b) Find the other stationary point and the vertical difference between the stationary points. (3)

(Total for Question 16 is 7 marks)

17.

A rectangular sheet has area $48\ \text{cm}^2$. Its sides are x cm and $48/x$ cm, with $x > 0$.

(a) Find the perimeter $P(x)$ and its derivative. (3)

(b) Find the minimum perimeter in exact form and justify it. (4)

(Total for Question 17 is 7 marks)

18. P2-C07-R18 · P2 · Chapter 7

The curve is $y = x^5 - 5x^4$.

(a) Find the stationary points. (3)

(b) Use a sign table for y' to classify them. (4)

(Total for Question 18 is 7 marks)

A storage box has square base of side x m and height h m. It has no lid and its total material area is $75\ \text{m}^2$.

(a) Express the volume as $V(x) = \frac{75}{4}x - \frac{1}{4}x^3$ and state the permitted interval. (4)

(b) Find the maximum possible volume and dimensions. (4)

(Total for Question 19 is 8 marks)

$f(x) = 2x^3 + 3x^2 - 12x + 7.$

(a) Find the intervals where f is decreasing. (3)

(b) Find the equation of the tangent at each stationary point. (3)

(Total for Question 20 is 6 marks)

A model for daily profit is $P = 60q - 3q^2 - 100$, where q is the number of items made and q is a non-negative integer.

(a) Use differentiation to find the optimal production level and maximum profit. (4)

(b) Find all integer values of q for which profit is at least £197. (3)

(Total for Question 21 is 7 marks)

The curve C is $y = x^3 + 3/x$ for $x > 0$.

(a) Find its stationary point exactly. (3)

(b) Prove that C has no local maximum for $x > 0$. (3)

(Total for Question 22 is 6 marks)

23.

A rectangular poster has printed area 96 cm^2. Margins of 2 cm are added on each vertical edge and 3 cm on each horizontal edge. The printed width is $x > 0$.

(a) Show that the total poster area is $A = 120 + 6x + 384/x$. (3)

(b) Find the printed width and height giving minimum total area. (3)

(Total for Question 23 is 6 marks)

$C : y = ax^3 + bx^2 + 4$, where $a \neq 0$. It has stationary points at $x = 0$ and $x = 2$, and passes through $(2, 0)$.

(a) Find a, b. (4)

(b) Classify each stationary point and find the greatest value on $[-1, 3]$. (3)

(Total for Question 24 is 7 marks)

A square-based closed box has volume $32\ \text{cm}^3$. Its base side is $x > 0$ cm.

(a) Find its total surface area in terms of x. (3)

(b) Find the dimensions of the box with least surface area and the least area. (4)

(Total for Question 25 is 7 marks)

$f(x) = x^3 - 3kx$, where $k > 0$.

(a) Find the stationary points in terms of k. (3)

(b) The difference between their y-coordinates is 32. Find k. (3)

(Total for Question 26 is 6 marks)

A curve is $y = x^4 - 4ax^2$, where $a > 0$.

(a) Find the three stationary x-coordinates. (3)

(b) Find a if the local minimum value is -36. (3)

(Total for Question 27 is 6 marks)

A rectangular enclosure has area $72\ \text{m}^2$ and one internal fence parallel to its width. Let its width be $x > 0$ m.

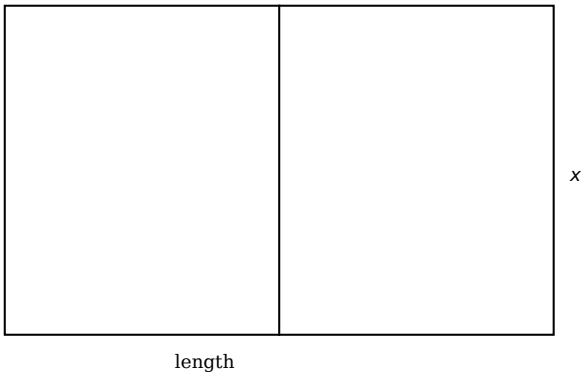


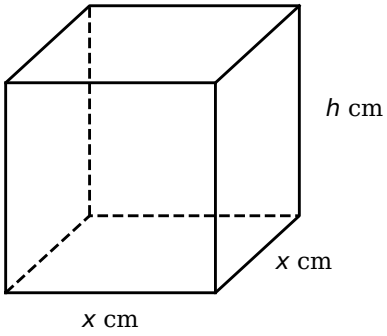
Figure 28

(a) Find the total fence length $L(x)$. (3)

(b) Find the dimensions giving minimum fence length and this minimum exactly. (4)

(Total for Question 28 is 7 marks)

A container has a square horizontal base of side x cm and height h cm. It has a base and four vertical sides but no lid. Its internal volume is 108 cm^3. The material used for its base costs twice as much per square centimetre as the material for its sides. Thickness and joins are ignored.



Open top; square base. Not to scale.

Figure 29

(a) Show that the total material cost is proportional to $C(x) = 2x^2 + 432/x$. (3)

(b) For $x > 0$,

(i) find the dimensions giving minimum cost. (3)

(ii) justify that the stationary value is a minimum. (2)

(Total for Question 29 is 8 marks)

The gradient of a curve is $dy/dx = 3x^2 - 12x + 9$. The curve passes through $A(0, 2)$.

(a) Find the equation of the curve. (2)

(b) For this curve, find

(i) the coordinates of both stationary points and determine their nature. (4)

(ii) the exact area between the curve and the line $y = 2$ for $0 \leq x \leq 3$. (2)

(Total for Question 30 is 8 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C07-R01 · 6 marks

(a) · 3 marks

$(0,0)$ and $(3,-27/4)$.

$y\prime = x^3 - 3x^2 = x^2(x-3)$；零点$0,3$，代入曲线得两坐标。

M1	求导并分解
A1	两横坐标
A1	两纵坐标

(b) Using the sign of the first derivative,

(i) · 2 marks

$(3,-27/4)$ is a minimum; $(0,0)$ is stationary but not a turning point.

x^2除零点外为正，所以导数符号主要由$x-3$决定：在0两侧均负，无增减改变；在3左负右正，故为极小。

M1	正确区间符号
A1	两种分类与理由

(ii) · 1 marks

$(-\infty,0)$ and $(0,3)$.

要求导数严格小于零，需$x<3$且$x\neq 0$；分写成两个开区间。

B1	严格负导数区间

Question 2

P2-C07-R02 · 6 marks

(a) · 4 marks

$(0, 3)$ local maximum; $(\pm 2, -13)$ local minima.

f′=4x(x−2)(x+2)，f″=12x²−16，在0负，在±2正。

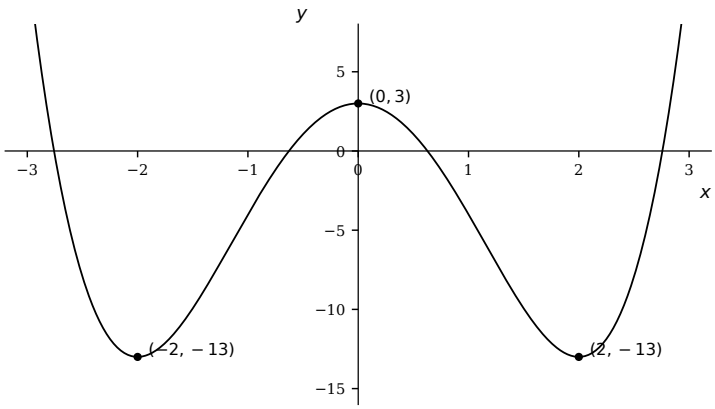
M1	一阶零点
M1	回代
M1	二阶符号
A1	三点分类

(b) · 2 marks

$(-2, 0)$ and $(2, \infty)$.

导数4x(x²−4)>0，在−2到0和2以上为正。

M1	导数符号分段
A1	两区间



Suggested sketch

Question 3

P2-C07-R03 · 6 marks

(a) · 3 marks

$A = 20x - x^2$; stationary at $x = 10$, area $100\ \text{m}^2$.

长20−x，导数20−2x=0。

M1	正确面积模型
M1	求导找点
A1	面积

(b) · 3 marks

$A'' = -2 < 0$ and $A' = 20 - 2x > 0$ on $0 < x < 10$, so maximum at 10.

二阶导负，且直到可行端点都递增，最大点是允许的正方形。

M1	二阶负或全区间正导数
M1	使用可行端点
A1	最大性

Question 4

P2-C07-R04 · 6 marks

(a) · 3 marks

$x = 2$, $C = 12$.

C′=2x−16/x²=0给x³=8，代C。

M1	正确负幂求导
M1	解正根
A1	点

(b) · 3 marks

$C' = 2(x^3 - 8)/x^2$ is negative for $0 < x < 2$ and positive for $x > 2$.

分母正，导数符号由x³−8决定，先降后升遍及整个正域。

M1	正确导数因式符号
M1	两侧符号
A1	全局最小

Question 5

P2-C07-R05 · 6 marks

(a) · 3 marks

$(1,4)$ and $(2,3)$.

f′=6(x−1)(x−2)，代原式。

M1	导数因式
M1	求零点
A1	两点

(b) · 3 marks

Greatest 8 at $x=3$; least -1 at $x=0$.

比较f0=−1、f1=4、f2=3、f3=8；必须包括端点。

M1	列出驻点及端点候选
M1	正确四值
A1	最大最小及位置

Question 6

P2-C07-R06 · 6 marks

(a) · 3 marks

$a = -3, b = -9.$

导函数首项$3x^2$，已知两根所以$y'=3(x+1)(x-3)=3x^2-6x-9$；比较2a、b。

M1	导数根构造
M1	比较系数
A1	a与b

(b) · 3 marks

$(-1, 5)$ local maximum; $(3, -27)$ local minimum.

$y''=6x-6$，-1时负、3时正；原函数代值。

M1	两纵坐标
M1	二阶符号
A1	两点类型

Question 7

P2-C07-R07 · 8 marks

(a) · 4 marks

$V = x(12-2x)(18-2x);\ 0 < x < 6.$

底边分别减2x，高x，三尺寸均须正。

M1	正确三尺寸
M1	展开
A1	多项式
B1	域

(b) · 4 marks

$x = 5 - \sqrt{7}$ cm; the other stationary root $5 + \sqrt{7} > 6$ is invalid, and $V'' = 24x - 120 < 0$ at the valid root.

V′=12x²−120x+216=12(x²−10x+18)，求两根，检查域及二阶导。

M1	正确求导方程
M1	两根与域筛选
M1	最大值检验
A1	精确x

Question 8

P2-C07-R08 · 5 marks

(a) · 2 marks

$(1, 5)$.

$f'=3(x-1)^2$为0只在1，代曲线得5。

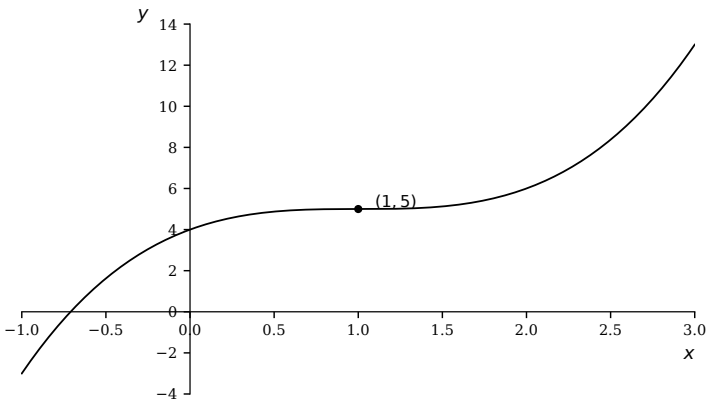
M1	求导并找根
A1	点

(b) · 3 marks

$f'(x) > 0$ for $x < 1$ and $x > 1$; the curve increases on both sides.

二阶导在1为0不能作极值判定；导数正号不变，函数没有转升为降或反之。

M1	两侧导数符号
M1	符号不变
A1	非极值结论



Suggested sketch

Question 9

P2-C07-R09 · 6 marks

(a) · 3 marks

$x = 4$.

A′=6x^{−1/2}−(3/2)x^{1/2}=0，乘2√x得12−3x=0。

M1	分数幂导数
M1	解正域方程
A1	x

(b) · 3 marks

Maximum 16; $A'' = -3x^{-3/2} - (3/4)x^{-1/2} < 0$.

二阶导在整个正域负；代x4得24−8=16。

M1	二阶导
M1	负性
A1	最大值

Question 10

P2-C07-R10 · 8 marks

(a) · 3 marks

$R = 300p - 20p^2$; $p = 7.5$.

收入单价乘销量，R′=300−40p=0。

M1	收入模型
M1	求导解驻点
A1	价格

(b) At this price, find

(i) · 2 marks

£1125.

q=150，收入7.5×150。

M1	模型代值
A1	收入

(ii) · 3 marks

150 items; $R'' = -40 < 0$.

代需求式；二阶导严格负，且价格在物理域。

B1	需求
M1	二阶导负
A1	最大性

Question 11

P2-C07-R11 · 5 marks

(a) · 2 marks

$-2 < x < 2$.

$y'=3x^2-12=3(x-2)(x+2)$，向上二次式两根间负。

M1	求导与根
A1	区间

(b) · 3 marks

Maximum 17 at $x=-2$; minimum -15 at $x=2$.

比较$y(-3)=10, y(-2)=17, y2=-15, y3=-8$。

M1	端点与驻点候选
M1	正确代值
A1	两极值与位置

Question 12

P2-C07-R12 · 7 marks

(a) · 3 marks

$h = 54/r^2$, hence the stated formula.

体积$\pi r^2 h$固定，先消h再代给定表面积式。

M1	体积条件消元
M1	代面积式
A1	简式

(b) · 4 marks

$r = 3$ cm, $h = 6$ cm; $S'' = 4\pi + 216\pi/r^3 > 0$.

$S'=4\pi r-108\pi/r^2=0$给$r^3=27$；二阶导正，再用体积求h。

M1	正确求导及正根
M1	最小性
A1	半径
A1	高

Question 13

P2-C07-R13 · 8 marks

(a) · 4 marks

$a=-8, b=5$ would give $f(2)=-11$, so these stated conditions are inconsistent.

f0给b5，f′2=32+4a=0给a−8；代f2=16−32+5=−11不等于−3。

M1	用原点值
M1	用驻点导数
M1	回检纵坐标
A1	一致性结论

(b) · 4 marks

$(0,5)$ maximum; $(\pm 2,-11)$ minima.

此时f=x^4−8x^2+5，f′=4x(x^2−4)，f″=12x^2−16，逐一判号。

M1	全部零点
M1	纵坐标
M1	二阶符号
A1	分类

Question 14

P2-C07-R14 · 7 marks

(a) · 3 marks

$A = x(30 - 2x),\ 0 < x < 15.$

两宽用2x，剩余长30−2x，几何尺寸正。

M1	三侧栅栏关系
M1	面积
A1	域

(b) · 4 marks

Area $112.5\ \text{m}^2$; width 7.5 m, length 15 m.

A′=30−4x=0给7.5，A″=−4证明最大；回求长和面积。

M1	求导解驻点
M1	最大性
A1	两尺寸
A1	面积

Question 15

P2-C07-R15 · 7 marks

(a) · 4 marks

$\left(\sqrt[3]{2}, 3\sqrt[3]{4}\right)$.

y′=2x−4/x²=0给x³=2；此时4/x=2x²，所以y=3x²。

M1	正确导数
M1	正根
M1	化简纵坐标
A1	点

(b) · 3 marks

-2 at $x = 1$.

梯度函数g=2x−4/x²，其导数g′=2+8/x³>0，所以g在区间递增，最小在1。

M1	区分梯度函数与原函数
M1	正二阶导说明梯度递增
A1	最小梯度及位置

Question 16

P2-C07-R16 · 7 marks

(a) · 4 marks

$c = 9$; $f''(1) = -6 < 0$.

f′1=3−12+c=0给9；再检二阶导6x−12。

M1	驻点条件
A1	c
M1	二阶负
A1	最大性

(b) · 3 marks

Other point $(3, 0)$; vertical difference 4.

导数3(x−1)(x−3)；f1=4，f3=0。

M1	另一零点
M1	原式代值
A1	点与纵差

Question 17

P2-C07-R17 · 7 marks

(a) · 3 marks

$P = 2x + 96/x$; $P' = 2 - 96/x^2$.

两边长之和乘2，再逐项求导。

M1	正确周长
M1	负幂求导
A1	导式

(b) · 4 marks

$16\sqrt{3}$ cm, at $x = 4\sqrt{3}$; $P'' = 192/x^3 > 0$.

令x^2=48取正根，另一边同长；二阶导正；周长四倍边长。

M1	解正驻点
M1	二阶最小性
M1	代周长
A1	精确值

Question 18

P2-C07-R18 · 7 marks

(a) · 3 marks

$(0,0)$ and $(4,-256)$.

$y'=5x^3(x-4)$，零点0、4；代原式。

M1	因式化导数
M1	两点代值
A1	坐标

(b) · 4 marks

$(0,0)$ local maximum; $(4,-256)$ local minimum.

$x<0$时两因子负故导数正；$0<x<4$正。0处二阶导零，须用符号变化。

M1	三段导数符号
M1	0处正转负
M1	4处负转正
A1	分类

Question 19

P2-C07-R19 · 8 marks

(a) · 4 marks

$x^2+4xh=75;\ V=x^2h=(75x-x^3)/4,\ 0<x<5\sqrt{3}.$

从材料面积消h，h正给$x^2<75$。

M1	正确无盖面积
M1	消元
A1	V式
B1	物理域

(b) · 4 marks

$x=5$ m, $h=5/2$ m; maximum $125/2$ m^3.

$V'=(75-3x^2)/4=0$给$x5$，$V''=-3x/2<0$，再求h和V。

M1	正驻点
M1	最大性
A1	尺寸
A1	体积

Question 20

P2-C07-R20 · 6 marks

(a) · 3 marks

$-2 < x < 1$.

f′=6(x+2)(x−1)，在两根间负。

M1	求导并分解
M1	符号
A1	区间

(b) · 3 marks

$y = 27$ at $x = -2$; $y = 0$ at $x = 1$.

f−2=−16+12+24+7=27；f1=2+3−12+7=0，驻点切线水平。

M1	两纵坐标
M1	水平切线性质
A1	两方程

Question 21

P2-C07-R21 · 7 marks

(a) · 4 marks

$q = 10$; maximum £200.

$P'=60-6q=0$，二阶$-6<0$，驻点10为允许整数，代值200。

M1	正确求导与驻点
M1	最大性及整数合法
A1	产量
A1	利润

(b) · 3 marks

$q = 9, 10, 11$.

$P=200-3(q-10)^2 \geqslant 197$给平方$\leqslant 1$，取三个非负整数。

M1	二次利润门槛
M1	解范围并取整数
A1	全部产量

Question 22

P2-C07-R22 · 6 marks

(a) · 3 marks

$(1, 4)$.

y′=3x²−3/x²=0给x⁴=1，正域取1。

M1	负幂导式
M1	解域内根
A1	点

(b) · 3 marks

$y'' = 6x + 6/x^3 > 0$ for $x > 0$, so its only stationary point is a minimum; the derivative changes from negative to positive.

二阶导正使梯度严格递增，唯一驻点为最小，不存在其他驻点最大。

M1	二阶导正
M1	唯一驻点或导数符号
A1	无最大结论

Question 23

P2-C07-R23 · 6 marks

(a) · 3 marks

$(x+4)(96/x+6)=120+6x+384/x$.

总宽加4，总高加6，展开并合并96+24。

M1	正确外尺寸
M1	展开
A1	A式

(b) · 3 marks

Width 8 cm, height 12 cm; $A''=768/x^3>0$.

A′=6−384/x²=0给x8，再高96/8=12；二阶导正。

M1	求正驻点
M1	最小性
A1	两尺寸

Question 24

P2-C07-R24 · 7 marks

(a) · 4 marks

$a = 1, b = -3$.

导数x(3ax+2b)，在2零给6a+2b=0；点条件8a+4b+4=0。

M1	驻点条件
M1	点条件
M1	联立
A1	系数

(b) · 3 marks

$(0, 4)$ maximum, $(2, 0)$ minimum; greatest 4 at $x = 0$ and $x = 3$.

二阶6x−6给两类型；端点f−1=0、f3=4，所以最大4也在x=3取得。

M1	二阶分类
M1	检查端点
A1	最大值及所有位置

Question 25

P2-C07-R25 · 7 marks

(a) · 3 marks

$S = 2x^2 + 128/x$.

h=32/x²，两底2x²加四侧4xh。

M1	体积消h
M1	两底四侧
A1	S式

(b) · 4 marks

$x = \sqrt[3]{32}$ cm, $h = x$; least area $6\,32^{2/3}$ cm^2.

S′=4x−128/x²=0给x³32；S″=4+256/x³正，h=x。

M1	正驻点
M1	二阶最小性
A1	尺寸
A1	精确面积

Question 26

P2-C07-R26 · 6 marks

(a) · 3 marks

$(-\sqrt{k}, 2k^{3/2})$ and $(\sqrt{k}, -2k^{3/2})$.

f′=3x²−3k=0，x=±√k，利用x²=k化纵坐标−2kx。

M1	正确驻点根
M1	化简纵坐标
A1	两点

(b) · 3 marks

$k = 4$.

纵差4k^{3/2}=32给k^{3/2}=8，k=8^{2/3}=4。

M1	正确纵差方程
M1	指数求解
A1	k

Question 27

P2-C07-R27 · 6 marks

(a) · 3 marks

$x = 0, \pm\sqrt{2a}.$

y′=4x(x²−2a)，三不同实根由a正保证。

M1	正确导数分解
M1	正根条件
A1	三x

(b) · 3 marks

$a = 3.$

在x²=2a时y=4a²−8a²=−4a²；−4a²=−36，a正取3。

M1	求极小纵坐标
M1	使用参数正号
A1	a

Question 28

P2-C07-R28 · 7 marks

(a) · 3 marks

$L = 3x + 144/x.$

外框两宽两长加一条内部宽，长度72/x。

M1	三宽两长计数
M1	面积消元
A1	L式

(b) · 4 marks

Width $4\sqrt{3}$ m, length $6\sqrt{3}$ m; minimum $24\sqrt{3}$ m.

L′=3−144/x²=0给x²48，正根4√3；L″=288/x³正，代回。

M1	求正驻点
M1	最小性
A1	两尺寸
A1	最小长度

Question 29

P2-C07-R29 · 8 marks

(a) · 3 marks

$h = 108/x^2$; $C = 2x^2 + 432/x$.

以侧面每平方厘米单价为单位，底面成本$2x^2$，四侧面成本$4xh$；由体积代入h。

M1	用体积消去h
M1	正确加权面积
A1	目标式

(b) For $x > 0$,

(i) · 3 marks

$x = \sqrt[3]{108}$ cm, $h = \sqrt[3]{108}$ cm.

$C\prime = 4x - 432/x^2 = 0$给$x^3 = 108$；由$h = 108/x^2$可知$h = x$。

M1	求导并设为零
A1	最优x
A1	对应h

(ii) · 2 marks

$C\prime\prime(x) = 4 + 864/x^3 > 0$ for $x > 0$.

二阶导数在所有正x处严格为正，特别在临界点处为正，所以得到最小成本。

M1	二阶导数
A1	正确最小值结论

Question 30

P2-C07-R30 · 8 marks

(a) · 2 marks

$y = x^3 - 6x^2 + 9x + 2$.

逐项积分并用$y(0) = 2$确定常数。

M1	正确积分
A1	常数

(b) For this curve, find

(i) · 4 marks

$(1, 6)$ maximum; $(3, 2)$ minimum.

导数$3(x-1)(x-3)$，零点$1, 3$；$y'' = 6x - 12$在两处分别为负、正，代入求纵坐标。

M1	找出临界横坐标
A1	两点坐标
M1	用二阶导数分类
A1	两种性质

(ii) · 2 marks

$27/4$.

$y - 2 = x(x-3)^2 \geq 0$，所以面积$\int_0^3 (x^3 - 6x^2 + 9x)\,dx = [x^4/4 - 2x^3 + 9x^2/2]_0^3 = 27/4$。

M1	正确上减下积分
A1	精确面积

Pearson Edexcel International A Level · Pure Mathematics 2

Chapter 8 — Integration
Original Practice Questions

30 complete questions · Questions followed by answers and suggested mark schemes

- Answer all parts and show sufficient working.
- Give exact answers unless a degree of accuracy is requested.
- Angles are in radians unless degrees are explicitly indicated.
- Diagrams are not necessarily drawn to scale.
- The marks for each subpart are shown in brackets.

P2 第8章 · 定积分
英文题目 · 中文解析 · 建议评分细则
本册30题；P1、P2、P3全套25章共750题。

内部复核稿，未完成外部教师逐题审阅。
Original practice material. Not a Pearson examination paper.

The curve C has equation $y = (x - 1)(x - 5)$. In Figure 1, region R lies between C and the x-axis. A horizontal line $y = -3$ divides R into a lower region L and an upper region U.

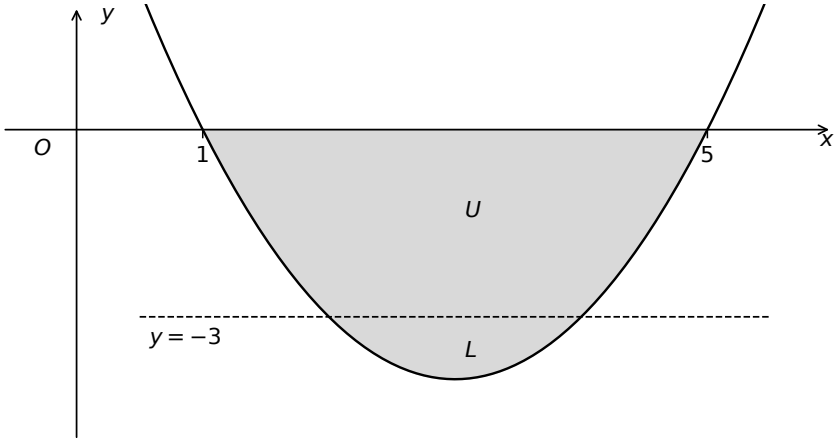


Figure 1

(a) Find where $y = -3$ meets C. (2)

(b) Find the exact area of L. (3)

(c) Find the exact area of U. (3)

(Total for Question 1 is 8 marks)

A parabolic arch above a horizontal baseline is modelled by $y = kx(6 - x)$ for $0 \leq x \leq 6$, where $k > 0$. Its cross-sectional area is 36 m^2, with x, y measured in metres. Figure 2 shows the part above the horizontal line $y = 5$.

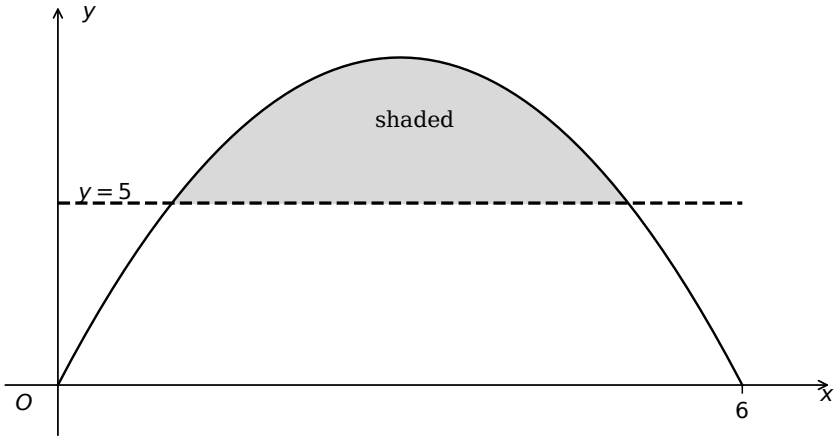


Figure 2

(a) Find k. (3)

(b) Find the width of the arch at height 5 m. (2)

(c) Find the exact fraction of the cross-sectional area above height 5 m. (3)

(Total for Question 2 is 8 marks)

Figure 3 shows the curves $y = x^3$ and $y = 4x$. They enclose two finite regions, one on each side of the y-axis.

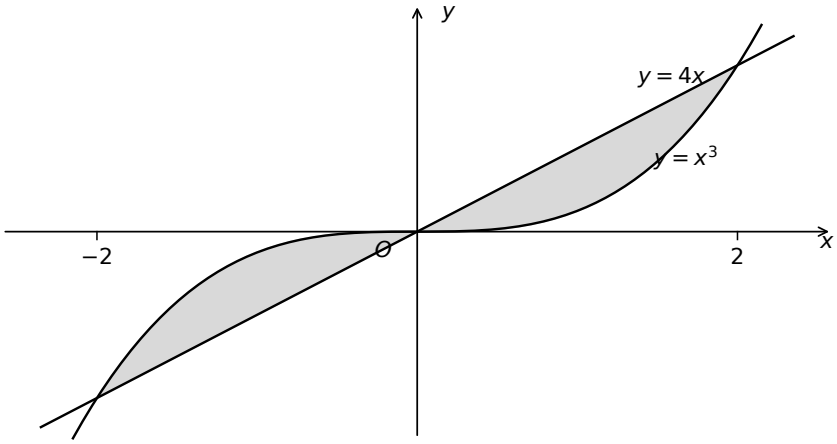


Figure 3

(a) Find all three points of intersection. (3)

(b) Find the total area of the two finite regions. (3)

(c) A student uses $\int_{-2}^{2}(4x - x^3)dx$ and obtains zero. Explain why this does not give the required total area. (2)

(Total for Question 3 is 8 marks)

The line l is tangent to $y = \sqrt{x}$ at $P(4, 2)$. Figure 4 shows the finite region bounded by l, the curve and the y-axis.

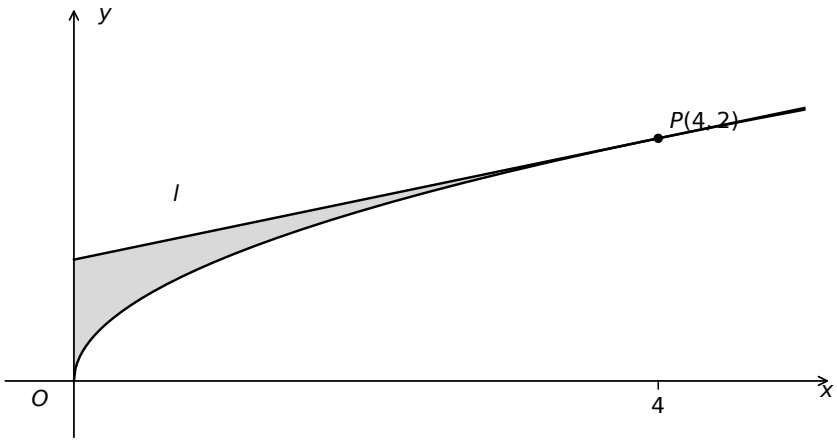


Figure 4

(a) Find the equation of l. (2)

(b) Find the exact shaded area. (3)

(c) Without calculating a new integral, find the area bounded by $y = \sqrt{x} + 3$, its tangent at $x = 4$, and the y-axis. Give a reason. (2)

(Total for Question 4 is 7 marks)

Figure 5 shows $y = \sqrt{x}$, the line $y = x - 2$, and the x-axis. The shaded finite region is bounded by all three.

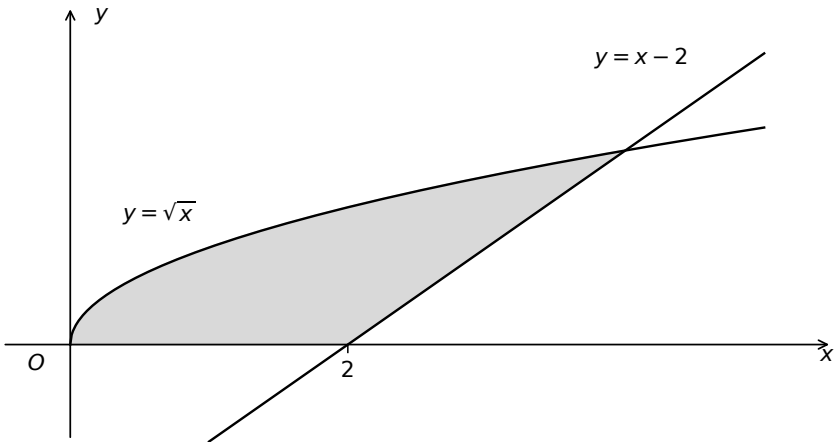


Figure 5

(a) Find the intersection of the curve and line, checking the sign condition before squaring. (3)

(b) Find the exact shaded area. (3)

(c) Explain why $\int_0^4 [\sqrt{x} - (x - 2)]dx$ gives too large a result. (2)

(Total for Question 5 is 8 marks)

For $x \geq 0$, $f(x) = 3x^2 - 12x + 9$. A positive number b satisfies $\int_0^b f(x)dx = 0$.

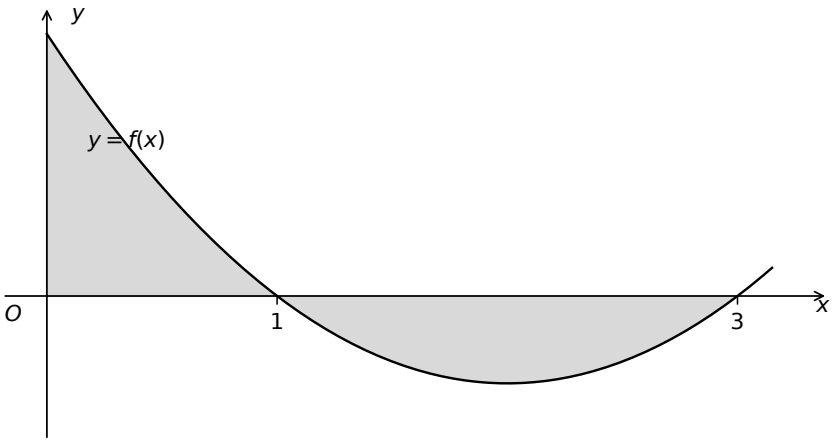


Figure 6

(a) Find b, showing all algebraic steps. (3)

(b) Find the total area between the graph and the x-axis on $0 \leq x \leq b$. (3)

(c) Explain how your answers to (a) and (b) can both be correct. (2)

(Total for Question 6 is 8 marks)

For $0 \leq x \leq 2$, $f(x) = x^2 + k$, where $k > 0$. Let $I = \int_0^2 f(x)dx$ and let T be its estimate using two equal trapezia.

(a) Find I and T in terms of k, and hence show that $T - I$ is independent of k. (3)

(b) The relative error is defined as $(T - I)/I$.

(i) Find k if this relative error is 5%. (2)

(ii) determine whether the relative error increases or decreases as k increases, giving a reason. (2)

(Total for Question 7 is 7 marks)

The table gives values of a continuous function.

x	0	1	2	3	4
$f(x)$	0	2	h	5	0

The trapezium estimate of $\int_0^4 f(x)dx$ using four equal strips is 12.

(a) Find h. (2)

(b) Using only $x = 0, 2, 4$, find a second trapezium estimate. (2)

(c) A student claims that the two estimates prove $10 \leq \int_0^4 f(x)\,dx \leq 12$. State whether this is justified by the table alone, giving one reason. (2)

(Total for Question 8 is 6 marks)

A hall roof has cross-section $y = k(9 - x^2)$ for $-3 \leq x \leq 3$, where x, y are in metres. Its greatest height above the baseline is 2.7 m. The hall has constant cross-section along its 20 m length. Figure 9 shows the roof and the horizontal level $y = 1.2$.

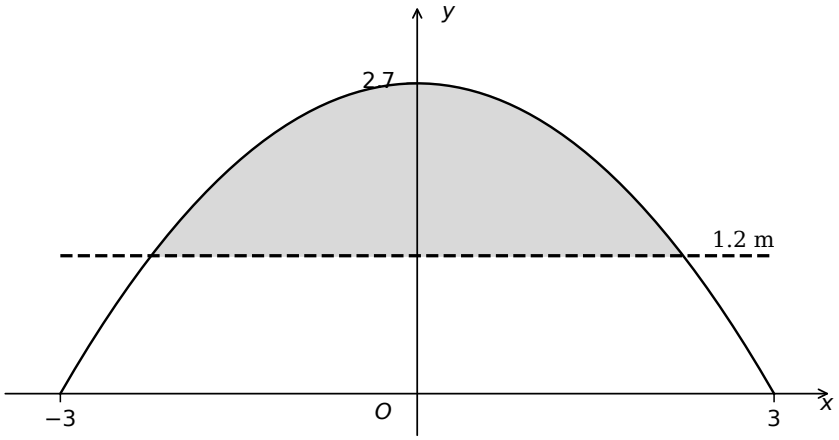


Figure 9

(a) Find k and the cross-sectional area below the roof and above the baseline. (3)

(b) Find the volume of the hall below the roof and above the baseline. (2)

(c) Find the exact cross-sectional area above $y = 1.2$. (3)

(Total for Question 9 is 8 marks)

A channel cross-section has depths, in metres, measured below a horizontal water surface:

x **(m)**	0	2	4	6	8
depth	0	0.8	1.6	1.4	0

For this question, model the bed by straight line segments joining consecutive measurement points. The water surface is then lowered by 0.2 m.

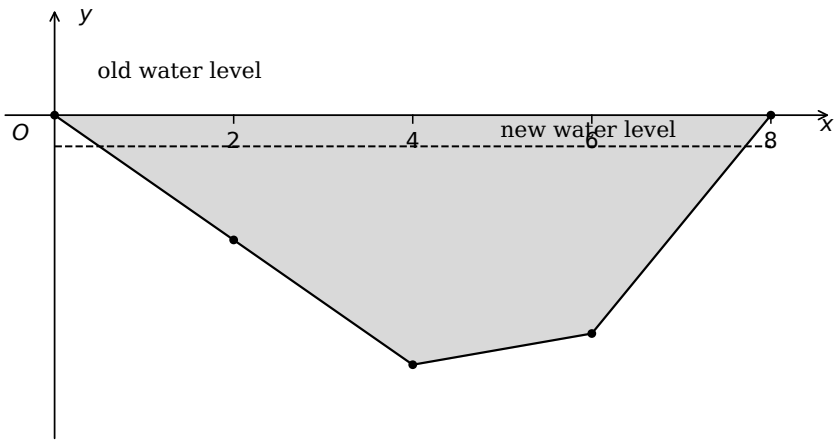


Figure 10

(a) Find the original cross-sectional water area. (2)

(b) The water level is lowered by 0.2 m, while the piecewise-linear bed model remains unchanged.

(i) Find the new positions of the left and right water edges. (3)

(ii) Find the new cross-sectional area. Explain why simply subtracting $8(0.2)$ from the old area is not exact. (4)

(Total for Question 10 is 9 marks)

Let $I = \int_1^4 \sqrt{x}\, dx$.

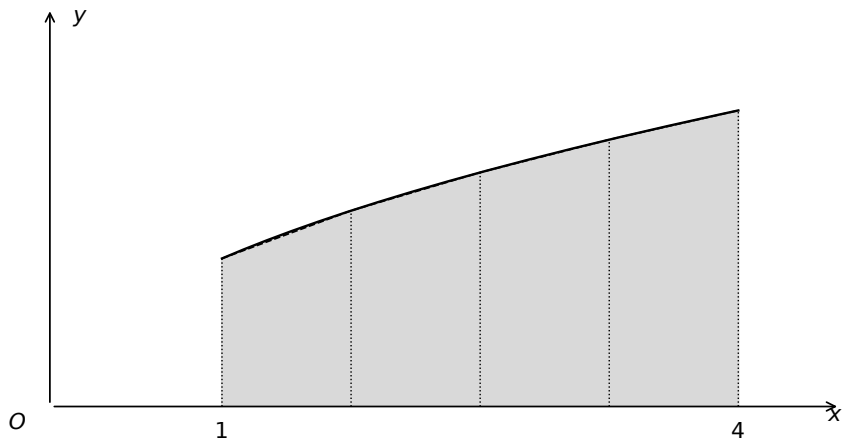


Figure 11

(a) Use four equal trapezia to estimate I, giving the result to 4 decimal places. Keep unrounded function values in your calculation. (3)

(b) Find I exactly and hence the absolute error of your estimate, to 3 significant figures. (3)

(c) State whether the trapezium estimate in part (a) is an overestimate or an underestimate of the integral. Give a reason based on the position of each chord relative to the curve. (2)

(Total for Question 11 is 8 marks)

The curve $y = x^2 - 4$ lies at or below the x-axis on $0 \leq x \leq 2$. Figure 12 shows two equal trapezia between this curve and the axis.

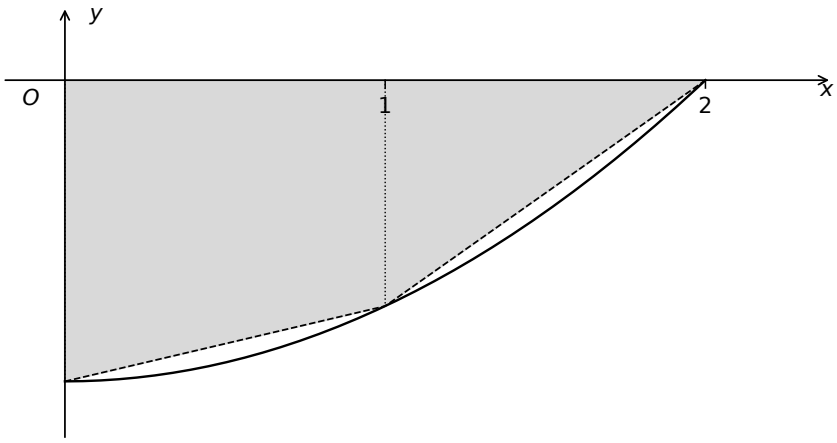


Figure 12

(a) Use these two trapezia to estimate the positive area between the curve and the axis. (3)

(b) Find the exact area and determine whether the estimate is too large or too small. (3)

(c) For the signed integral $\int_0^2 (x^2 - 4)dx$, is the trapezium estimate an overestimate or an underestimate? Give numerical evidence. (2)

(Total for Question 12 is 8 marks)

The horizontal line $y = h$, where $h > 0$, and the curve $y = x^2$ enclose a finite region R, as shown in Figure 13.

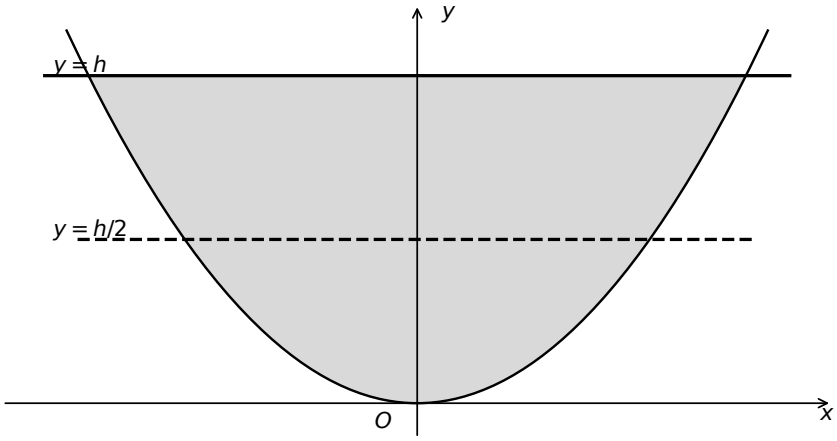


Figure 13

(a) Show that the area of R is $\frac{4}{3}h^{3/2}$. (3)

(b) The area of R is 36. Find h and the two points of intersection. (2)

(c) Find the fraction of R lying above $y = h/2$. (3)

(Total for Question 13 is 8 marks)

The upper boundary of a moulded strip is modelled by $y = ax^2 + b$ for $0 \le x \le 2$. Lengths are in millimetres. The cross-section has vertical sides at $x = 0, 2$ and base $y = 0$. At the right edge it is 5 mm thick, and its cross-sectional area is $6\ \text{mm}^2$.

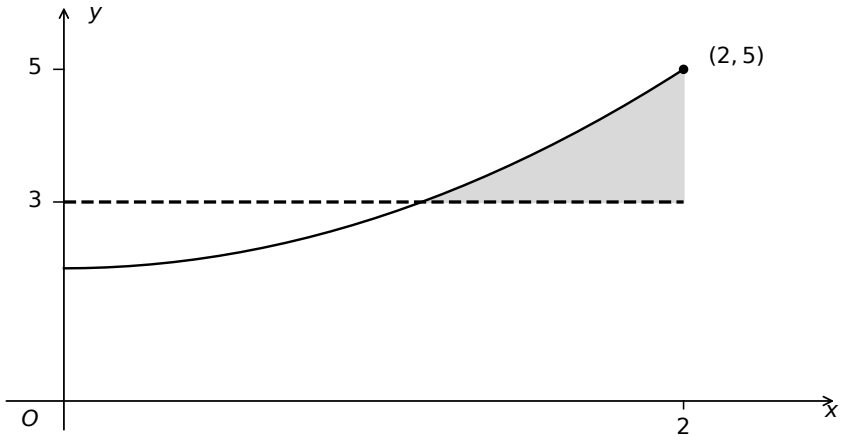


Figure 14

(a) Form two equations in a, b and find them. (3)

(b) Find the position at which the strip first reaches height 3 mm. (2)

(c) Find the exact cross-sectional area of material lying above the level $y = 3$. (3)

(Total for Question 14 is 8 marks)

For $1 \leq x \leq 4$, the curve C has equation $y = (x^2 + 4)/\sqrt{x}$. The chord joins its endpoints at $x = 1$ and $x = 4$. Figure 15 shows that the chord lies above the curve between these endpoints.

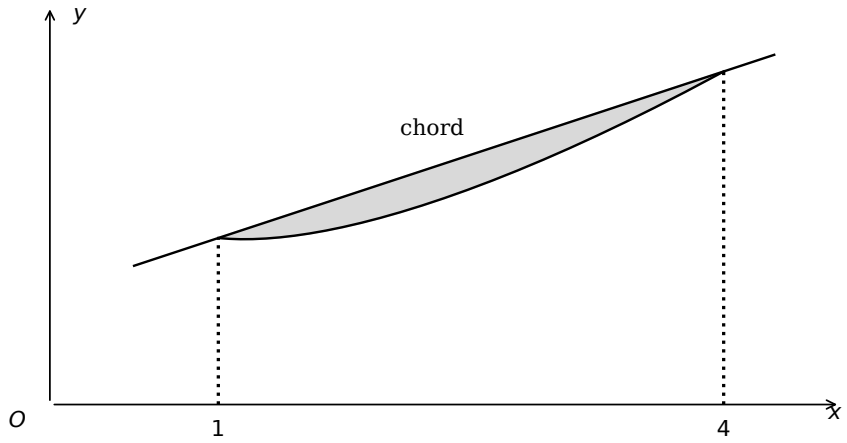


Figure 15

(a) Find the equation of the chord. (3)

(b) Find $\int_1^4 y\,dx$ exactly. (3)

(c) Find the exact area between the chord and curve. (2)

(Total for Question 15 is 8 marks)

The curve $y = 2^x$ on $0 \leq x \leq 3$ is shown in Figure 16 inside the rectangle $0 \leq x \leq 3$, $0 \leq y \leq 8$.

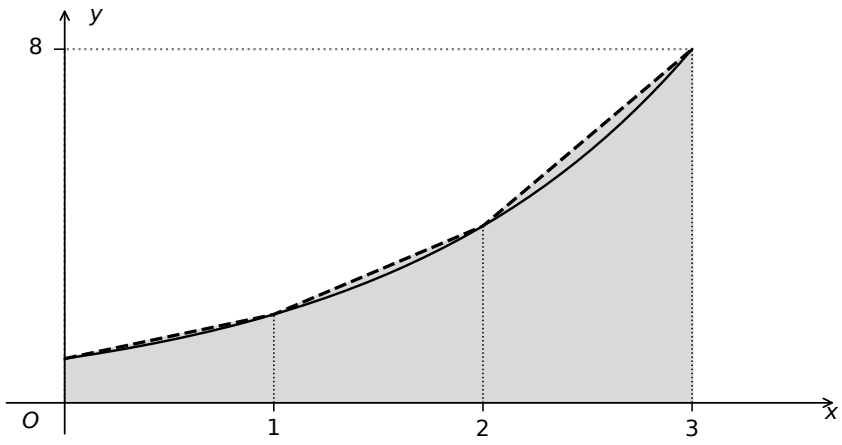


Figure 16

(a) Use three equal trapezia to estimate the area below the curve and above the x-axis. (3)

(b) Hence estimate the area inside the rectangle but above the curve. (2)

(c) The curve lies below each chord joining consecutive tabulated points. State whether your answer to (b) overestimates or underestimates the true area. (2)

(Total for Question 16 is 7 marks)

The boundary of a panel is modelled by $f(x) = (x + 2)^2$ for $-2 \leq x \leq 0$, and $f(x) = 4 - 2x$ for $0 < x \leq 2$. Its base is the x-axis. A vertical line $x = a$, where $0 < a < 2$, divides the panel into two equal areas.

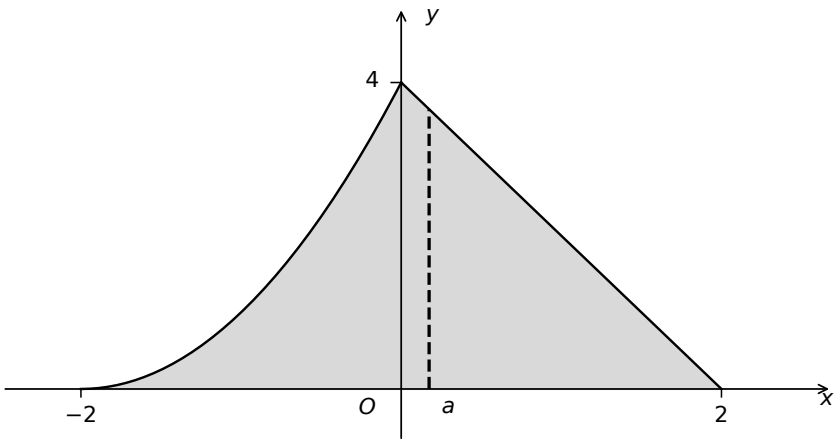


Figure 17

(a) Find the total panel area. (3)

(b) Show that $a^2 - 4a + 2/3 = 0$. (3)

(c) Find a exactly, explaining your choice of root. (3)

(Total for Question 17 is 9 marks)

The line $y = 2x + 3$ and curve $y = x^2$ enclose a finite region. The y-axis divides it into regions L and R, as shown in Figure 18.

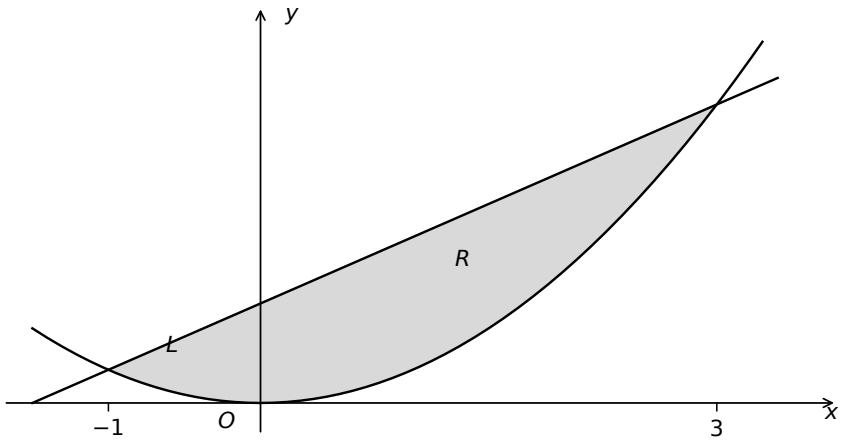


Figure 18

(a) Find their two points of intersection. (2)

(b) Find the exact area of L. (3)

(c) Find the ratio of the areas $L : R$ in simplest integer form. (2)

(Total for Question 18 is 7 marks)

Figure 19 shows a continuous curve $y = f(x)$ on $[-2, 4]$. It is above the x-axis for $-2 < x < 0$ and below it for $0 < x < 4$. The two shaded areas are 7 and 3 square units, respectively. No formula for f is given.

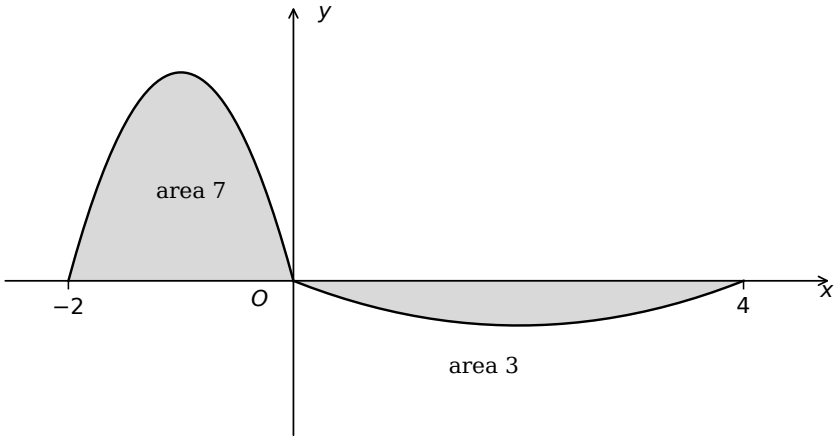


Figure 19

(a) Find $\int_{-2}^{4} f(x)dx$ and $\int_{0}^{4} f(x)dx$. (2)

(b) Using a horizontal scale factor, find $\int_{-1}^{2} f(2x)dx$. Do not use an unknown antiderivative. (3)

(c) Find $\int_{-4}^{2} f(-x)dx$, explaining the graph transformation. (2)

(Total for Question 19 is 7 marks)

A panel has a straight base of length 6 m. Perpendicular heights of its top boundary are measured as follows:

distance x (m)	0	1	3	6
height y (m)	1	2	4	3

For a model, join consecutive measured points by straight lines.

(a) State why the widths of the trapezia must be treated separately for these data. (2)

(b) Find the modelled panel area. (3)

(c) Both faces require 0.4 litres of coating per square metre per face. Coating is sold in 5-litre tins. Find the minimum number of tins required under the model. (2)

(Total for Question 20 is 7 marks)

Let $f_k(x) = x^2 - 2kx + k$ on $0 \le x \le 1$, where k is real. Figure 21 shows several members of this family.

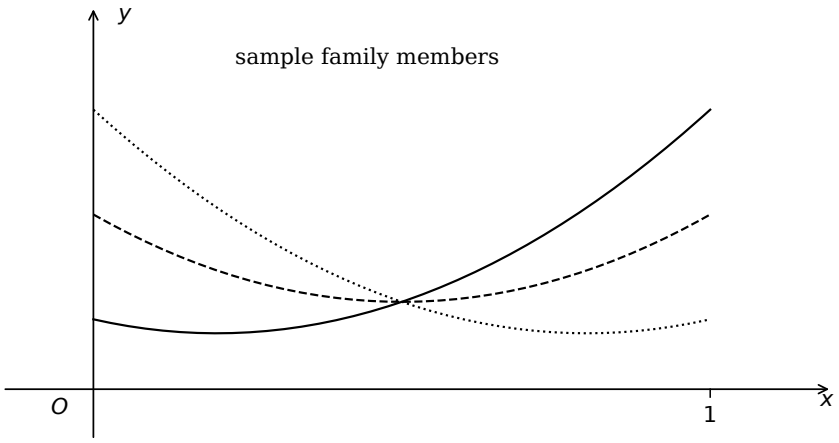


Figure 21

(a) Show that $\int_0^1 f_k(x)dx = 1/3$ for every k. (3)

(b) Find the complete range of k for which $f_k(x) \ge 0$ throughout $[0, 1]$. (3)

(c) Using one trapezium, find the area estimate when $0 \le k \le 1$, and its absolute error. Explain why both are independent of k. (3)

(Total for Question 21 is 9 marks)

The curve C has equation $y = x^2 - 4x + 5$. The line l is tangent to C at the point with $x = 3$. The shaded region in Figure 22 lies between C, l and the lines $x = 1, x = 5$.

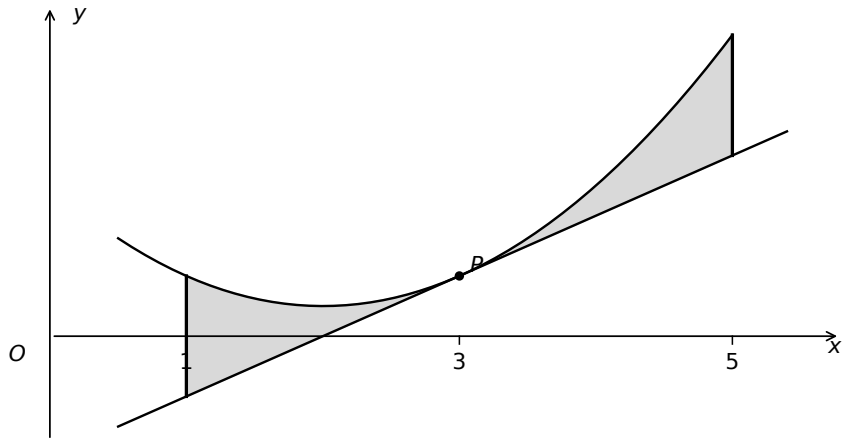


Figure 22

(a) Find l and show that the vertical separation of C above l is $(x - 3)^2$. (4)

(b) Find the total shaded area exactly. (3)

(c) Without evaluating another integral, explain why $x = 3$ bisects the shaded area. (2)

(Total for Question 22 is 9 marks)

Figure 23 gives a continuous curve $y = f(x)$ with $f(x) \geq 0$ on $[0, 2]$ and $f(x) \leq 0$ on $[2, 5]$. It is known that $\int_0^2 f(x)dx = 5$ and $\int_0^5 f(x)dx = -1$. Define $g(x) = f(x) + k$.

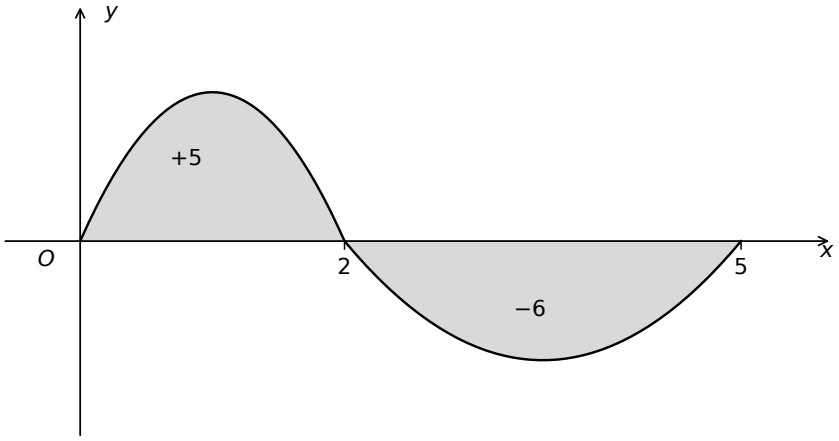


Figure 23

(a) Find the positive area between f and the x-axis on $[2, 5]$. (2)

(b) Find k if $\int_2^5 g(x)dx = 0$. (3)

(c) For the value of k found in part (b), explain why $\int_2^5 g(x)\,dx = 0$ does not determine the total positive area between the graph of g and the x-axis. (2)

(Total for Question 23 is 7 marks)

A planting area is bounded by $y = 4$, $y = 1 + \sqrt{x}$ and the y-axis, as shown in Figure 24. Coordinates are in metres. A project is viable only if the planting area is at least $9.5\ \text{m}^2$.

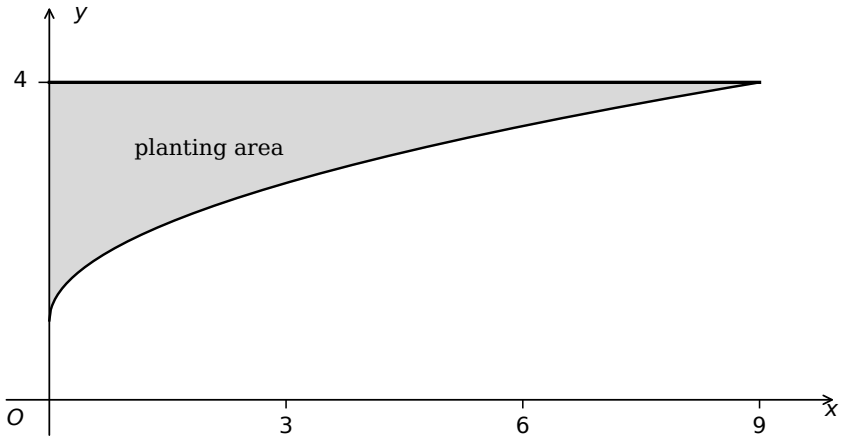


Figure 24

(a) Using the points $x = 0, 3, 6, 9$ on the curved boundary, obtain a trapezium-based estimate of the planting area. (4)

(b) Use integration to find the exact planting area. (3)

(c) State the correct project decision and explain why using the estimate alone could lead to the wrong decision. (2)

(Total for Question 24 is 9 marks)

On $0 \leq x \leq 4$, two profiles are $y = x^2$ and $y = k\sqrt{x}$, with $k > 0$. The areas beneath the two profiles over this interval are equal. Figure 25 shows the profiles after k has been determined.

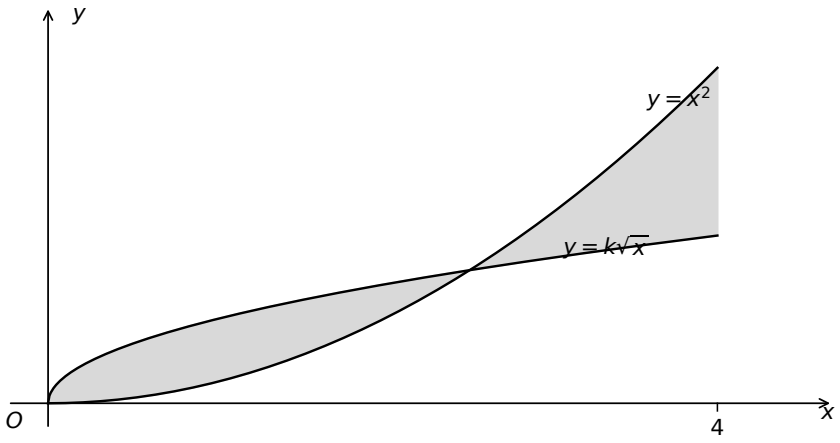


Figure 25

(a) Find k. (3)

(b) Find the positive x-coordinate at which the profiles cross. (2)

(c) Find the total area between the profiles on $[0, 4]$. (4)

(Total for Question 25 is 9 marks)

The curve C is $y = x(4 - x)$ for $0 \leq x \leq 4$. A horizontal line at height h, where $0 < h < 4$, cuts off a region between the line and the top of the curve.

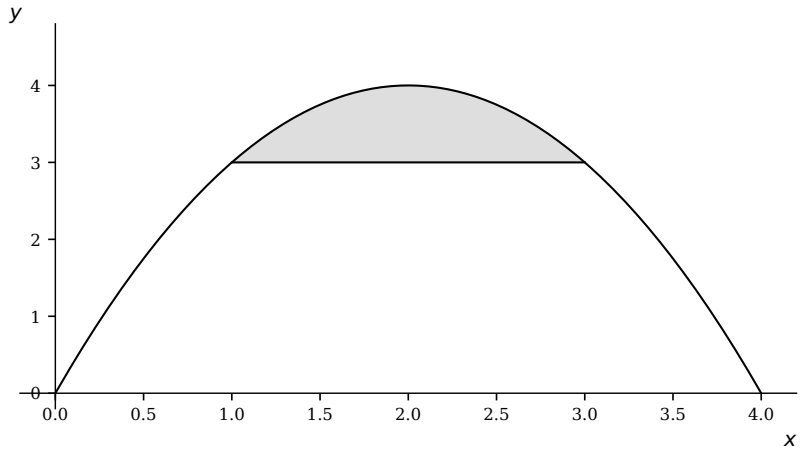


Figure 26

(a) Show that its intersection coordinates have $x = 2 \pm \sqrt{4 - h}$. (3)

(b) Given that the chord has length 2, find

(i) the value of h, (2)

(ii) the exact area above the chord and below C. (3)

(Total for Question 26 is 8 marks)

The table gives values of a continuous positive function f on $[0, 4]$.

x	0	1	2	3	4
$f(x)$	2.0	2.8	3.4	3.8	4.0

Write $I = \int_0^4 f(x)dx$.

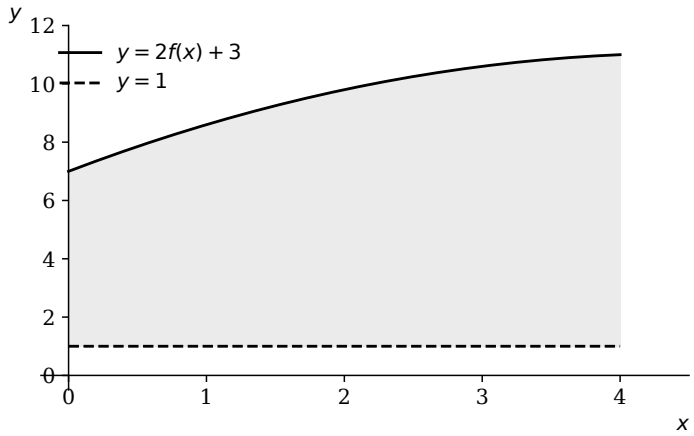


Figure 27

(a) Use the trapezium rule with all the tabulated values to estimate I. (2)

(b) For $0 \leq x \leq 4$, the upper curve is $y = 2f(x) + 3$ and the lower line is $y = 1$.

(i) Use your answer to (a) to estimate the area between these graphs. (2)

(ii) The graph of f lies above each chord joining consecutive tabulated points. Determine whether the estimate in (b)(i) is an overestimate or an underestimate, giving a reason. (2)

(Total for Question 27 is 6 marks)

A positive curve $y = f(x)$ is defined for $0 \leq x \leq 6$. A trapezium-rule estimate using equally spaced ordinates gives $\int_0^6 f(x)dx \approx 18.4$. In the rectangle $0 \leq x \leq 6$, $0 \leq y \leq 5$, the curve divides the rectangle into two regions. The curve lies entirely inside the rectangle.

(a) Estimate the ratio of the area below the curve to the area above the curve, in simplest integer form. (2)

(b) The curve lies below each straight chord used by the trapezium rule.

(i) State whether the estimated area above the curve is too large or too small. (2)

(ii) Determine whether the ratio in (a) is too large or too small, explaining your answer. (2)

(Total for Question 28 is 6 marks)

The curve C has equation $y = x^2 - 4x + 6$. Its tangent at $x = 1$ is l. The finite region R is bounded by C, l and the lines $x = 0$ and $x = 3$.

(a) Find an equation for l and show that the vertical separation between C and l is $(x - 1)^2$. (3)

(b) Find

(i) the exact area of R. (2)

(ii) the value of a, where $0 < a < 3$, such that $x = a$ divides R into two equal areas. (3)

(Total for Question 29 is 8 marks)

The curve $y = f(x)$ passes through $(1, 0)$ and has $f\prime(x) = 4x - 6$.

(a) Find $f(x)$ and the other point where the curve crosses the x-axis. (3)

(b) Let A be the total area between the curve and the x-axis on $0 \leq x \leq 2$.

(i) Find A exactly. (2)

(ii) Find $\int_0^2 f(x)dx$ and explain why this differs from A. (2)

(Total for Question 30 is 7 marks)

Answers and Suggested Mark Schemes

中文解答与建议评分点

M：方法分；A：准确分，通常依赖对应方法；B：独立得分点。
等价正确解法应获相应分数。评分点为原创建议，并非官方评分细则。

Question 1

P2-C08-R01 · 8 marks

(a) · 2 marks

$(2, -3)$ and $(4, -3)$.

联立$x^2-6x+5=-3$，得$(x-2)(x-4)=0$。

M1	联立并解二次式
A1	两交点

(b) · 3 marks

$4/3$ square units.

下方区域位于x=2至4，面积 $\int_2^4[-3-(x^2-6x+5)]dx=[-x^3/3+3x^2-8x]_2^4=4/3$。

M1	正确上下函数及积分界
M1	正确原函数
A1	精确面积

(c) · 3 marks

$28/3$ square units.

总区域在x=1至5且曲线在横轴下，面积 $-\int_1^5(x^2-6x+5)dx=32/3$；扣下方4/3得28/3。

M1	正确求完整负值区的面积
M1	减去L
A1	精确面积

Question 2

P2-C08-R02 · 8 marks

(a) · 3 marks

$k = 1$.

横截面积 $\int_0^6 k(6x - x^2)dx = 36k$，与36相等给k=1。

M1	用面积建立定积分方程
M1	正确积分及代界
A1	k

(b) · 2 marks

4 m.

解$6x-x^2=5$得x=1、5，因此宽度4，不是任一交点横坐标。

M1	解水平截线交点
A1	宽度

(c) · 3 marks

$8/27$.

上帽形面积 $\int_1^5 (6x - x^2 - 5)dx = 32/3$；除以总面积36得$8/27$。

M1	正确帽形积分
A1	帽形面积
A1	占原总面积的比例

Question 3

P2-C08-R03 · 8 marks

(a) · 3 marks

$(-2,-8),(0,0),(2,8)$.

联立得x(x−2)(x+2)=0，再代直线求纵坐标。

M1	联立并因式分解
A1	三个横坐标
A1	三交点

(b) · 3 marks

8 square units.

在0到2直线在上，面积 $\int_0^2 (4x - x^3)dx = 4$；左侧反转上下函数但面积同为4，总8。

M1	分清两区上下关系或用对称性
M1	正确积分
A1	总面积

(c) · 2 marks

The integrand changes sign at 0, so equal positive and negative signed areas cancel.

左侧$4x-x^3$为负，右侧为正；积分相抵不表示几何面积为0，必须分区后取正。

B1	指出符号改变
B1	有向面积抵消与面积的区别

Question 4

P2-C08-R04 · 7 marks

(a) · 2 marks

$y = x/4 + 1$.

导数$1/(2\sqrt{x})$在4为$1/4$，代点P得$y-2=(x-4)/4$。

M1	求导及切点斜率
A1	切线方程

(b) · 3 marks

$2/3$ square units.

0到4内直线在上，面积 $\int_0^4 (x/4 + 1 - x^{1/2})dx = [x^2/8 + x - 2x^{3/2}/3]_0^4 = 2/3$。

M1	设置上减下与积分界
M1	正确原函数
A1	精确面积

(c) · 2 marks

$2/3$: both boundaries are translated vertically by 3, so their separation is unchanged.

同时平移曲线和相应切线不会改变两者的纵向差及横向宽度，故面积保持不变。

B1	平移后面积正确
B1	两边界一起平移的理由

Question 5

P2-C08-R05 · 8 marks

(a) · 3 marks

$(4, 2)$.

√x=x−2要求x≥2。平方给$x^2-5x+4=0$，候选1和4；1使原式右边为负，故舍弃。

M1	写出平方前的符号限制
M1	解二次式
A1	只保留有效交点

(b) · 3 marks

$10/3$ square units.

曲线下0到4面积16/3，减去直线与横轴之间x=2到4的三角形面积2，得10/3。若分段积分，0至2下边界是横轴。

M1	正确分界x=2或减去三角形
M1	正确积分曲线
A1	面积

(c) · 2 marks

For $0 < x < 2$ the line is below the x-axis; that integral includes an extra triangle of area 2.

所求阴影在0至2以下边界为横轴，不是线y=x−2；错误积分额外包含横轴以下底2高2的三角形。

B1	识别错误区间和下边界
B1	额外面积2

Question 6

P2-C08-R06 · 8 marks

(a) · 3 marks

$b = 3$.

积分得$b^3-6b^2+9b=b(b-3)^2=0$；b>0排除0，故3。

M1	正确积分并代未知上限
M1	因式分解并解方程
A1	正上限

(b) · 3 marks

8 square units.

f=3(x−1)(x−3)，在0至1正、1至3负。原函数$F=x^3-6x^2+9x$，F(1)=4、F(0)=F(3)=0，两面积各4，总8。

M1	找到符号切换点1
M1	按符号分两积分
A1	两正面积和8

(c) · 2 marks

The signed integral is $4 - 4 = 0$, while total geometric area is $4 + 4 = 8$.

定积分计正负方向，几何面积取正；两个相等的面积可在积分中抵消。

B1	正负积分关系
B1	面积相加关系

Question 7

P2-C08-R07 · 7 marks

(a) · 3 marks

$I = 8/3 + 2k$, $T = 3 + 2k$, $T - I = 1/3$.

积分得$I = 8/3 + 2k$。表值为$k, 1 + k, 4 + k$，步长1，所以$T = \frac{1}{2}[k + (4 + k) + 2(1 + k)] = 3 + 2k$；相减为$1/3$，参数消去。

M1	精确积分
M1	两梯形估计
A1	误差常数

(b) The relative error is defined as $(T - I)/I$.

(i) · 2 marks

$k = 2$.

$(1/3)/(8/3 + 2k) = 1/20$，得$I = 20/3$，所以$k = 2$。

M1	正确相对误差方程
A1	参数

(ii) · 2 marks

Decreases.

绝对误差固定为正数$1/3$，而精确积分$8/3 + 2k$随k增加而增大，所以比值变小。无需用商法则。

B1	固定误差与增加分母
B1	正确方向

Question 8

P2-C08-R08 · 6 marks

(a) · 2 marks

$h = 5$.

步长1，估计½[0+0+2(2+h+5)]=h+7=12，故h=5。

M1	正确权重与步长
A1	h

(b) · 2 marks

10.

此时步长2，估计2/2[0+0+2×5]=10；不能继续用第一问的步长1。

M1	正确较粗步长与节点
A1	估计值

(c) · 2 marks

No. Values or shape between the tabulated points are not specified, so neither estimate is known to be an upper or lower bound.

连续不意味着曲线必在每条连接线同一侧；这些采样点之间可能向上或向下弯，不能把两种近似当作一对界。

B1	不能下此结论
B1	指出缺少曲线形状或界的依据

Question 9

P2-C08-R09 · 8 marks

(a) · 3 marks

$k = 3/10$; area $10.8\ \mathrm{m}^2$.

顶点x=0给9k=2.7；面积 $\int_{-3}^{3}(2.7 - 0.3x^2)dx = 10.8$。

M1	用最高点定k
M1	正确积分界与表达式
A1	横截面积

(b) · 2 marks

$216\ \mathrm{m}^3$.

截面恒定，体积=横截面积×长度=10.8×20；此处不使用旋转体积分。

M1	恒截面棱柱关系
A1	体积及单位

(c) · 3 marks

$2\sqrt{5}\ \mathrm{m}^2$.

截线满足$2.7-0.3x^2=1.2$，故x=$\pm\sqrt{5}$。面积 $\int_{-\sqrt{5}}^{\sqrt{5}}(1.5 - 0.3x^2)dx = 2\sqrt{5}$。

M1	求截线左右交点
M1	正确上减下的积分
A1	精确面积

Question 10

P2-C08-R10 · 9 marks

(a) · 2 marks

$7.6\ \mathrm{m}^2$.

分段直线模型下梯形求和给准确模型面积：2/2[0+0+2(0.8+1.6+1.4)]=7.6。

M1	正确步长与梯形权重
A1	原面积

(b) The water level is lowered by 0.2 m, while the piecewise-linear bed model remains unchanged.

(i) · 3 marks

$x = 1/2$ m and $x = 54/7$ m.

左岸首段深度=0.4x，达到0.2处x=0.5；右岸最后段深度=0.7(8−x)，达到0.2处x=8−2/7=54/7。

M1	用左岸线性深度关系
M1	用右岸线性深度关系
A1	两个新边界

(ii) · 4 marks

$851/140 \approx 6.08\ \mathrm{m}^2$.

初算7.6−1.6=6，把已露出的两岸也当成负水深；需加回两小三角形，面积½×0.5×0.2=0.05和½×(2/7)×0.2=1/35。总6+1/20+1/35=851/140。

M1	从旧面积减矩形或重新分段求和
M1	正确处理新干燥的两岸
A1	新面积
B1	说明固定8m宽扣法的错误

Question 11

P2-C08-R11 · 8 marks

(a) · 3 marks

$T = \frac{3}{8}[3 + 2(\sqrt{7}/2 + \sqrt{10}/2 + \sqrt{13}/2)] \approx 4.6551.$

步长(4−1)/4=3/4，内部节点7/4、5/2、13/4，函数值分别√7/2、√10/2、√13/2；端点值1和2。

B1	正确步长和内部节点
M1	正确梯形公式
A1	估计及精度

(b) · 3 marks

$I = 14/3$; error ≈ 0.0116.

原函数2x^(3/2)/3，代界得14/3。误差是|精确值−未舍入T|，不是用近似积分本身作分母。

M1	正确原函数及定积分
A1	精确积分
A1	绝对误差与精度

(c) · 2 marks

The curve lies above its chords, so the trapezia underestimate the integral.

平方根图像向下弯，每段连接线在曲线下；图中所有高度为正，因此梯形面积总和小于真实面积。

B1	曲线与弦的上下关系
B1	正确低估结论

Question 12

P2-C08-R12 · 8 marks

(a) · 3 marks

5 square units.

用正深度4、3、0求梯形和½[4+0+2×3]=5；若先求有向积分得到−5，最后再取正。

M1	正确处理负函数值为正深度
M1	两梯形权重
A1	正面积估计

(b) · 3 marks

$16/3$ square units; the estimate is too small by $1/3$.

几何面积 $\int_0^2 (4-x^2)dx = 8 - 8/3 = 16/3$，减估计5得1/3。

M1	设置正确面积积分
A1	精确面积
A1	比较及误差

(c) · 2 marks

Overestimate: $-5 > -16/3$.

有向积分越负越小；虽正面积被低估，但其负值−5反而大于真实有向积分−16/3。

B1	正确有向比较
B1	结论与数值依据

Question 13

P2-C08-R13 · 8 marks

(a) · 3 marks

$\int_{-\sqrt{h}}^{\sqrt{h}}(h-x^2)dx=\frac{4}{3}h^{3/2}$.

先由交点x=±√h确定积分界；利用对称或直接代入原函数hx−x^3/3，得到目标。

M1	以h表达两积分界
M1	正确上减下及原函数
A1	推出目标式

(b) · 2 marks

$h=9;\ (-3,9),(3,9)$.

4h^(3/2)/3=36得h^(3/2)=27，取正h=9；交点横坐标±3。

M1	解正参数的幂方程
A1	h与两点

(c) · 3 marks

$1-1/(2\sqrt{2})$.

先求截线下方：它恰好是y=x^2与y=h/2围成的同形区域，面积(4/3)(h/2)^(3/2)，占整个R的1/(2√2)。所求为截线上方，因此必须用1减去这个比例。

M1	正确识别截线下方的同形区域
M1	计算下部比例后取补集
A1	精确上部比例

Question 14

P2-C08-R14 · 8 marks

(a) · 3 marks

$4a+b=5,\ 8a/3+2b=6;\ a=3/4,b=2.$

右端点给4a+b=5；截面积为0到2的积分，得到8a/3+2b=6，联立。

M1	点条件方程
M1	积分面积方程
A1	正确参数

(b) · 2 marks

$x=2/\sqrt{3}\ \text{mm}.$

解$3x^2/4+2=3$，得$x^2=4/3$，在$[0,2]$取正根；该模型在$x\geqslant 0$单调递增。

M1	解厚度方程并限制范围
A1	有效位置

(c) · 3 marks

$4/(3\sqrt{3})\ \text{mm}^2.$

该部分仅在$x=2/\sqrt{3}$到2存在，积分 $\int_{2/\sqrt{3}}^{2}(3x^2/4-1)dx=[x^3/4-x]_{2/\sqrt{3}}^{2}$。

M1	正确新积分界及减基准厚度
M1	正确积分
A1	精确面积和单位

Question 15

P2-C08-R15 · 8 marks

(a) · 3 marks

$y = 5x/3 + 10/3$.

两个端点为(1,5)、(4,10)，斜率5/3，代点确定截距10/3。

M1	正确计算两端点
M1	斜率与点斜式
A1	方程

(b) · 3 marks

$102/5$.

先写成x^(3/2)+4x^(−1/2)，原函数2x^(5/2)/5+8√x，代界得62/5+8=102/5。

M1	化成可积分幂次
M1	正确原函数
A1	精确定积分

(c) · 2 marks

$21/10$ square units.

弦下是宽3、两端高度5和10的梯形，面积45/2；减曲线下面积102/5，得21/10。

M1	正确求弦下梯形面积或积分
A1	正面积差

Question 16

P2-C08-R16 · 7 marks

(a) · 3 marks

10.5 square units.

节点0、1、2、3的高度1、2、4、8，步长1，估计½[1+8+2(2+4)]=10.5。本单元不需对2^x作解析积分。

B1	四个正确节点高度
M1	梯形公式
A1	估计

(b) · 2 marks

13.5 square units.

矩形面积24，减下方估计10.5，得到13.5。

M1	矩形减曲线下方估计
A1	结果

(c) · 2 marks

Underestimates: the below-curve area was overestimated, so its complement was underestimated.

梯形在曲线上方使所扣面积偏大，固定矩形减去偏大量就得到偏小补集。

B1	指出曲线下面积的误差方向
B1	正确推出补集误差方向

Question 17

P2-C08-R17 · 9 marks

(a) · 3 marks

$20/3$ square units.

左段积分8/3，右段为底2高4的三角形面积4，总20/3。接点x=0两段均高4。

M1	分别处理两段定义
A1	两部分面积
A1	总面积

(b) · 3 marks

$8/3 + 4a - a^2 = 10/3 \Rightarrow a^2 - 4a + 2/3 = 0.$

左半目标是总面积的一半10/3，必须包括x<0的全部8/3以及0到a的直线下面积。

M1	设定一半总面积
M1	包括已存在的左曲线区域
A1	推出二次式

(c) · 3 marks

$a = 2 - \sqrt{30}/3.$

两根2±√(10/3)；正号大于2违背题设，故选负号。

M1	解二次式
A1	精确有效根
B1	明确排除另一根的范围理由

Question 18

P2-C08-R18 · 7 marks

(a) · 2 marks

$(-1, 1), (3, 9)$.

联立$x^2-2x-3=0=(x+1)(x-3)$，代回$y=x^2$。

M1	联立并解二次式
A1	两点

(b) · 3 marks

$5/3$ square units.

左区在-1至0，直线在上，积分 $\int_{-1}^{0}(2x+3-x^2)dx=[x^2+3x-x^3/3]_{-1}^{0}=5/3$。

M1	正确上下函数与界
M1	积分
A1	精确面积

(c) · 2 marks

$5:27$.

右区积分0到3，原函数在3为9，因此$R=9$；$L:R=(5/3):9=5:27$。横轴左右宽度之比不是面积比。

M1	求右区面积
A1	正确最简比

Question 19

P2-C08-R19 · 7 marks

(a) · 2 marks

4 and -3, respectively.

左区有向贡献+7，右区−3；整段积分7−3=4，右段为−3。

M1	正确赋予有向面积符号
A1	两个积分

(b) · 3 marks

2.

y=f(2x)在水平方向压缩至原来的一半，两区有向面积均减半：7/2−3/2=2；所给界正好是原横坐标的一半。

M1	正确横向压缩倍数
M1	同步映射区间
A1	积分值

(c) · 2 marks

4.

y=f(−x)是纵轴反射，正区面积仍7、负区仍3，只是交换左右位置；积分界变为−4到2，故总积分仍4。

M1	纵轴反射及新区间
A1	保留正负面积并得到4

Question 20

P2-C08-R20 · 7 marks

(a) · 2 marks

The consecutive widths are $1, 2, 3$ m, not equal.

梯形法同一步长公式要求横坐标等间隔；此表的三段间隔不同，必须逐个梯形计算。

B1	三个间隔正确
B1	指出等间隔条件不满足

(b) · 3 marks

$18\ \text{m}^2$.

三个梯形分别½×1×(1+2)=1.5、½×2×(2+4)=6、½×3×(4+3)=10.5，相加18。

M1	每段使用自己的宽度
A1	三面积计算
A1	合计与单位

(c) · 2 marks

3 tins.

两面用量2×18×0.4=14.4L；14.4/5=2.88，需要向上取整为3，不是按最近整数估算面积。

M1	同时计两面和每面涂覆率
A1	最少整罐数

Question 21

P2-C08-R21 · 9 marks

(a) · 3 marks

$[x^3/3 - kx^2 + kx]_0^1 = 1/3.$

逐项积分，上限处−k与+k相消；不能据图形高低直觉断言每个参数都改变面积。

M1	正确含参数原函数
M1	代界并消去参数
A1	不变量

(b) · 3 marks

$0 \le k \le 1.$

端点非负给k⩾0与1−k⩾0，是必要条件；这时 $f_k = (x-k)^2 + k(1-k) \ge 0$，所以也是充分条件。

M1	两端点给必要范围
M1	配方证明全段非负
A1	完整充分必要范围

(c) · 3 marks

Estimate $1/2$; error $1/6$.

两端高度k与1−k之和恒1，宽1，所以梯形面积½；真积分1/3，绝对误差1/6。两者均由前述不变量给出。

M1	用端点高度和求梯形
A1	估计及误差
B1	独立于k的解释

Question 22

P2-C08-R22 · 9 marks

(a) · 4 marks

$l: y = 2x - 4$; $C - l = x^2 - 6x + 9 = (x-3)^2$.

求导2x−4，x=3时斜率2，点为(3,2)，得切线。原函数减切线后配方。

M1	求切线
A1	切线正确
M1	两函数相减
A1	完全平方关系

(b) · 3 marks

$16/3$ square units.

差函数全段非负，可直接积分 $\int_1^5 (x-3)^2 dx$，或展开为x^2−6x+9；结果16/3。

M1	正确积分界与差函数
M1	正确积分代界
A1	精确面积

(c) · 2 marks

The separation $(x-3)^2$ is symmetric about $x = 3$, and the two vertical boundaries are equally distant from it.

两条原边界不必各自关于x=3对称；只需垂直差函数对称且左右积分宽度相同，就能保证两块面积相等。

B1	使用差函数的对称性
B1	同时说明界限等距

Question 23

P2-C08-R23 · 7 marks

(a) · 2 marks

6 square units.

区间可加性给 $\int_2^5 f = -1 - 5 = -6$，函数非正，所以几何面积为6。

M1	用积分可加性
A1	由非正函数转成正面积

(b) · 3 marks

$k = 2$.

移高k在宽3区间增加有向积分3k，因此−6+3k=0，得k=2。

M1	常数移高产生3k
A1	正确积分方程
A1	k

(c) · 2 marks

No. Its signed integral is zero, but the sizes of its positive and negative regions are not individually known.

仅知道正负有向面积相等，没有各自数值；移高后过轴位置也未知，因此无法确定总几何面积。

B1	不能确定
B1	说明缺少正负分区或面积信息

Question 24

P2-C08-R24 · 9 marks

(a) · 4 marks

$45/2 - 3\sqrt{3} - 3\sqrt{6} \approx 9.96\ \text{m}^2$.

先估曲线下方面积，步长3，估计3/2[1+4+2(1+√3+1+√6)]；从宽9、高4的矩形36中扣除。

M1	正确节点高度与步长
M1	梯形下方面积
M1	矩形补集
A1	估计面积

(b) · 3 marks

$9\ \text{m}^2$.

曲线与y=4在x=9相交，面积 $\int_0^9 (3 - \sqrt{x})dx = 27 - 18 = 9$。

M1	正确差函数及界
M1	正确积分
A1	精确面积

(c) · 2 marks

Not viable: $9 < 9.5$, although the estimate is above 9.5.

估计约9.96会误判达标；它是上方补集的高估，准确积分9低于规定门槛。

B1	正确决定
B1	估计可能误导的数值依据

Question 25

P2-C08-R25 · 9 marks

(a) · 3 marks

$k = 4$.

两面积分别 $\int_0^4 x^2 dx = 64/3$ 与 $k\int_0^4 \sqrt{x} dx = 16k/3$，相等给4。

M1	正确列两个积分面积
A1	两积分正确
A1	k

(b) · 2 marks

$a = 4^{2/3}$.

解$x^2=4\sqrt{x}$，除$\sqrt{x}$只在x>0合法，得x^(3/2)=4，故a=4^(2/3)；原点也是交点但此问只要正点。

M1	在正数范围化幂方程
A1	正交点横坐标

(c) · 4 marks

$32/3$ square units.

在0至a根式曲线在上，此块面积 $\int_0^a (4\sqrt{x} - x^2) dx = (8/3)a^{3/2} - a^3/3 = 16/3$。两原总面积相等使两差区域同面积，所以总为32/3。

M1	正确第一分区积分
M1	使用a^(3/2)=4与a^3=16
A1	单块16/3
B1	利用原总面积相等得到另一块同大

Question 26

P2-C08-R26 · 8 marks

(a) · 3 marks

$x^2 - 4x + h = 0 \Rightarrow (x-2)^2 = 4 - h.$

将曲线与水平线联立并配方。

M1	联立
M1	配方
A1	两根

(b) Given that the chord has length 2, find

(i) · 2 marks

$h = 3.$

两横坐标差为$2\sqrt{(4-h)}=2$，得$h=3$。

M1	弦长关系
A1	h

(ii) · 3 marks

$4/3$ square units.

此时交点1、3，积分$\int_1^3[x(4-x)-3]dx$，原函数$-x^3/3+2x^2-3x$，两端代值相减。

M1	正确差与积分界
M1	原函数
A1	精确面积

Question 27

P2-C08-R27 · 6 marks

(a) · 2 marks

13.0.

步长1，$T=\frac{1}{2}[2+4+2(2.8+3.4+3.8)]=13$。

M1	步长与端点权重
A1	估计值

(b) For $0 \le x \le 4$, the upper curve is $y=2f(x)+3$ and the lower line is $y=1$.

(i) · 2 marks

34.0.

竖直间距$2f(x)+2$，面积$2I+8$，用$I \approx 13$得34；不用重新列表。

M1	面积转为2I+8
A1	估计值

(ii) · 2 marks

Underestimate.

原曲线高于各弦，梯形和低估I。对面积$2I+8$，乘以正数2并加常数不改变误差方向，所以仍为低估。

B1	原梯形误差方向
B1	变换后仍低估

Question 28

P2-C08-R28 · 6 marks

(a) · 2 marks

$46:29$.

下方面积估计18.4，上方面积$30-18.4=11.6$；比$184:116=46:29$。

M1	用矩形减积分
A1	最简比

(b) The curve lies below each straight chord used by the trapezium rule.

(i) · 2 marks

Too small.

梯形高估曲线下方面积，而上方面积是固定矩形减去该值，所以低估上方面积。

B1	下方面积高估
B1	余面积低估

(ii) · 2 marks

Too large.

分子被高估、正的分母被低估，两种影响都使下方与上方之比偏大。

B1	识别分子与分母的误差
B1	比值方向

Question 29

P2-C08-R29 · 8 marks

(a) · 3 marks

$l: y=-2x+5$; separation $(x-1)^2$.

$y\prime=2x-4$，切点$(1,3)$，切线$y=-2x+5$；相减x^2-2x+1。

M1	切线方程
A1	切线
A1	平方差距

(b) Find

(i) · 2 marks

3.

$\int_0^3(x-1)^2dx=[(x-1)^3/3]_0^3=3$。

M1	非负差积分
A1	面积

(ii) · 3 marks

$a=1+\sqrt[3]{7/2}$.

左部面积$[(a-1)^3+1]/3=3/2$，得$(a-1)^3=7/2$；实根在所要求区间。

M1	等面积方程
M1	正确积分上下限
A1	参数

Question 30

P2-C08-R30 · 7 marks

(a) · 3 marks

$f(x) = 2x^2 - 6x + 4$; $(2, 0)$.

积分得$2x^2 - 6x + c$，代入$(1, 0)$得$c = 4$，所以$f(x) = 2(x - 1)(x - 2)$。

M1	重建曲线
A1	常数与因式
A1	另一个截距

(b) Let A be the total area between the curve and the x-axis on $0 \le x \le 2$.

(i) · 2 marks

2.

在$[0, 1]$函数非负、在$[1, 2]$非正。原函数$2x^3/3 - 3x^2 + 4x$在$1, 2$分别为$5/3, 4/3$；总面积$5/3 + (5/3 - 4/3) = 2$。

M1	在零点分段取正面积
A1	总面积

(ii) · 2 marks

$4/3$.

有向积分$4/3$，将轴下方的$1/3$作为负数；几何总面积则将其作为正数，所以与2不同。

M1	有向积分
A1	正确解释差异